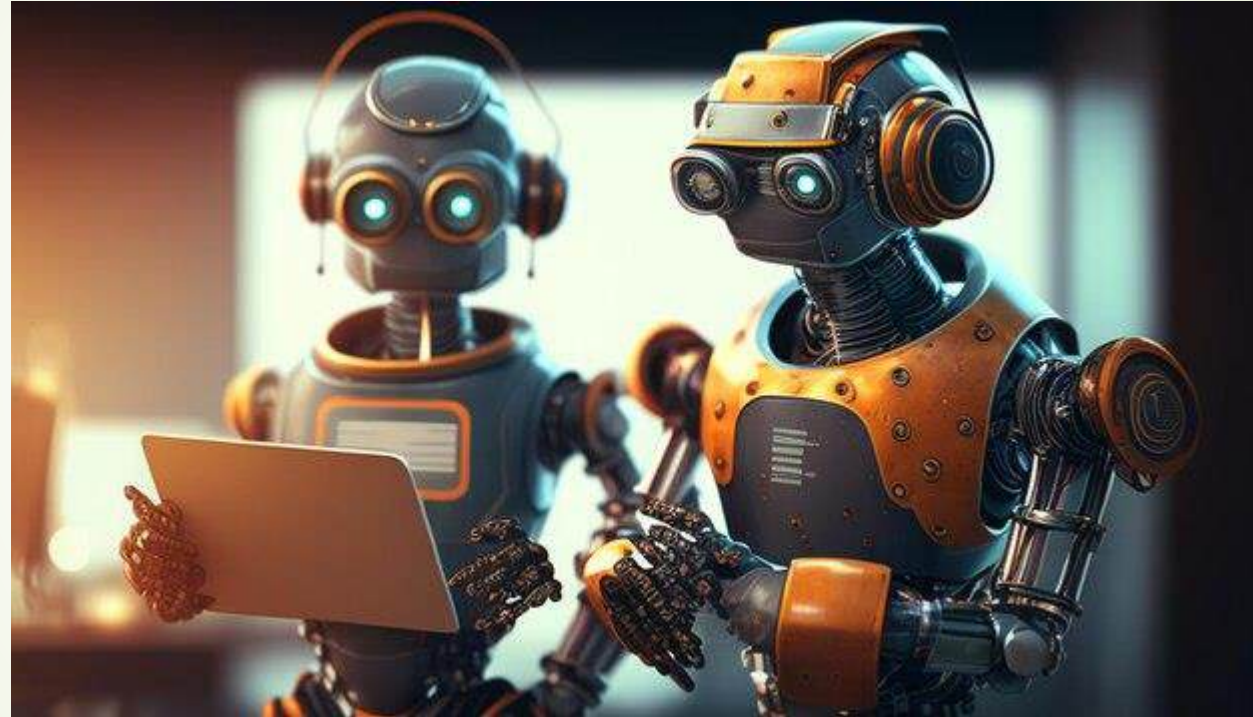


ROBOTICS

Topic 1: Introduction to Robots and Robotics



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Introduction to Robots and Robotics

A Few Questions



- ❖ What is a robot?
- ❖ What is robotics?
- ❖ Why do we study robotics?
- ❖ How can we teach a robot to perform a particular task?
- ❖ What are possible applications of robots?
- ❖ Can a human being be replaced by a robot? etc



Definitions

- ❖ The term: **robot** has come from the Czech word: **robota**, which means **forced** or slave laborer
- ❖ In 1921, **Karel Capek**, a Czech playwright used the term robot first in his drama **Rossum's Universal Robots (R. U. R)**
- ❖ According to **Karel Capek**, a robot is a machine look-wise similar to a human being

Robot has been defined in various different ways:

1. According to the **Oxford Dictionary**,

A machine capable of carrying out a complex series of actions automatically, especially one programmable by a computer

2. According to ISO (**international Organization for Standardization**): A robot is an automatically controlled, reprogrammable, multipurpose manipulator either fixed in place or mobile for use in industrial automation applications



3. According to Robot Institute of America (RIA)

It is a reprogrammable multi-functional manipulator designed to move materials, parts, tools or specialized devices through variable programmed motions for the performance of a variety of tasks.



Robotics



- ❖ It is a science, which deals with the issues related to design, manufacturing, usages of robots
- ❖ In 1942, the term **robotics** was introduced by Isaac Asimov in his story named **Runaround**
- ❖ In robotics, we use the fundamentals of **Physics**, **Mathematics**, **Mechanical Engg.**, **Electronics Engg.**, **Electrical Engg.**, **Computer Sciences** etc



3 Hs in Robotics

3 Hs of Human beings are copied into Robotics, such as

❖ Hand

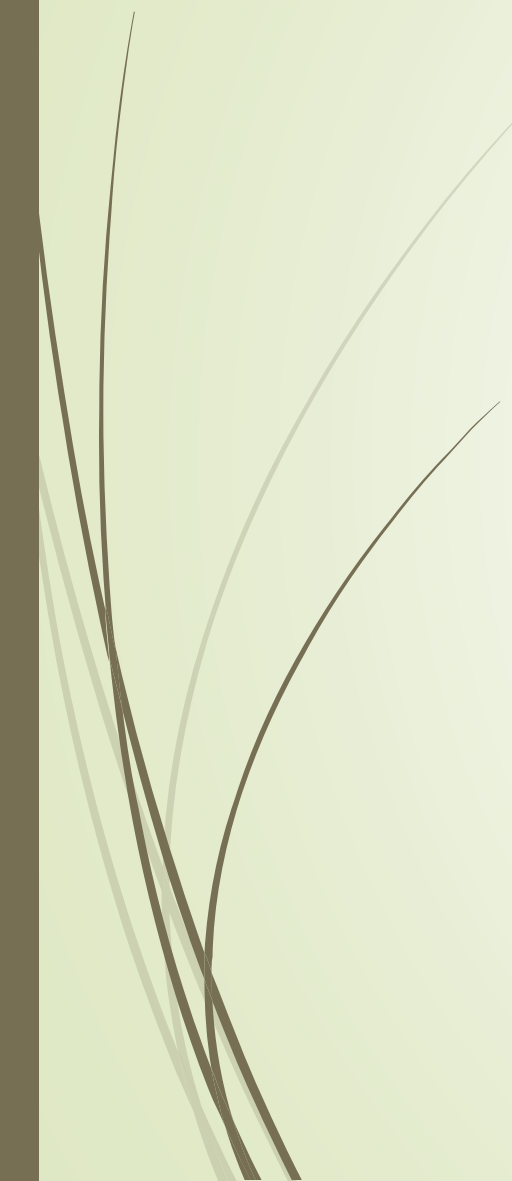
the hand of a human copied in the form of a manipulator to mimic the mechanical hand of humans

❖ Head

trying to copy the intelligence of humans

❖ Heart

giving the humanly emotions to robots





Motivation

To cope with increasing demands of a dynamic and competitive market, modern manufacturing methods should satisfy the following requirements:

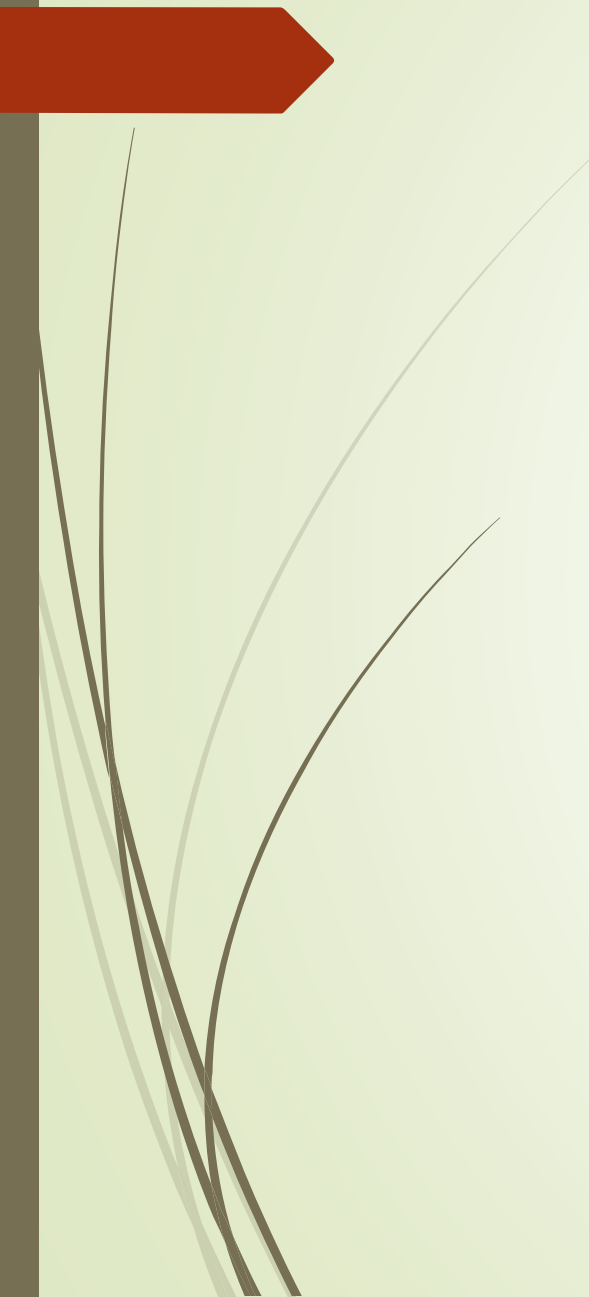
- ❖ Reduced production cost
- ❖ Increased productivity
- ❖ Improved product quality

Notes:

1. Automation can help to fulfil the above requirements
2. Automation : either Hard or Flexible automation
3. Robotics is an example of flexible automation

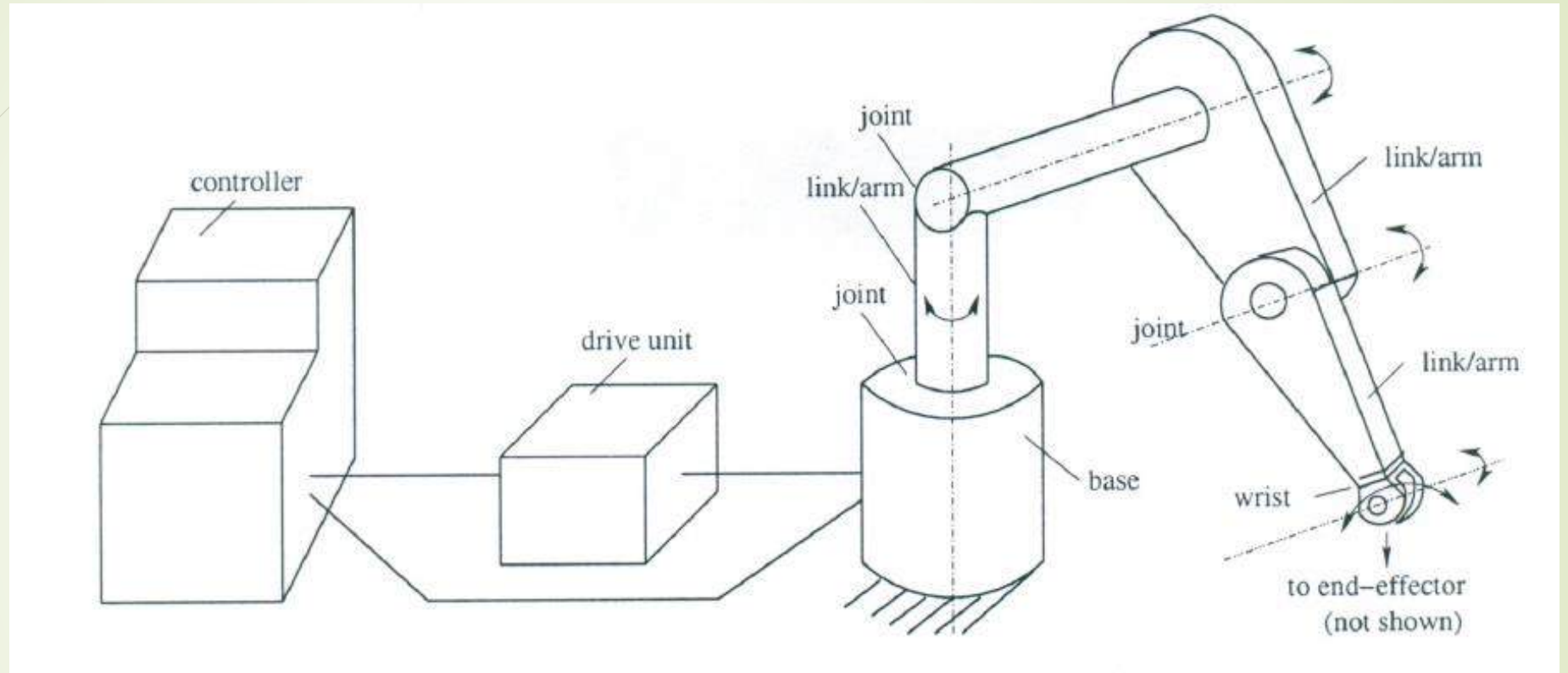
A brief History of Robotics

Year	Events and Development
1954	First patent on manipulator by George Devol , the father of robot
1956	Joseph Engelberger started the first robotics company: Unimation
1962	General Motors used the manipulator: Unimate in die-casting application
1967	General Electrical Corporation made a 4-legged vehicle
1969	1. SAM was built by the NASA, USA 2. Shakey, an intelligent mobile robot, was built by Stanford Research Institute (SRI)
1970	1. Victor Scheinman demonstrated a manipulator known as Stanford Arm 2. Lunokhod I was built and sent to the moon by USSR 3. ODEX 1 was built by Odetics



Year	Events and Development
1973	Richard Hohn of Cincinnati Milacron Corporation manufactured T³ (The Tomorrow Tool) robot
1975	Raibert at CMU, USA , built a one-legged hopping machine, the first dynamically stable machine
1978	Unimation developed PUMA (Programmable Universal Machine for Assembly)
1983	Odetics introduced a unique experimental six-legged device
1986	ASV (Adaptive Suspension Vehicle) was developed at Ohio State University, USA
1997	Pathfinder and Sojourner was sent to the Mars by the NASA, USA
2000	Asimo humanoid robot was developed by Honda
2004	The surface of the Mars was explored by Spirit and Opportunity

A Robotic System



Various Components

1. Base,
2. Links and Joints,
3. End-effector / gripper,
4. Wrist,
5. Drive / Actuator
6. Controller
7. Sensors



Interdisciplinary Areas in Robotics

Mechanical Engineering

- ❖ **Kinematics**: Motion of robot arm without considering the forces and /or moments
- ❖ **Dynamics**: Study of the forces and/or moments
- ❖ **Sensing**: Collecting information of the environment

Computer Science

- ❖ **Motion Planning**: Planning the course of action
- ❖ **Artificial Intelligence**: To design and develop suitable brain for the robots

Electrical and Electronics Engg.

- ❖ **Control schemes** and **hardware** implementations

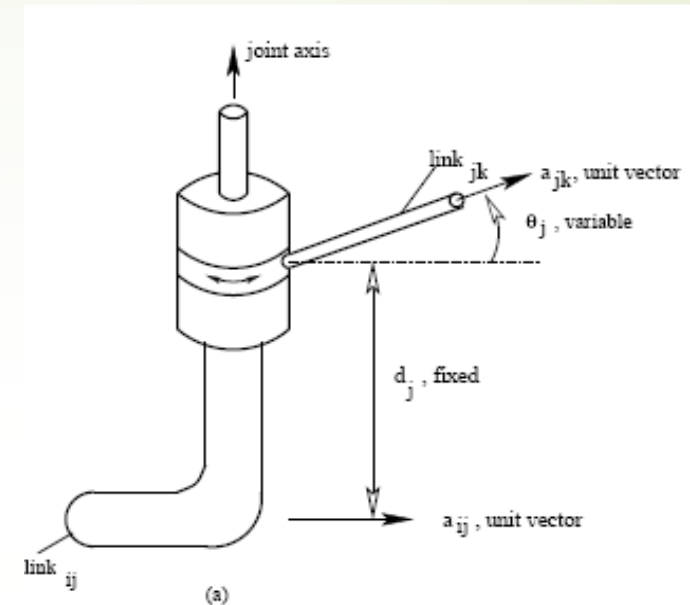
General Sciences

- ❖ **Physics**
- ❖ **Mathematics**

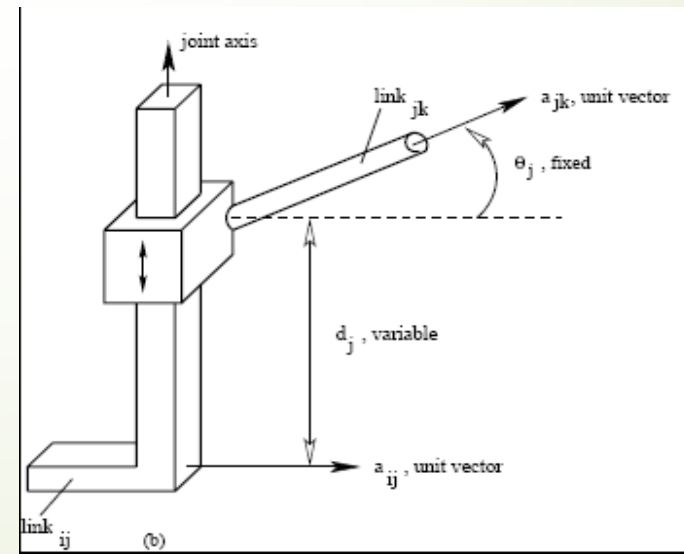
Connectivity / Degrees of Freedom of a Joint

It indicates the number of rigid (bodies) that can be connected to a fixed rigid body through the said joint

Joints with One dof Revolute Joint (R)

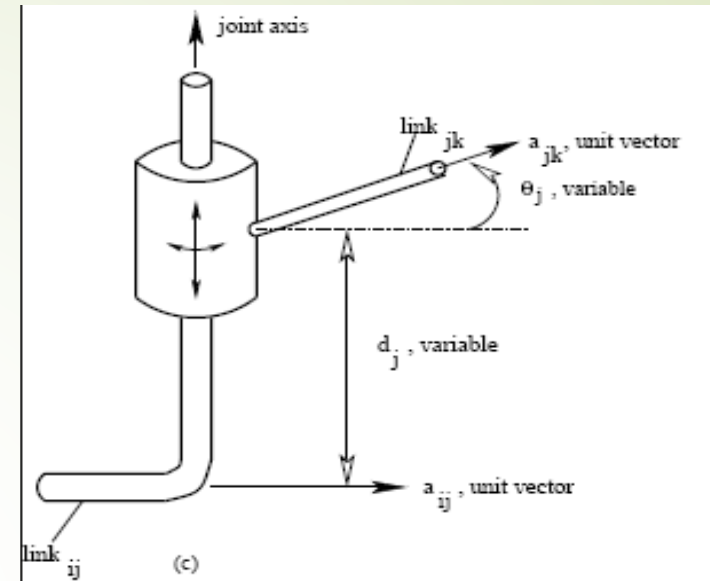


Prismatic Joint (P)

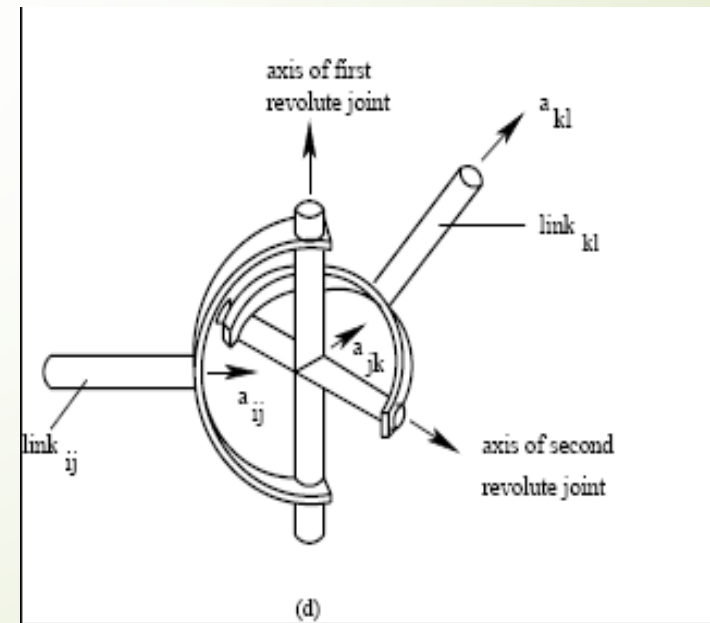


Joints with Two dof

Cylindrical Joint (C)

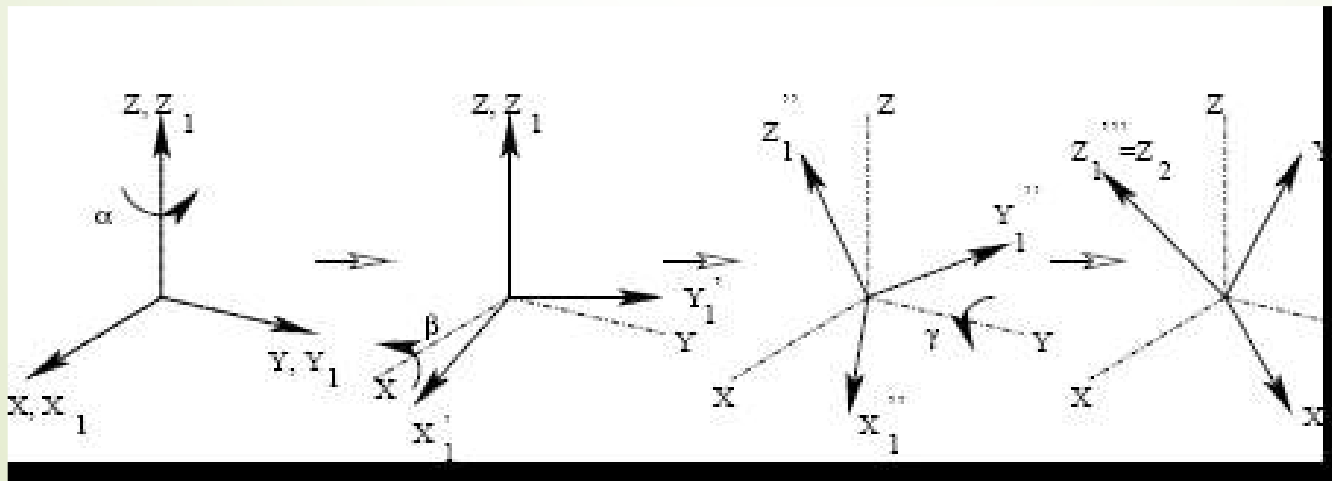
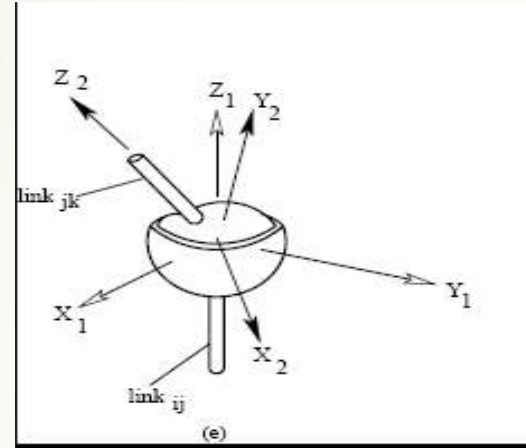


Hooke Joint or Universal Joint (U)



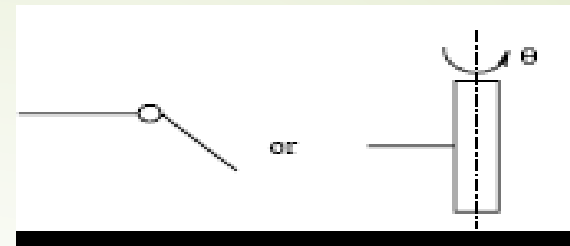
Joints with Three dof

Ball and Socket Joint / Spherical Joint (S¹)

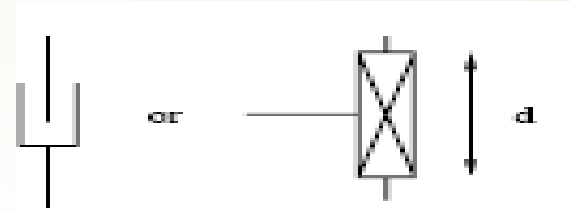


Representation of the Joints

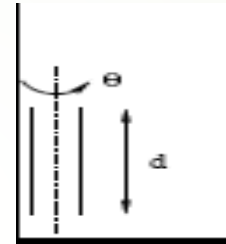
Revolute joint (R)



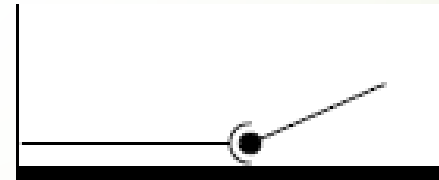
Prismatic joint (P)



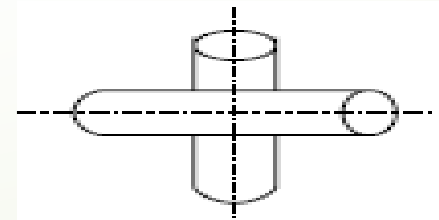
Cylindrical joint (C)



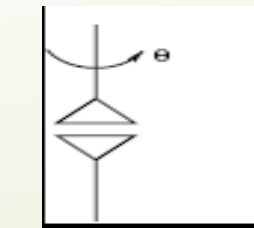
Spherical joint (S)



Hooke joint (U)



Twisting joint (T)





Degrees of Freedom of a System

It is defined as the minimum number of independent parameters / variables / coordinates needed to describe a system completely

Notes

- A point in 2-D: 2 dof; in 3-D space: 3 dof
- A rigid body in 3-D: 6 dof
- Spatial Manipulator: 6 dof
- Planar Manipulator: 3 dof

• Redundant Manipulator

Either a Spatial Manipulator with more than 6 dof
or a Planar Manipulator with more than 3 dof

• Under-actuated Manipulator

Either a Spatial Manipulator with less than 6 dof
or a Planar Manipulator with less than 3 dof

Mobility/dof of Spatial Manipulator

Let us consider a manipulator with n rigid moving links and m joints

C_i : Connectivity of i -th joint; $i = 1, 2, 3, \dots, m$

No. of constraints put by i -th joint = $6 - C_i$

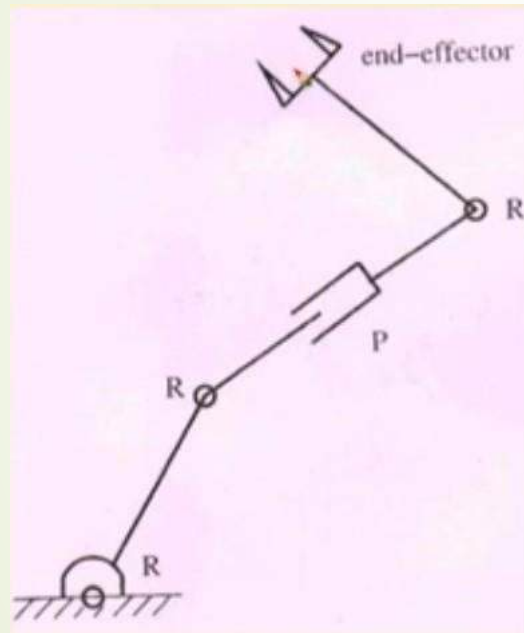
Total no. of constraints = $\sum_{i=1}^m (6 - C_i)$

Mobility of the manipulator $M = 6n - \sum_{i=1}^m (6 - C_i)$

It is known as **Grubler's criterion**.

Numerical Example

Serial Planar Manipulator



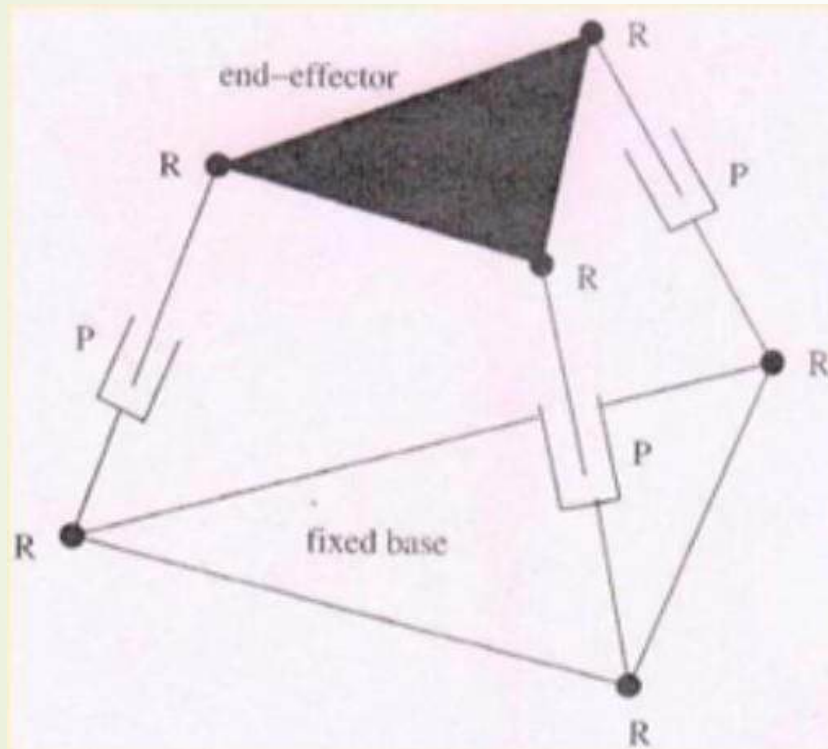
$$n=4, \quad m=4$$

$$C1 = C2 = C3 = C4 = 1$$

Mobility/ dof:

$$M = 3n - \sum_{i=1}^m (3 - C_i) = 3 \times 4 - 8 = 4$$

Parallel planar Manipulator



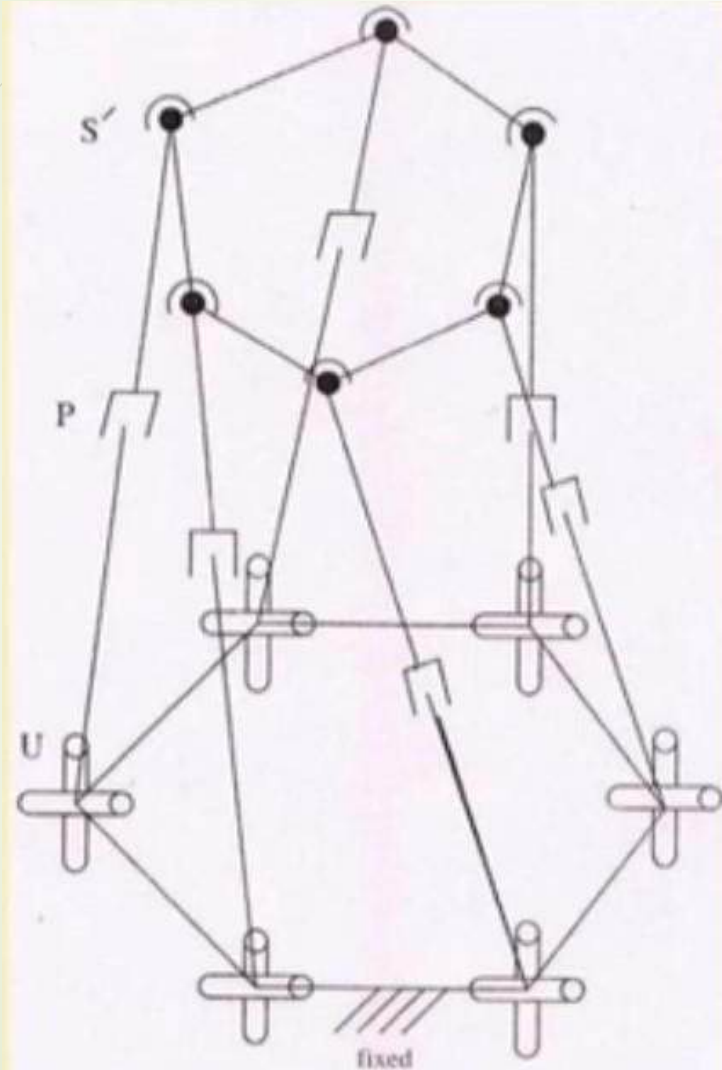
$$n=7, \quad m=9$$

$$C_i = 1, \text{ where } i = 1, \dots, 9$$

Mobility/ dof:

$$M = 3n - \sum_{i=1}^m (3 - C_i) = 3 \times 7 - 18 = 3$$

Parallel spatial Manipulator



$$n=13, \quad m=18$$

Mobility/ dof:

$$M = 6n - \sum_{i=1}^m (6 - C_i) = 6 \times 13 - 72 = 6$$

Classifications of Robots

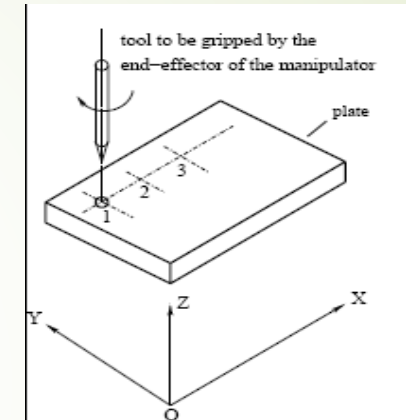
❖ Based on the Type of Tasks Performed

1. Point-to-Point Robots

Examples:

Unimate 2000

T³

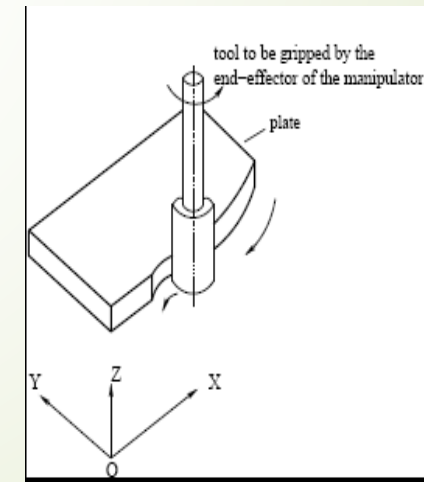


2. Continuous Path Robots

Examples

PUMA

CRS





❖ Based on the Type of Controllers

1. Non-Servo-Controlled Robots

❖ Open-loop control system

Examples: Seiko PN-100

- Less accurate and less expensive

2. Servo-Controlled Robots

❖ Closed-loop control system

Examples: Unimate 2000

PUMA

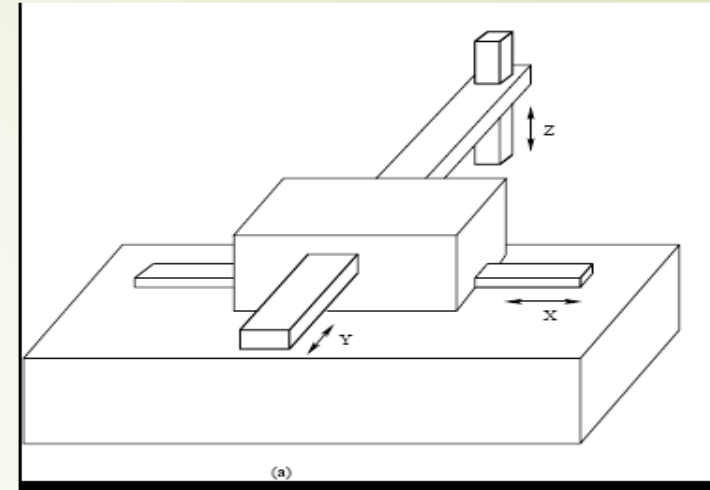
T3

- More accurate and more expensive

❖ Based on Configuration (coordinate system) of the Robot

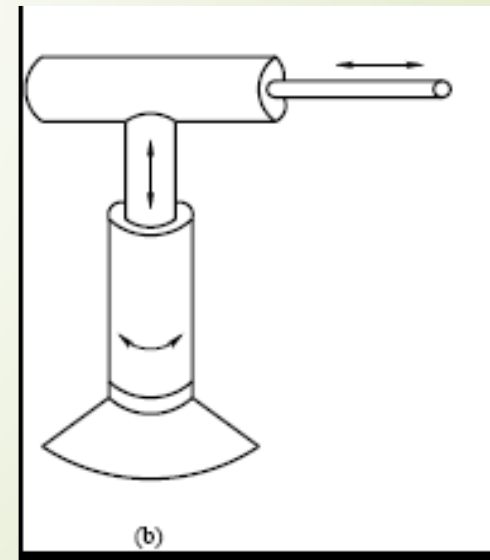
1. Cartesian Coordinate Robots

- Linear movement along three different axes
- Have either sliding or prismatic joints, that is, SSS or PPP
- Rigid and accurate
- Suitable for pick and place type of operations
- Examples: IBM's RS-1, Sigma robot



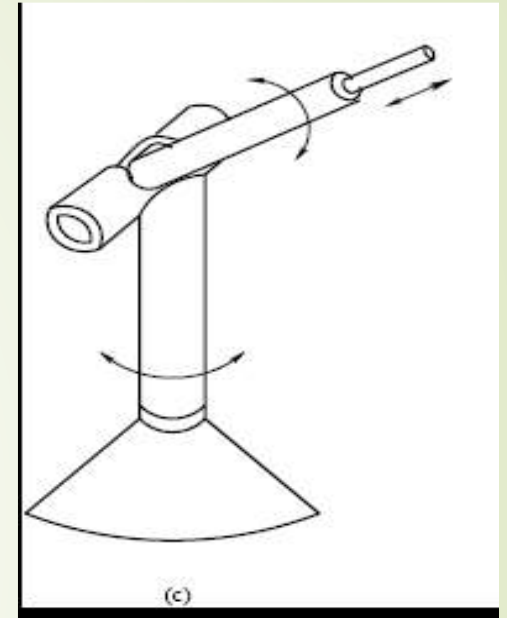
2. Cylindrical Coordinate Robots

- Two linear and one rotary movements
- Represented as TPP, TSS
- Used to handle parts/ objects in manufacturing
- Cannot reach the objects lying on the floor
- Poor dynamic performance
- Examples: Versatran 600



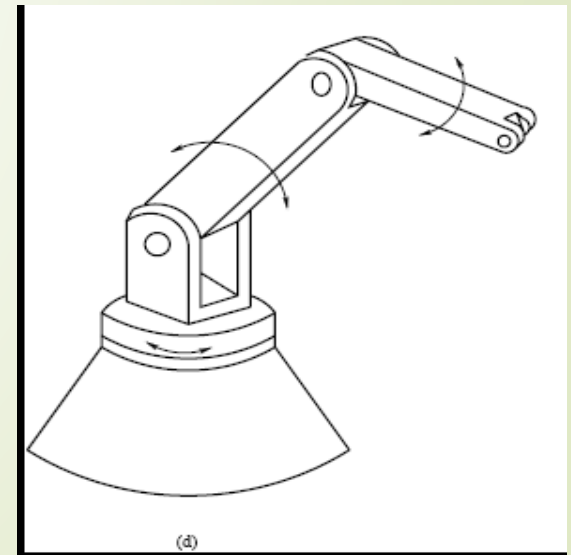
3. Spherical Coordinate or Polar Coordinate Robots

- One linear and two rotary movement
- Represented as TRP, TRS
- Suitable for handling parts/objects in manufacturing
- Can pick up objects lying on the floor
- Poor dynamic performance
- Examples: Unimate 2000B



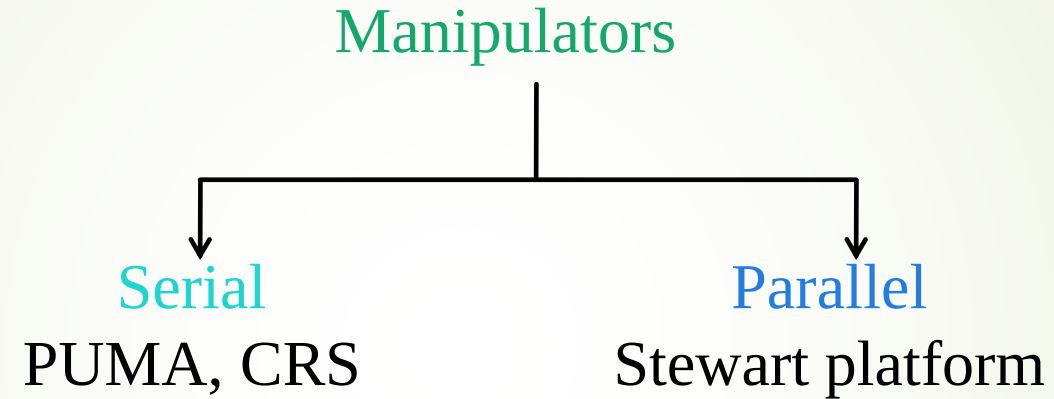
4. Revolute Coordinate or Articulated Coordinate Robots

- Rotary movement about three independent axes
- Represented as TRR
- Suitable for handling parts/components in manufacturing system
- Rigidity and accuracy may not be good enough
- Examples: T³, PUMA

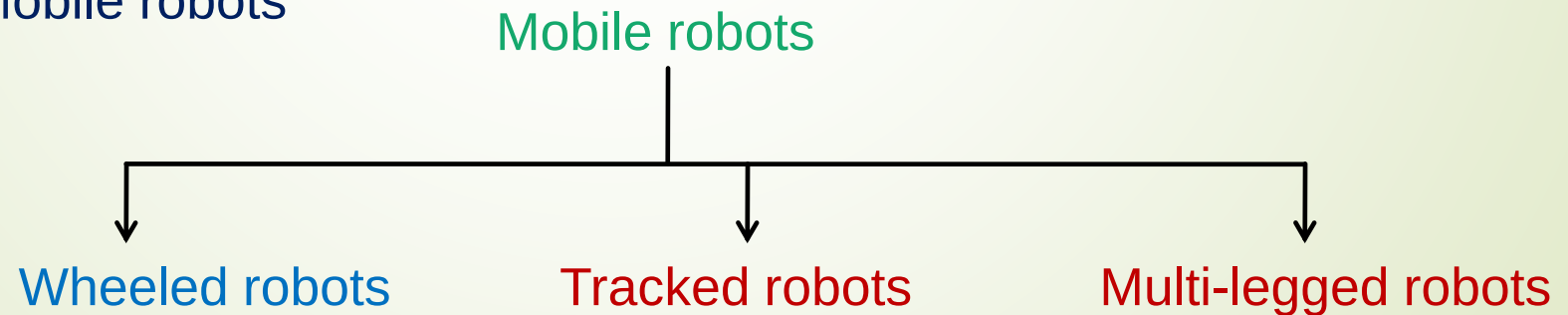


❖ Based on Mobility Levels

1. Robots with fixed base (also known as **manipulators**)

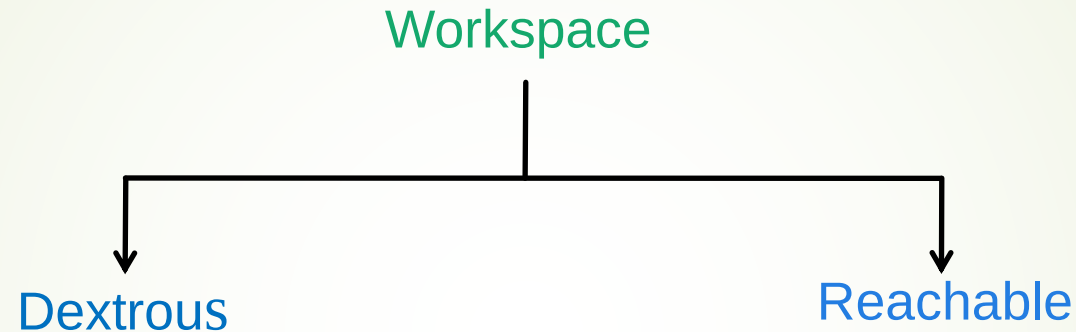


2. Mobile robots



Workspace of Manipulators

It is the volume of space that the end-effector of a manipulator can reach



Dextrous Workspace

It is the volume of space, which the robot's end-effector can reach with various orientations

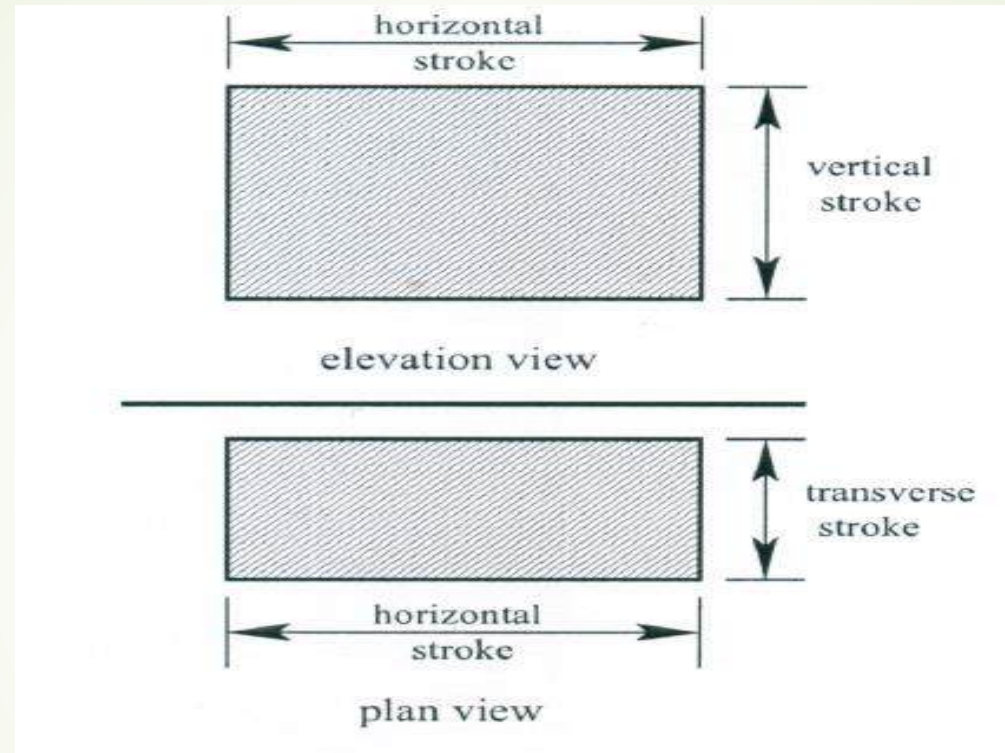
Reachable Workspace

It is the volume of space that the end-effector can reach with a minimum of one orientation

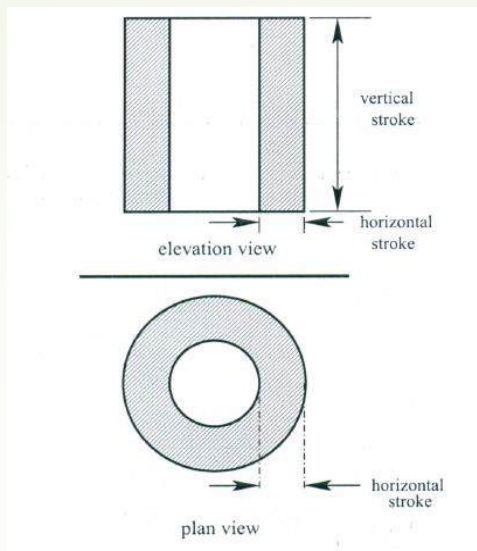
Note

Dextrous workspace is a subset of the reachable workspace

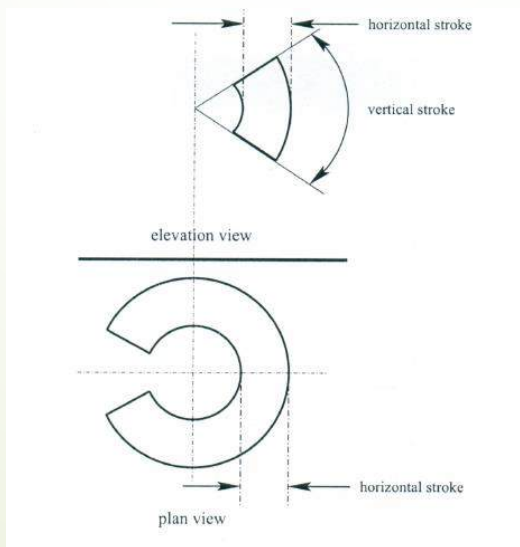
Workspace of Cartesian Coordinate Robot



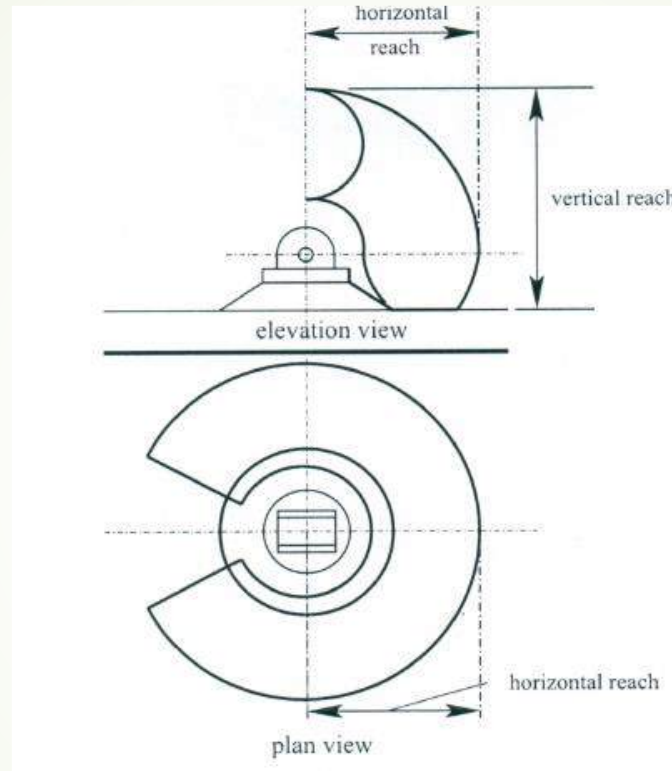
Workspace of Cylindrical Coordinate Robot



Workspace of Spherical Coordinate Robot



Workspace of Revolute Coordinate Robot

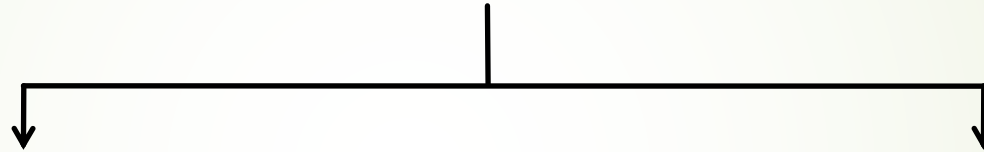


Resolution, Accuracy and Repeatability

Resolution

It is defined as the smallest allowable position increment of a robot

Resolution



Programming resolution

Smallest allowable position increment in robot programme
Basic Resolution Unit
BRU = 0.01 inch/0.1degree

Control resolution

Smallest change in position that the feedback device can measure say 0.36 degrees per pulse



Accuracy (mm)

It is the precision with which a computed point can be reached

Repeatability (mm)

It is defined as the precision with which a robot re-position itself to a previous taught point

Applications of Robots

❖ In Manufacturing Units

Advantages of Robots

1. Robots can work in hazardous and dirty environment
2. Can increase productivity after maintaining improved quality
3. Direct labour cost will be reduced
4. Material cost will be reduced
5. Repetitive tasks can be handled more efficiently



Application Areas

- 1.Arc Welding
- 2.Spot Welding
- 3.Spray Painting
- 4.Pick and Place Operation
- 5.Grinding
- 6.Drilling

❖ Under-Water Applications

Purposes

- 1.To explore various resources
- 2.To study under-water environment
- 3.To carry out drilling, pipe-line survey, inspection and repair of ships

Notes

- Robots are developed in the form of ROV (Remotely Operated Vehicle) and AUV (Autonomous Under-water Vehicle)
- Robots are equipped with navigational sensors, propellers/ thrusters, on-board softwares, and others

❖ Medical Applications

1. Telesurgery
2. Micro-capsule multi-legged robots
3. Prosthetic devices

❖ Space Applications

1. For carrying out on-orbit services, assembly job and interplanetary missions
2. Spacecraft deployment and retrieval, survey of outside space shuttle; assembly, testing, maintenance of space stations; transport of astronauts to various locations
3. Robo-nauts
4. Free-flying robots
5. Planetary exploration rovers



❖ In Agriculture

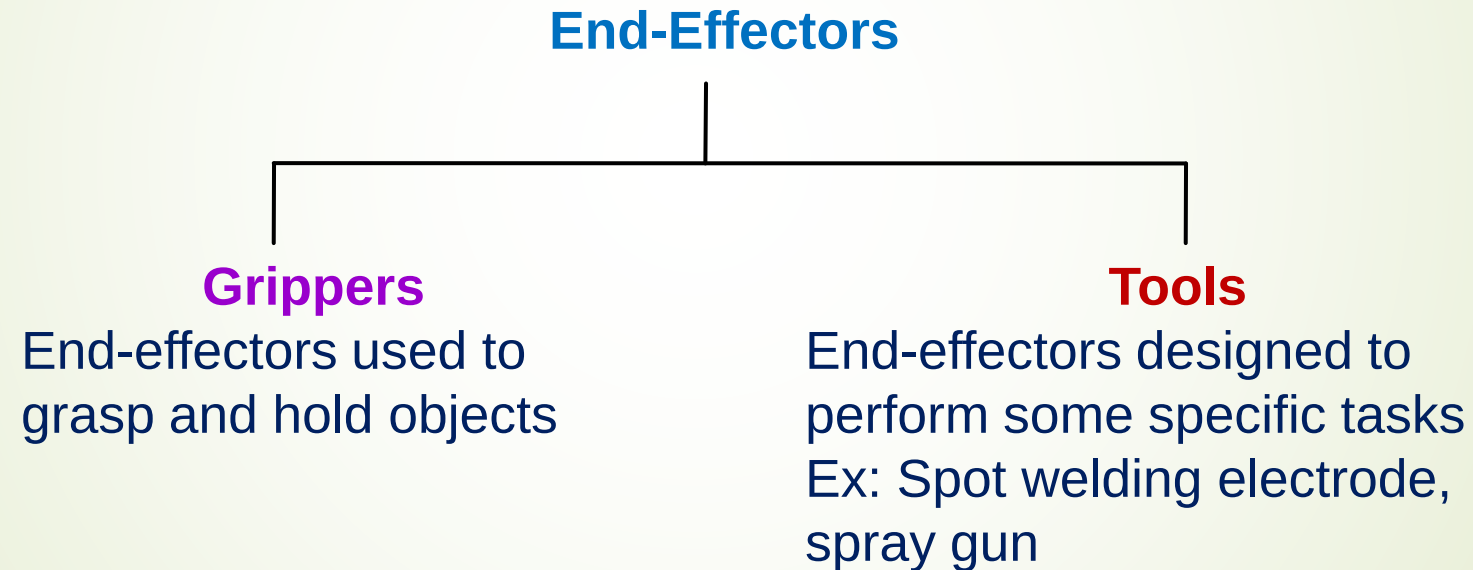
1. For spraying pesticides
2. For spraying fertilizers in liquid form
3. Cleaning weeds
4. Sowing seeds
5. Inspection of plants

Some Other Applications

- ❖ Replacement for maid-servant
- ❖ Garbage collection
- ❖ Underground coal mining
- ❖ Sewage-line cleaning
- ❖ Fire-fighting etc.

Robot End-Effectors

An end-effector is a device attached to the wrist of a manipulator for the purpose of holding materials, parts, tools to perform a specific task



Classification of Grippers

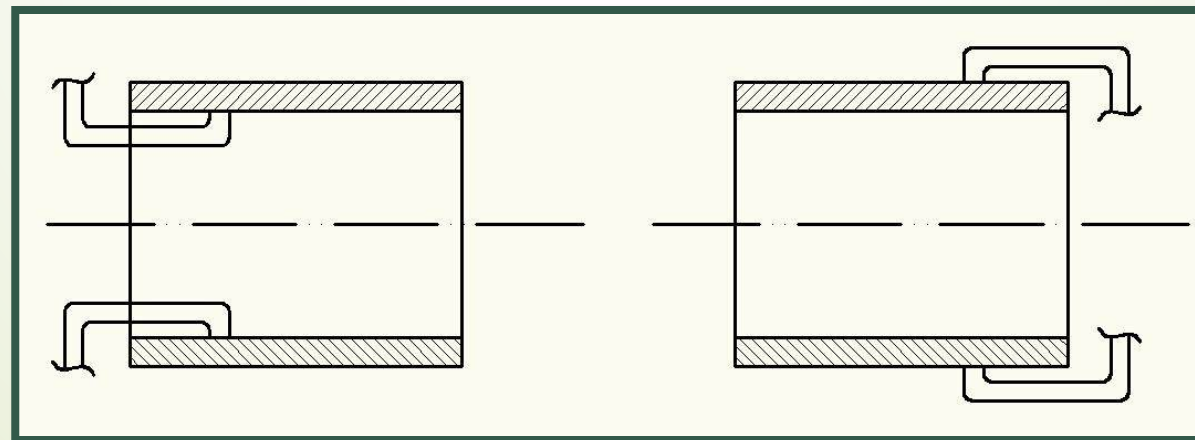
1. Single gripper and double gripper

Single gripper: Only one gripping device is mounted on the wrist

Double gripper: Two independent gripping devices are attached to the wrist

Example: Two separate grippers mounted on the wrist for loading and unloading applications

2. Internal gripper vs. External gripper



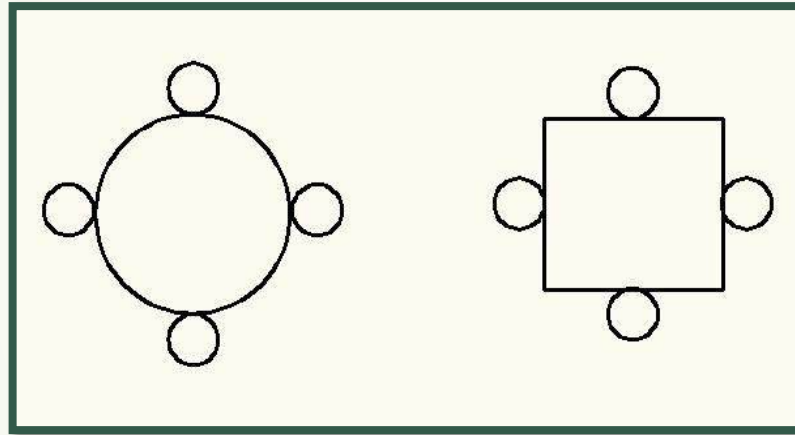
Internal gripper

External gripper

3. **Soft** gripper vs. **Hard** gripper

Hard gripper: Point contact between the finger and object

Soft gripper: Area (surface) contact between the finger and object



4. **Active** gripper vs. **Passive** gripper

Active gripper: Gripper with sensor(s)

Passive gripper: Gripper without sensor(s)

Ex: Remote Center Compliance (RCC)

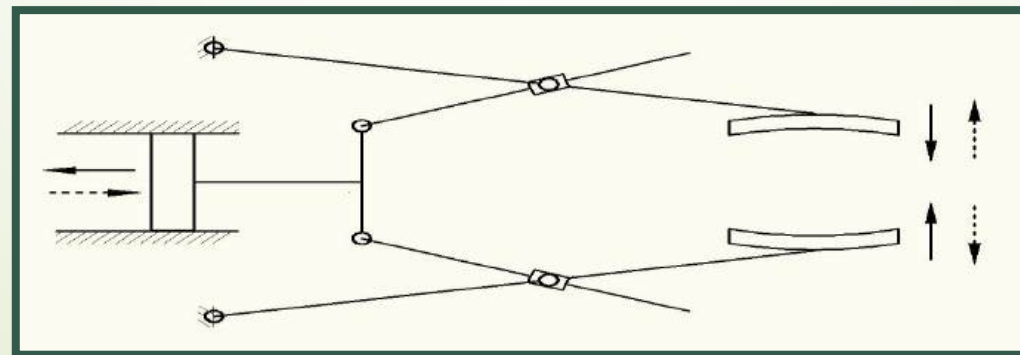
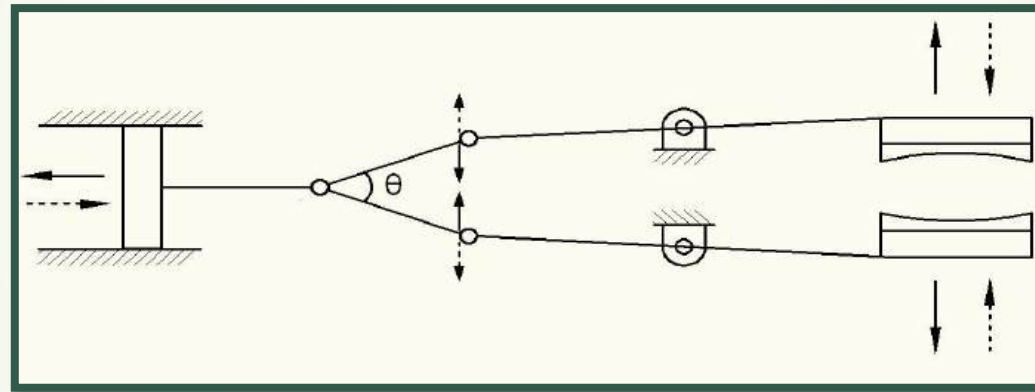
A Few Robot Grippers

1. Mechanical Grippers

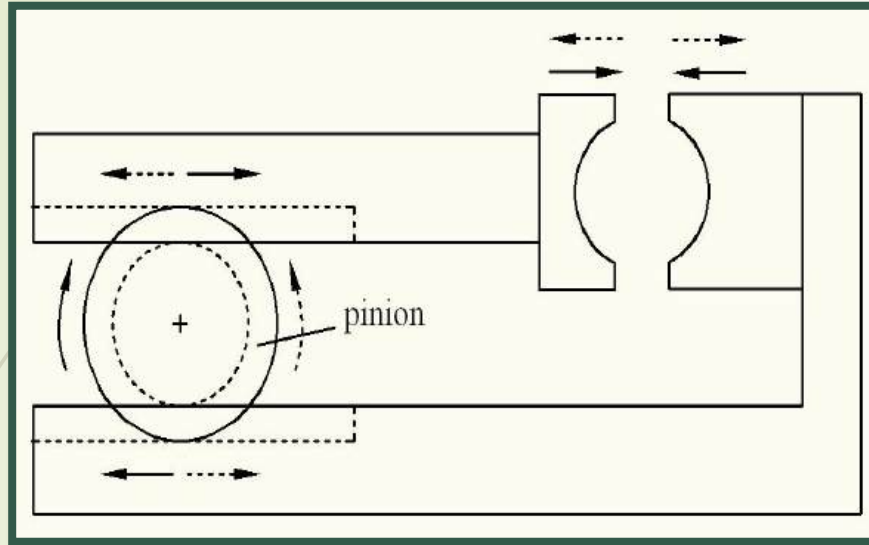
- Use mechanical fingers (jaws) actuated by some mechanisms
- Less versatile, less flexible and less costly

Examples

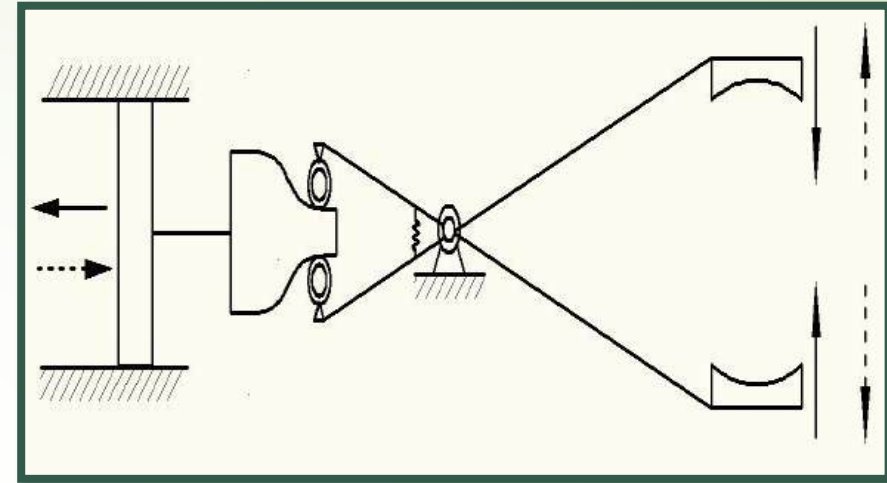
(i) Gripper with linkage actuation



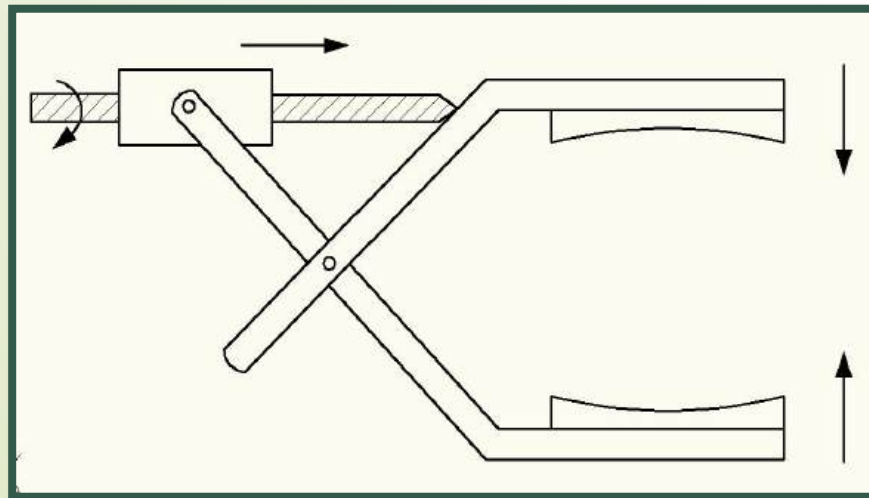
(ii) Gripper with rotary actuation



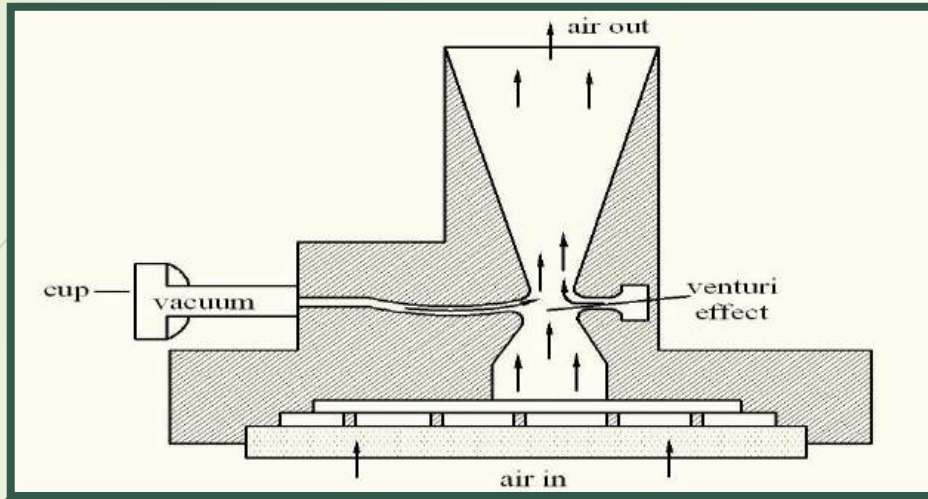
(iii) Gripper with cam actuation

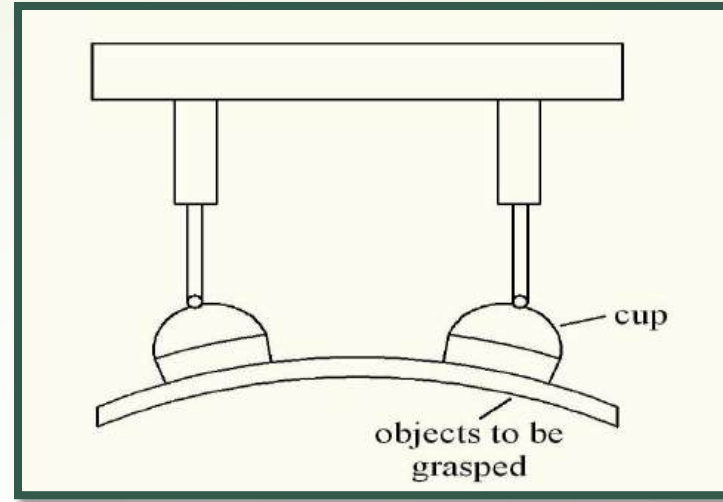


(iv) Gripper with screw actuation



2. Vacuum Gripper (used for thin parts)

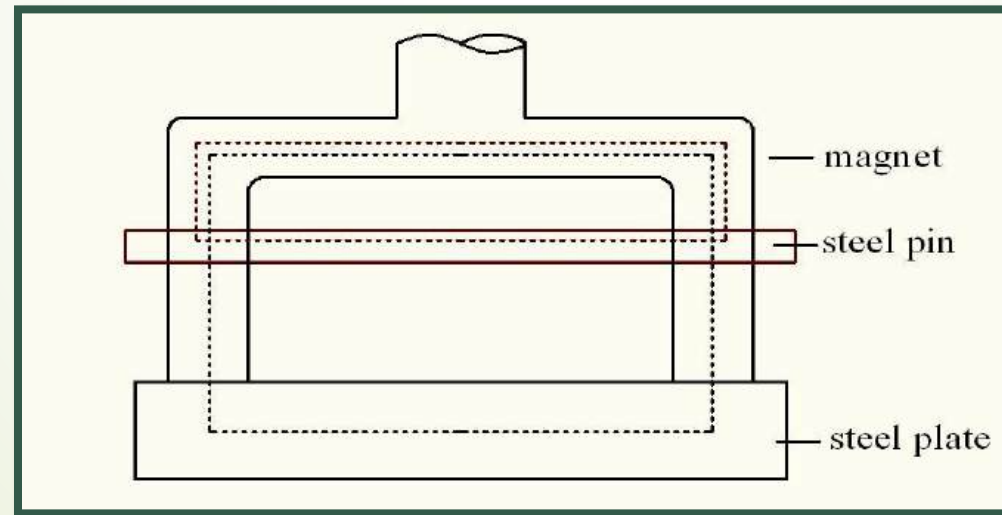


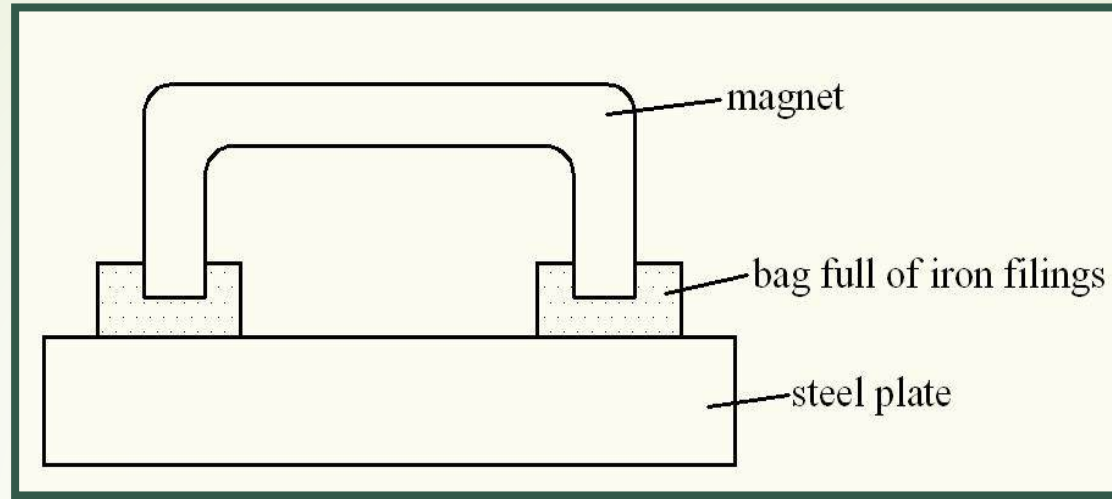


- Suction cup is made of elastic material like rubber or soft plastic
- When the object to be handled is soft, the cup should be made of hard substance
- Two devices can be used: Either Vacuum pump or venturi

3. **Magnetic Gripper** (for magnetic materials only. For example: various steels but not stainless steel)

- Can use either electro-magnets or permanent magnets
- Pick up time is less
- Can grip parts of various sizes
- Disadvantage: residual magnetism
- Stripping device: for separating the part from the permanent magnet
- For separating the part from electro-magnet, reverse the polarity





4. Adhesive Gripper

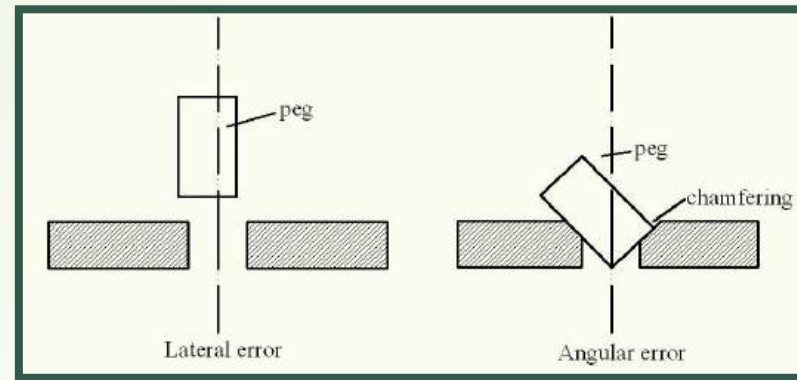
- Grasping action using adhesive substance
- To handle lightweight materials

5. Universal Gripper

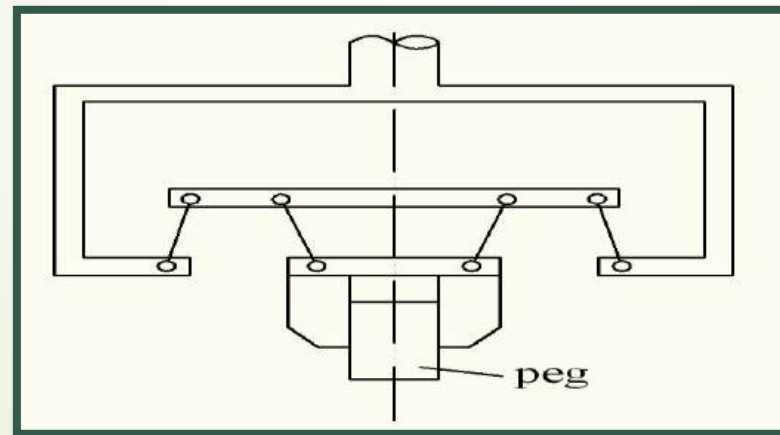
Example: Human gripper

Passive Gripper

Task: To insert a peg into a hole



Solution: Use Remote Center Compliance (RCC)



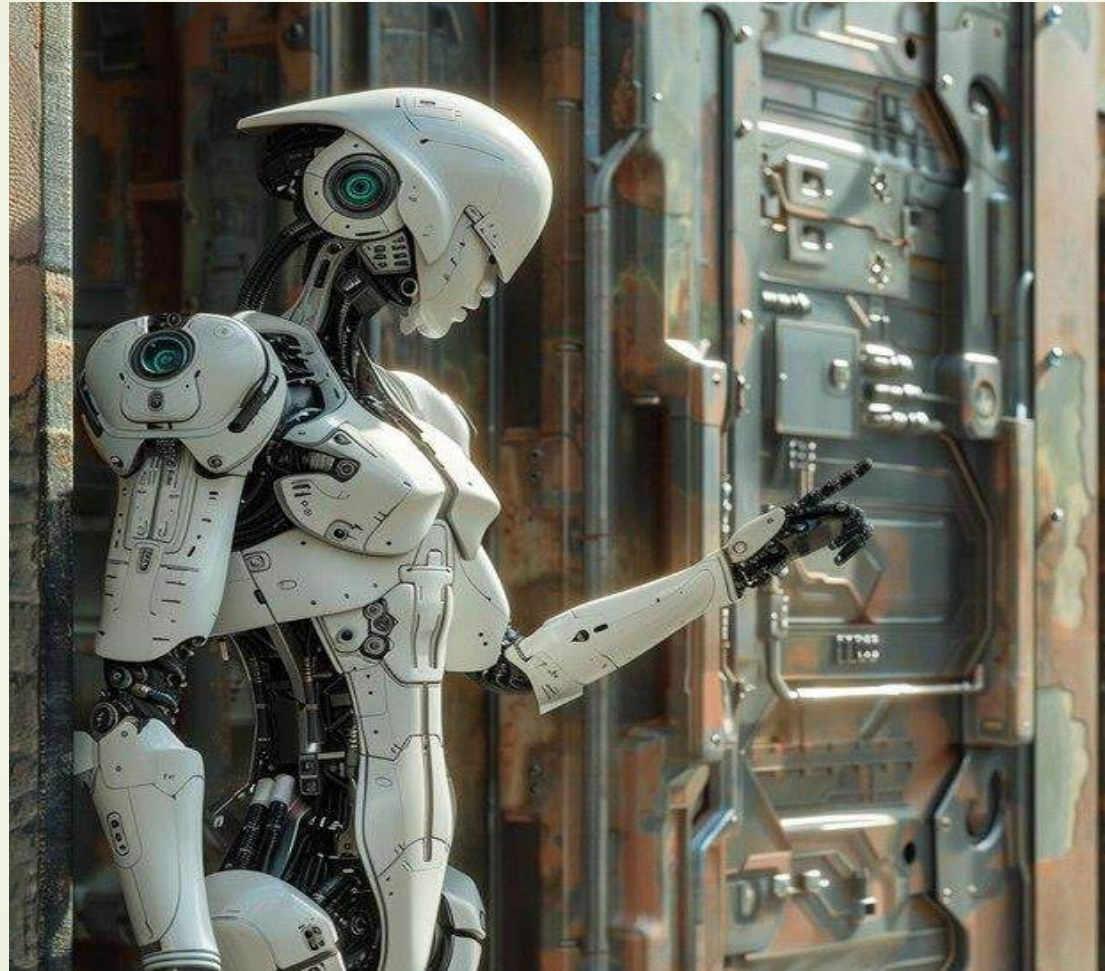
RCC is inappropriate for assembly of pegs in horizontal direction

Insertion angle must be less than 45 degrees

Cannot be used in chamferless insertion tasks

Robotics

Topic 2: Robot Kinematics

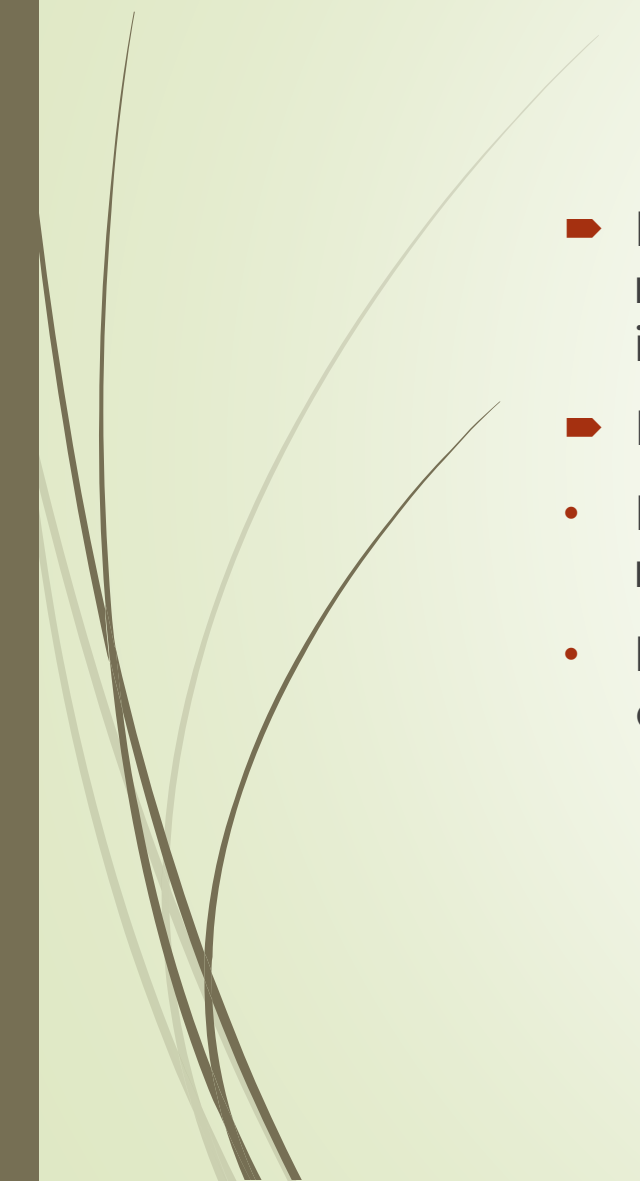


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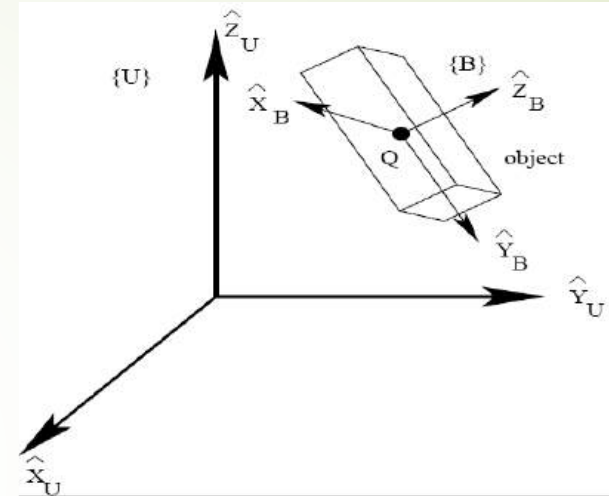
Dept. of Computer Science and Engineering, UMaT



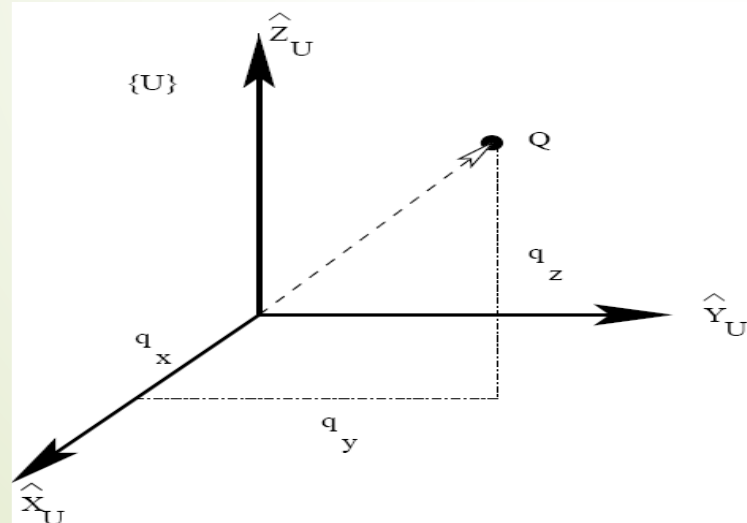
Kinematics

- **Kinematics** refers to the study of motion without considering forces. In robotics, robot kinematics focuses on how a robot moves and positions itself in space, specifically its position and orientation in a 3D environment.
 - Kinematics is categorized into:
 - **Forward Kinematics (FK):** Determines the position and orientation of the robot's end-effector given joint parameters.
 - **Inverse Kinematics (IK):** Determines joint parameters needed to achieve a desired position and orientation of the end-effector.
- 

Representation of an object in 3-D

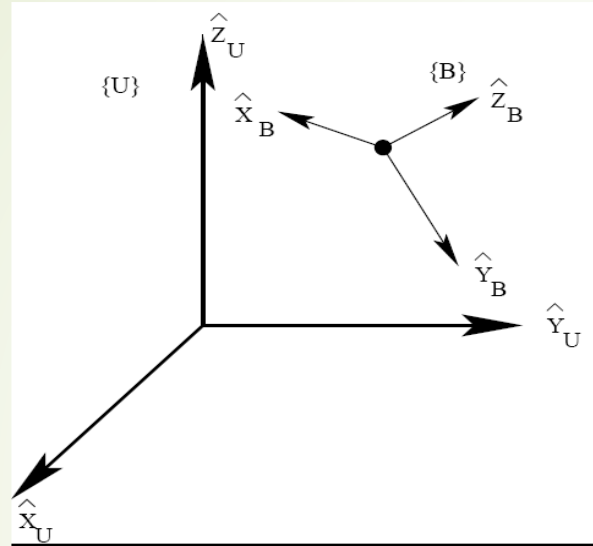


Representation of the position



$${}^U\bar{Q} = \begin{bmatrix} q_x \\ q_y \\ q_z \end{bmatrix} ; 3 \times 1 \text{ matrix}$$

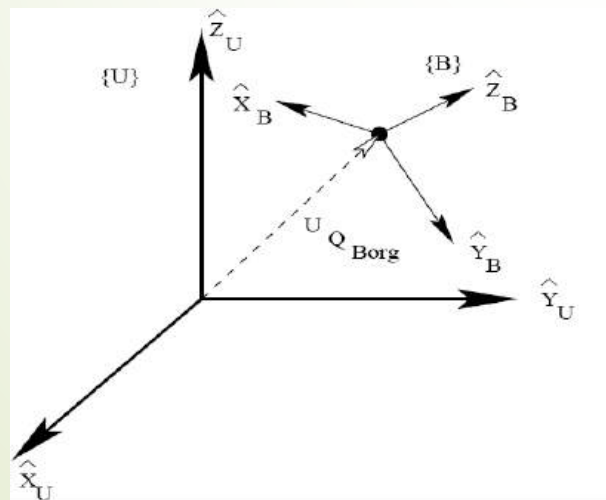
Representation of the Orientation



$${}^U_B \mathbf{R} = [{}^U \hat{X}_B \quad {}^U \hat{Y}_B \quad {}^U \hat{Z}_B]$$

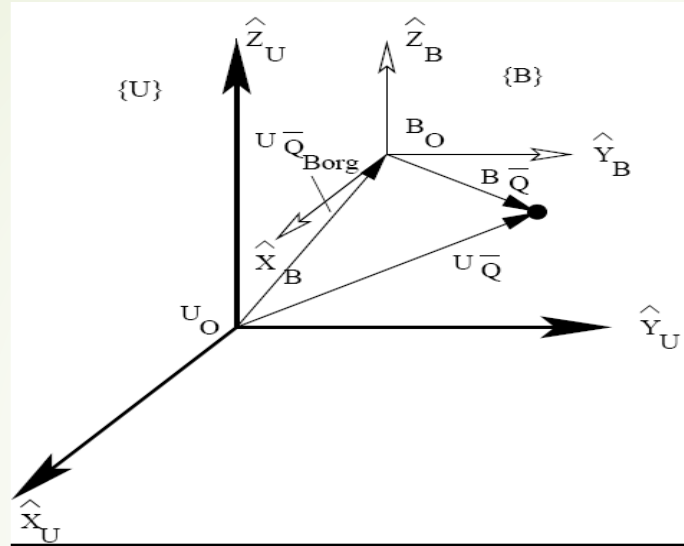
3x3 matrix

Frame Transformations



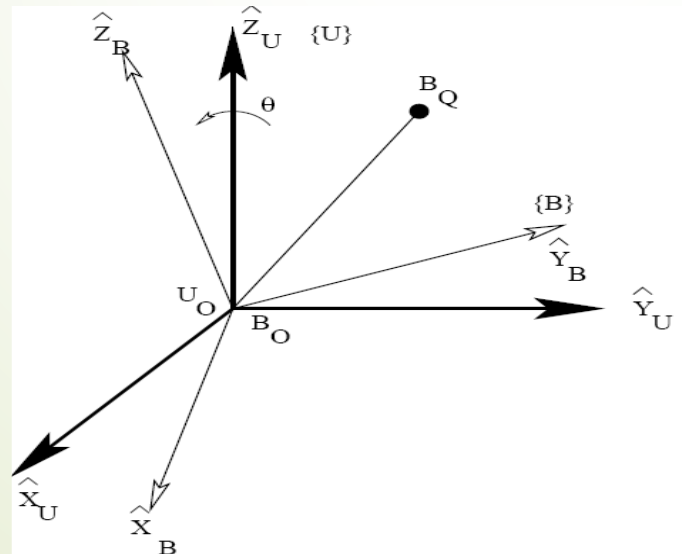
Frame: A set of four vectors carrying position and orientation information

Translation of a Frame



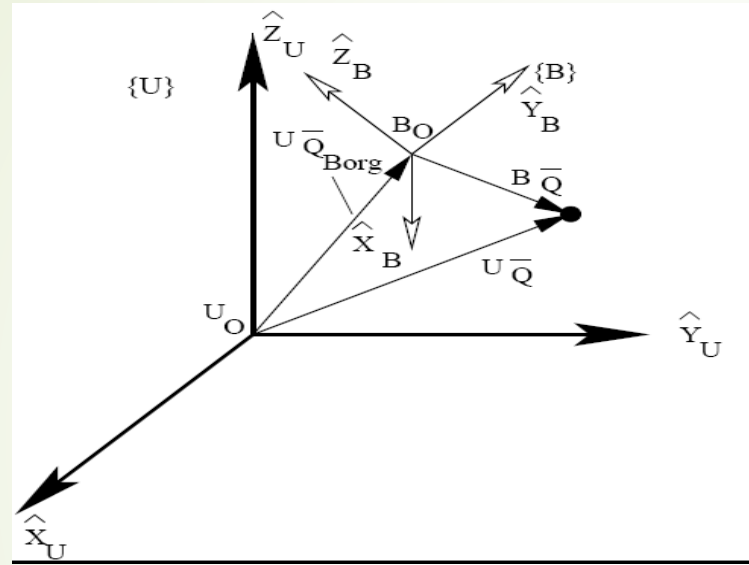
$${}^U\overline{Q} = {}^U\overline{Q}_{\text{Borg}} + {}^B\overline{Q}$$

Rotation of a Frame



$${}^U\overline{Q} = {}^U_B R {}^B\overline{Q}$$

Translation and Rotation of a Frame



$${}^U\bar{Q} = {}^U_B R {}^B\bar{Q} + {}^U\bar{Q}_{Borg}$$

$${}^U\bar{Q} = {}^U_B R {}^B\bar{Q} + {}^U\bar{Q}_{Borg}$$

$$\Rightarrow {}^U\bar{Q} = {}^U_B T {}^B\bar{Q}$$

where **T**: transformation

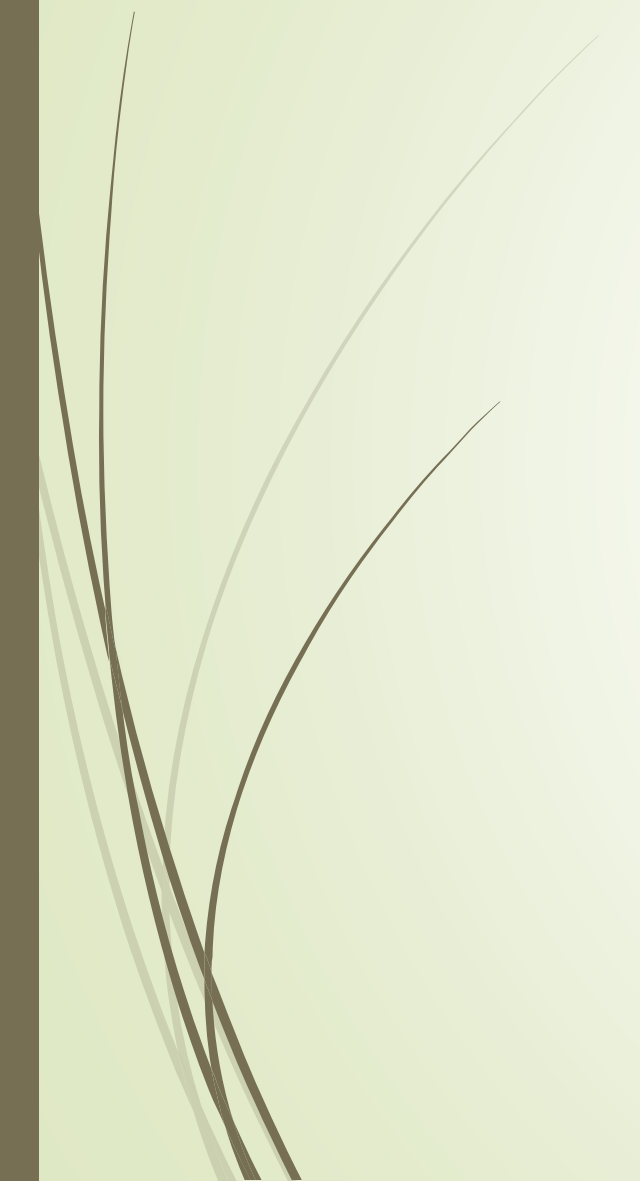


$$\Rightarrow \begin{bmatrix} {}^U\bar{Q}(3 \times 1) \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} = \left[\begin{array}{c|c} {}^U_B R(3 \times 3) & {}^U\bar{Q}_{\text{Borg}}(3 \times 1) \\ \text{---} & \text{---} \\ \text{---} & \text{---} \end{array} \right] \begin{bmatrix} {}^B\bar{Q}(3 \times 1) \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} {}^U\bar{Q}(3 \times 1) \\ \text{---} \\ \text{---} \\ 1 \end{bmatrix} = \left[\begin{array}{c|c} {}^U_B R(3 \times 3) & {}^U\bar{Q}_{\text{Borg}}(3 \times 1) \\ \text{---} & \text{---} \\ 0 & 0 & 0 & 1 \end{array} \right] \begin{bmatrix} {}^B\bar{Q}(3 \times 1) \\ \text{---} \\ \text{---} \\ 1 \end{bmatrix}$$

Let **[T]**: Homogeneous transformation matrix

$$[T] = \left[\begin{array}{c|c} {}^U_B R(3 \times 3) & {}^U\bar{Q}_{\text{Borg}}(3 \times 1) \\ \text{---} & \text{---} \\ 0 & 0 & 0 & 1 \end{array} \right]$$



Say

$$[T] = \begin{bmatrix} r_{11} & r_{12} & r_{13} & q_x \\ r_{21} & r_{22} & r_{23} & q_y \\ r_{31} & r_{32} & r_{33} & q_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Translation Operator

Trans : Translation of q units along x-direction

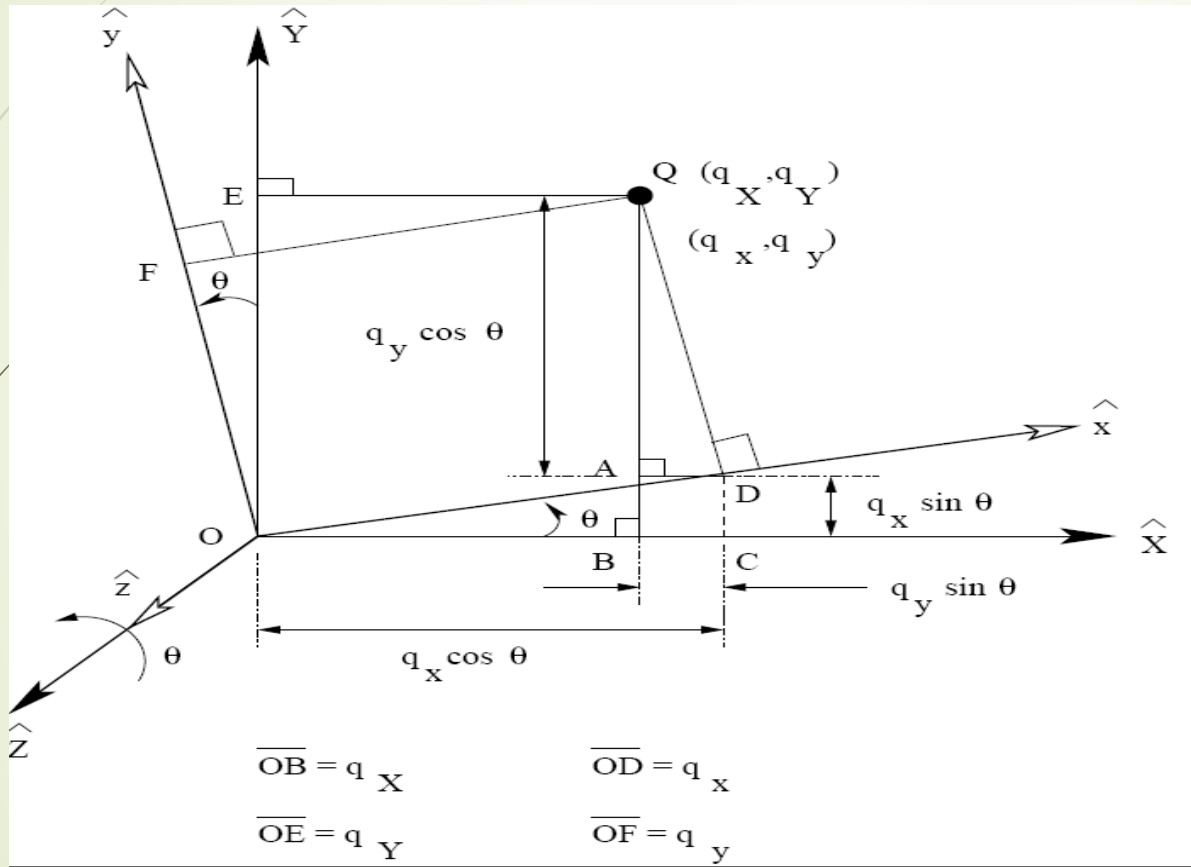
$$\text{Trans} = \begin{bmatrix} 1 & 0 & 0 & q \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note: Trans operators are commutative in nature

$$\text{Trans}(\hat{X}, q_x) \text{Trans}(\hat{Y}, q_y) = \text{Trans}(\hat{Y}, q_y) \text{Trans}(\hat{X}, q_x)$$

Rotational Operator

Rot : Rotation about $\hat{z}$ axis by an angle θ (anticlockwise sense)



$$\overline{DC} = q_x \sin \theta$$

$$\overline{AQ} = q_y \cos \theta$$

$$\overline{OC} = q_x \cos \theta$$

$$\overline{AD} = \overline{BC} = q_y \sin \theta$$


$$q_X = q_x \cos \theta - q_y \sin \theta + q_z X_0$$

$$q_Y = q_x \sin \theta + q_y \cos \theta + q_z X_0$$

$$q_Z = q_x X_0 + q_y X_0 + q_z X_1$$

In matrix form:

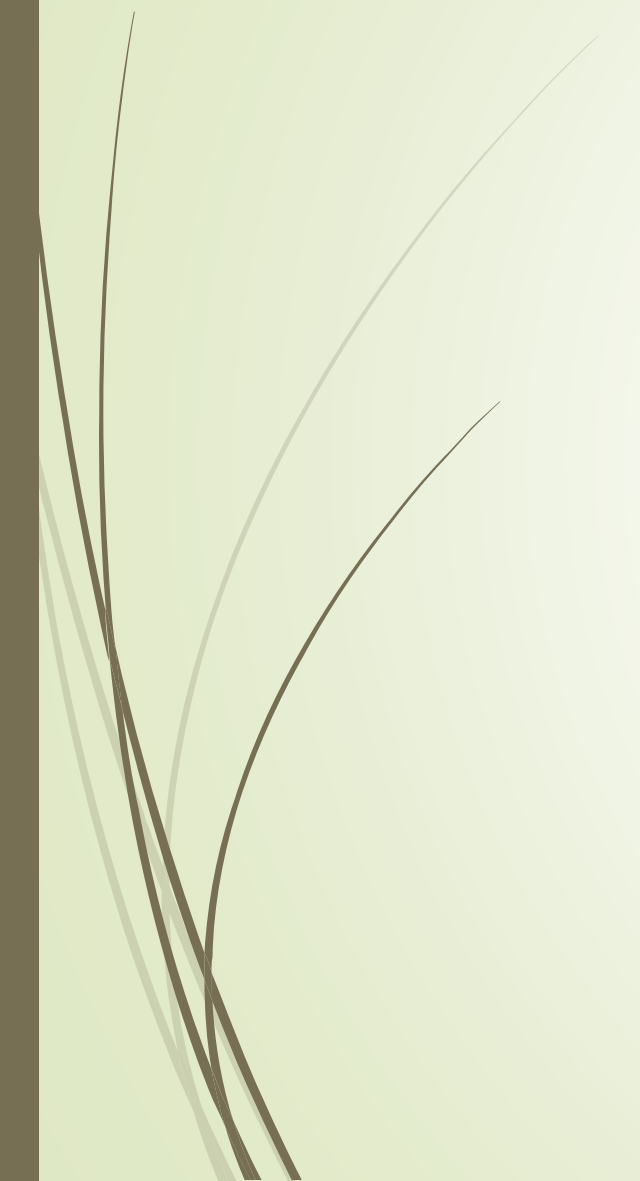
$$\begin{bmatrix} q_X \\ q_Y \\ q_Z \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} q_x \\ q_y \\ q_z \end{bmatrix}$$

$$\text{Rot}(\hat{Z}, \theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Similarly, we get

$$\text{Rot}(\hat{X}, \theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$\text{Rot}(\hat{Y}, \theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$


Properties of Rotation Matrix

- Each row/column of a rotation matrix is a unit vector
- Inner (dot) product of each row of a rotation matrix with each other row becomes equal to 0. The same is true for each column also.
- Rotation matrices are not commutative in nature

$$\text{ROT}(\hat{X}, \theta_1) \text{ROT}(\hat{Y}, \theta_2) \neq \text{ROT}(\hat{Y}, \theta_2) \text{ROT}(\hat{X}, \theta_1)$$

- Inverse of a rotation matrix is nothing but its transpose

$$\text{ROT}^{-1}(\hat{X}, \theta) = \text{ROT}^T(\hat{X}, \theta)$$

- ${}^A_B T = {}^B_A T^{-1}$

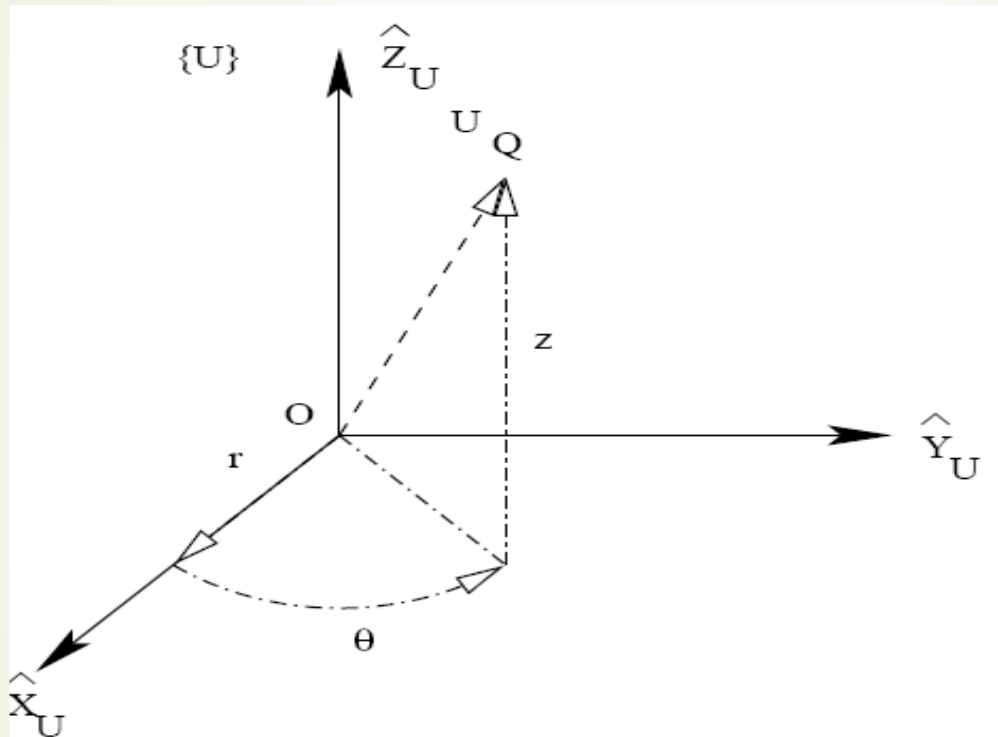
Composite Rotation Matrix

Composite rotation matrix representing a rotation of α angle about $\hat{Z}$ axis, followed by a rotation of β angle about $\hat{Y}$ axis, followed by a rotation of γ angle about $\hat{X}$ axis.

$$\text{ROT}_{\text{composite}} = \text{ROT}(\hat{X}, \gamma) \text{ROT}(\hat{Y}, \beta) \text{ROT}(\hat{Z}, \alpha)$$

Representations of Position in Other Than Cartesian Coordinate System

Cylindrical Coordinate System



Steps:

1. Starting from the origin O , translate by r units along $\hat{x}_U$ axis
2. Rotate in anti-clockwise sense about $\hat{x}_U$ axis by an angle θ
3. Translate along $\hat{z}_U$ axis by z units


$$[T]_{\text{composite}} = TRANS(\hat{Z}_U, z) ROT(\hat{Z}_U, \theta) TRANS(\hat{X}_U, r)$$

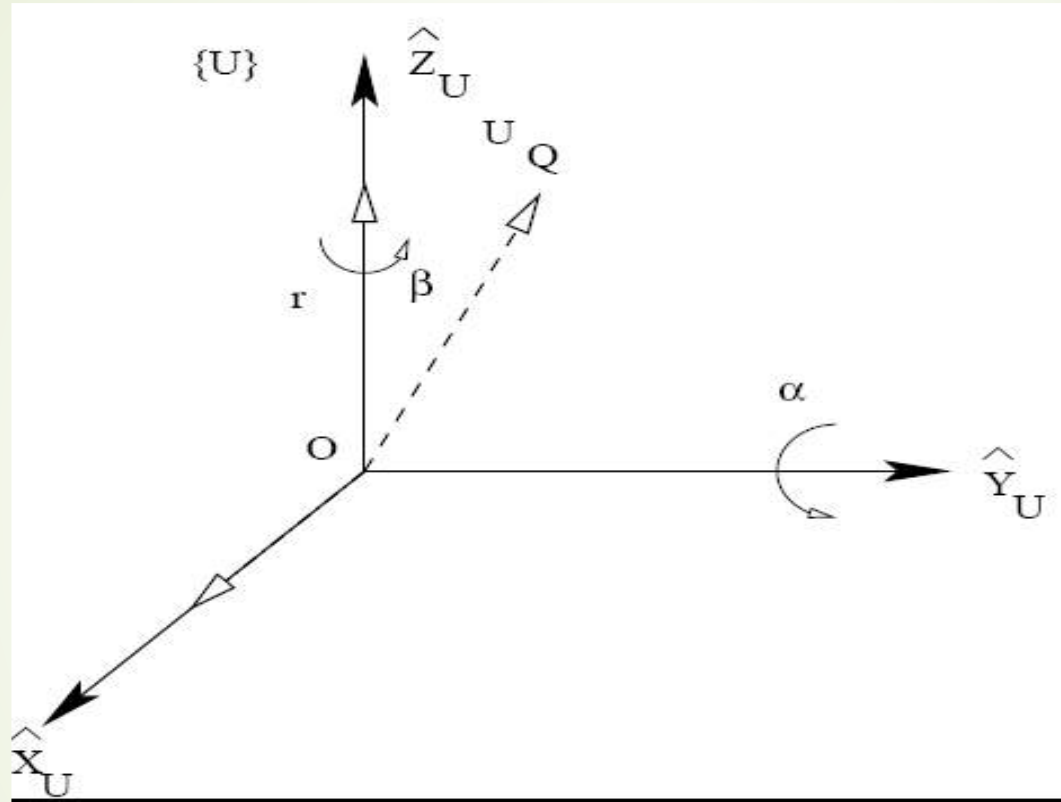
$$= \begin{bmatrix} \cos \theta & -\sin \theta & 0 & r \cos \theta \\ \sin \theta & \cos \theta & 0 & r \sin \theta \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

We get $q_x = r \cos \theta$

$$q_y = r \sin \theta$$

$$q_z = z$$

Spherical Coordinate System



Steps:

1. Starting from the origin O , translate along $\hat{z}_U$ axis by r units
2. Rotate in anti-clockwise sense about $\hat{z}_U$ axis by an angle α
3. Rotate in anti-clockwise sense about $\hat{x}_U$ axis by an angle β

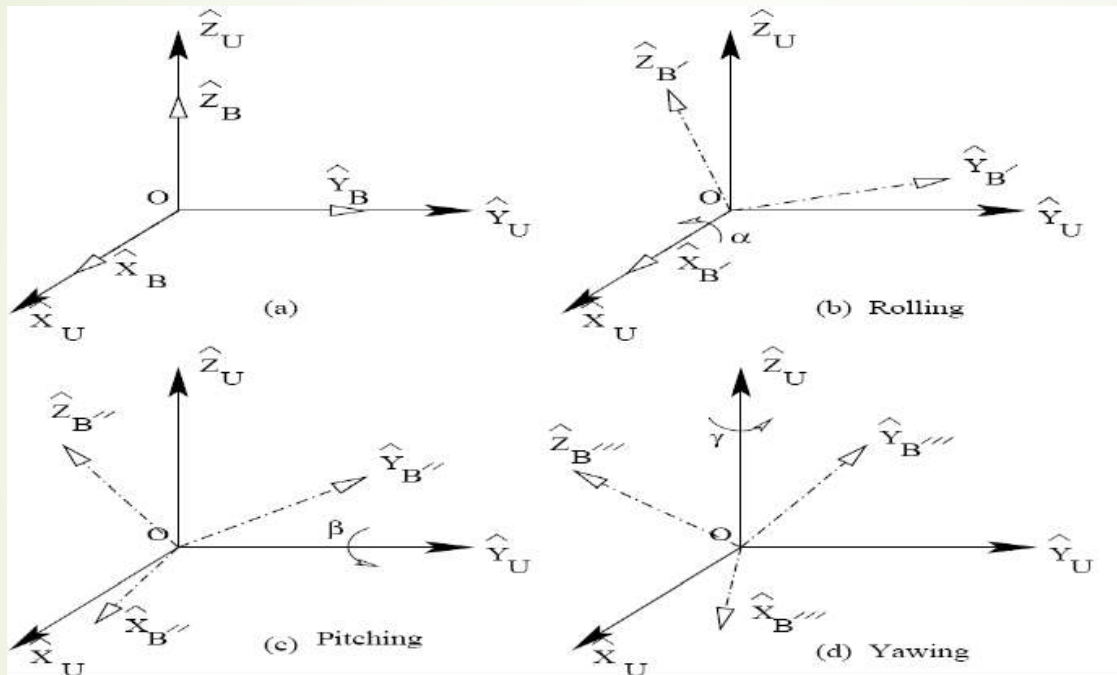

$$[T]_{\text{composite}} = ROT(\hat{Z}_U, \beta) ROT(\hat{Y}_U, \alpha) TRANS(\hat{Z}_U, r)$$

$$= \begin{bmatrix} \cos \alpha \cos \beta & -\sin \beta & \sin \alpha \cos \beta & r \sin \alpha \cos \beta \\ \cos \alpha \sin \beta & \cos \beta & \sin \alpha \sin \beta & r \sin \alpha \sin \beta \\ -\sin \alpha & 0 & \cos \alpha & r \cos \alpha \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

We get $q_x = r \sin \alpha \cos \beta$
 $q_y = r \sin \alpha \sin \beta$
 $q_z = r \cos \alpha$

Representations of Orientation in Other Than Cartesian Coordinate System

Roll, Pitch and Yaw Angles



Steps:

1. Rotate $\{B\}$ about
2. Rotate $\{B^I\}$ about
3. Rotate $\{B^{II}\}$ about

by an angle α
by an angle β
by an angle γ

rolling
pitching
yawing



$${}^U_B \mathbf{R}_{\text{composite : rpy}} = \text{ROT}(\hat{\mathbf{Z}}_U, \gamma) \text{ROT}(\hat{\mathbf{Y}}_U, \beta) \text{ROT}(\hat{\mathbf{X}}_U, \alpha)$$

$$= \begin{bmatrix} c\beta c\gamma & -c\alpha s\gamma + s\alpha s\beta c\gamma & s\alpha s\gamma + c\alpha s\beta c\gamma \\ c\beta s\gamma & c\alpha c\gamma + s\alpha s\beta s\gamma & -s\alpha c\gamma + c\alpha s\beta s\gamma \\ -s\beta & c\beta s\alpha & c\alpha c\beta \end{bmatrix}$$

We compare with

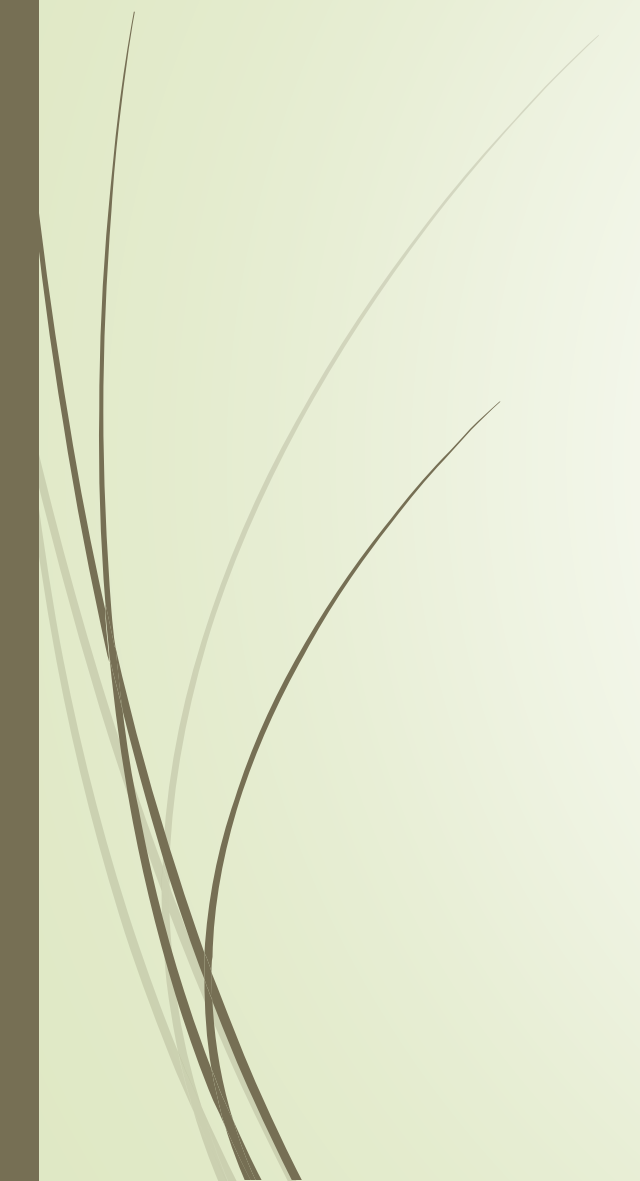
$${}^U_B \mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$



We get

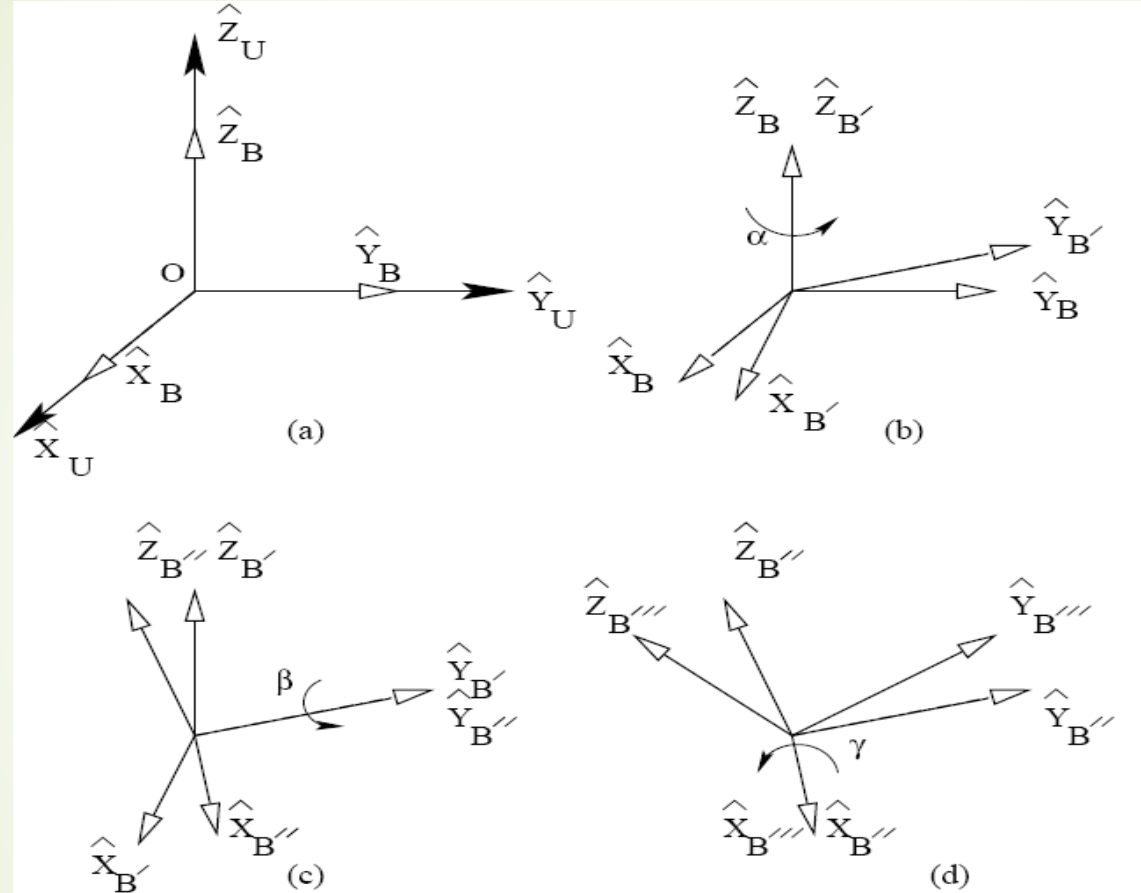
$$\alpha = \tan^{-1}\left(\frac{\mathbf{r}_{32}}{\mathbf{r}_{33}}\right)$$

$$\beta = \tan^{-1}\left(\frac{-\mathbf{r}_{31}}{\sqrt{\mathbf{r}_{11}^2 + \mathbf{r}_{21}^2}}\right)$$

$$\gamma = \tan^{-1}\left(\frac{\mathbf{r}_{21}}{\mathbf{r}_{11}}\right)$$


Using Euler Angles

$${}^U_B R = {}^B_U R^{-1}$$



Steps:

1. Rotate $\{B\}$ about $\hat{z}_B$ by an angle α in anti-clockwise sense
2. Rotate $\{B\}$ about $\hat{y}_{B'}$ by an angle β in anti-clockwise sense
3. Rotate $\{B\}$ about $\hat{x}_{B''}$ by an angle γ in anti-clockwise sense

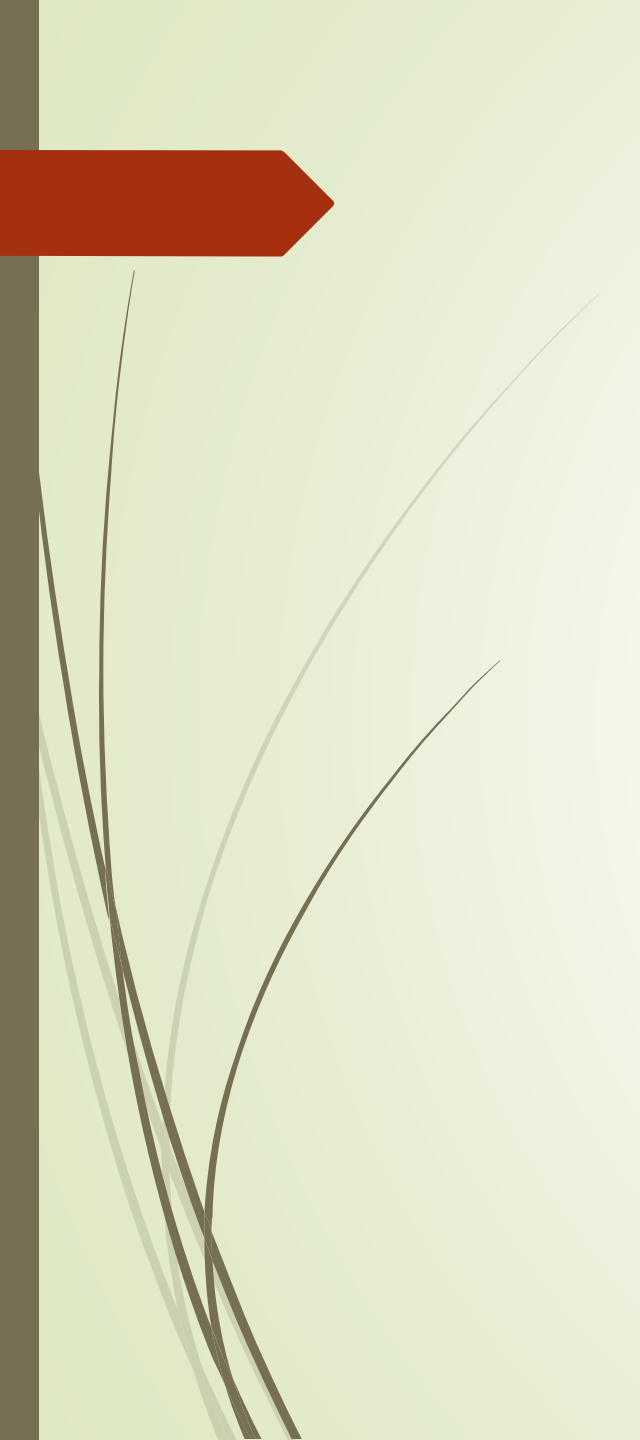


$${}^B_U R_{\text{Eulerangles}} = \text{ROT}(\hat{X}_{B''}, -\gamma) \text{ROT}(\hat{Y}_{B'}, -\beta) \text{ROT}(\hat{Z}_B, -\alpha)$$

$${}^U_B R = \begin{bmatrix} c\alpha c\beta & s\beta s\gamma c\alpha - s\alpha c\gamma & s\beta c\gamma c\alpha + s\alpha s\gamma \\ s\alpha c\beta & s\beta s\gamma s\alpha + c\alpha c\gamma & s\beta c\gamma s\alpha - s\gamma c\alpha \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

We compare with

$${}^U_B R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$


$$\alpha = \tan^{-1} \left(\frac{r_{21}}{r_{11}} \right)$$

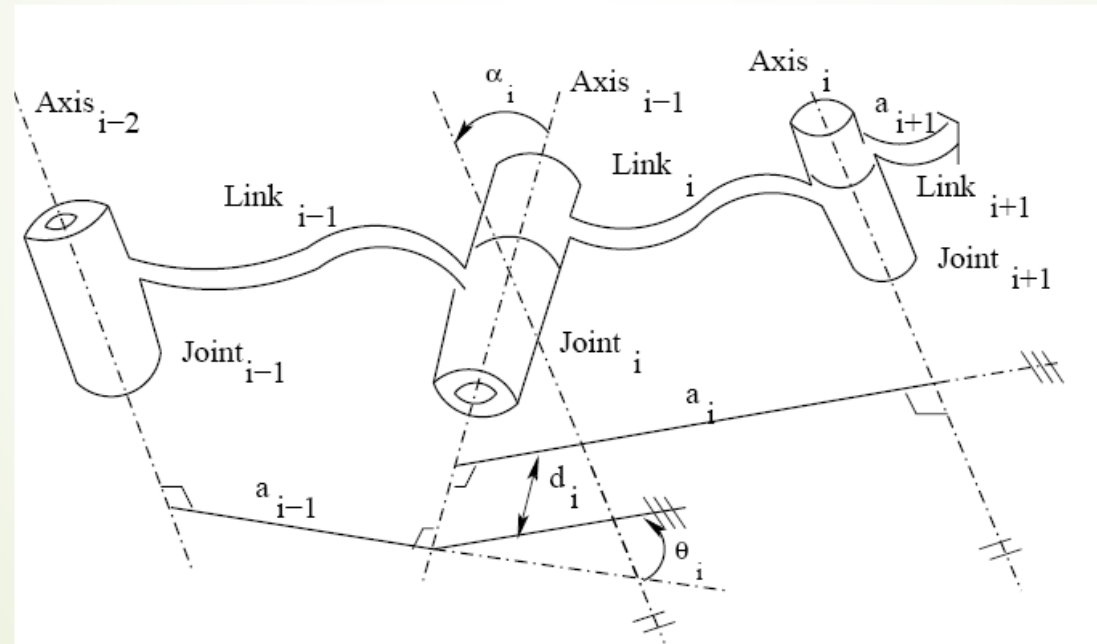
$$\beta = \tan^{-1} \left(\frac{-r_{31}}{\sqrt{r_{11}^2 + r_{21}^2}} \right)$$

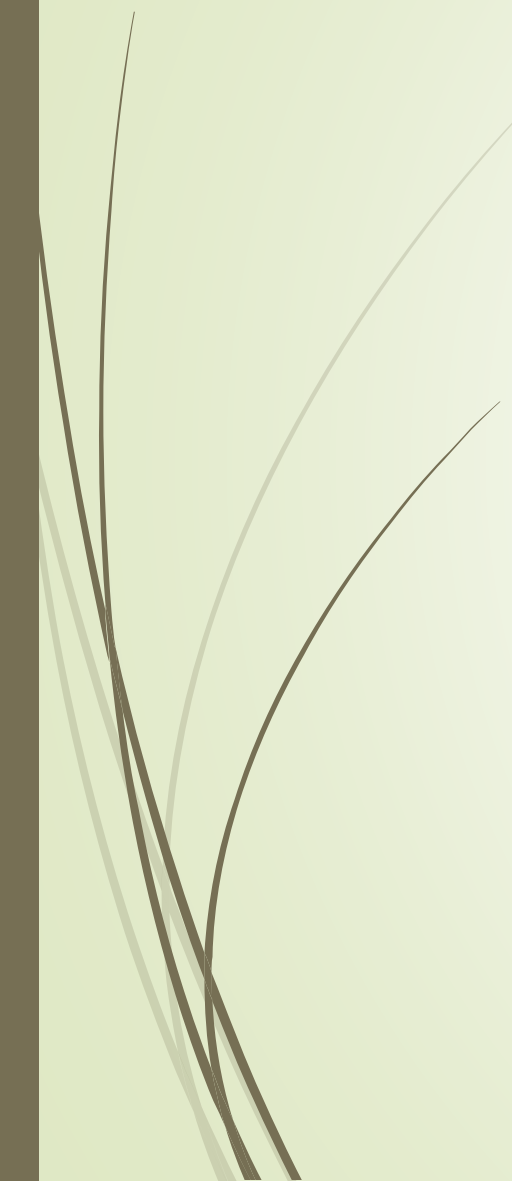
$$\gamma = \tan^{-1} \left(\frac{r_{32}}{r_{33}} \right)$$

Denavit-Hartenberg Notations

- Proposed in the year 1955

Link and Joint Parameters

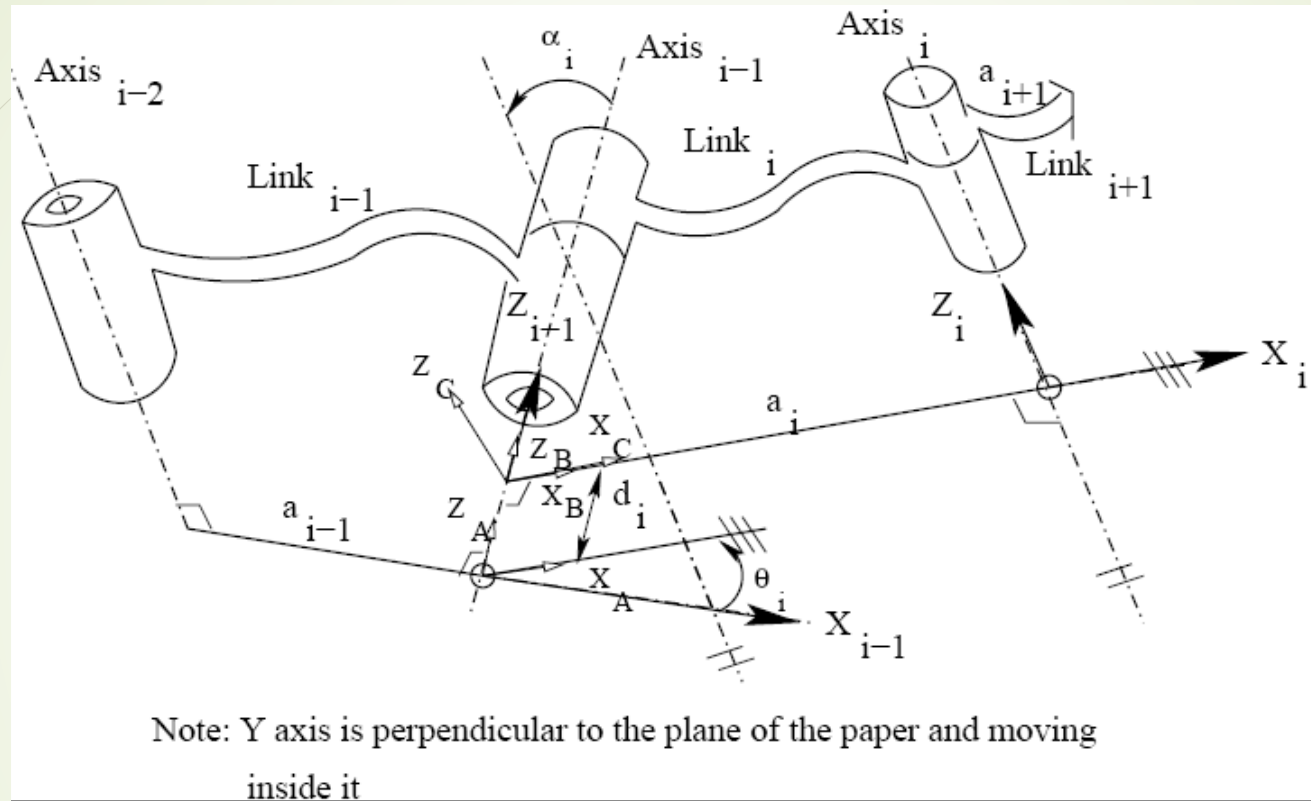


- 
- 
- **Length of link_i (a_i):** It is the mutual perpendicular distance between Axis _{$i-1$} and Axis _{i}
 - **Angle of twist of link_i (α_i):** It is defined as the angle between Axis _{$i-1$} and Axis _{i}
 - **Offset of link_i (d_i):** It is the distance measured from a point where a_{i-1} intersects the Axis _{$i-1$} to the point where a_i intersects the Axis _{$i-1$} measured along the said axis
 - **Joint Angle (θ_i):** It is defined as the angle between the extension of a_{i-1} and a_i measured about the Axis _{$i-1$}

Notes:

- **Revolute joint:** θ_i is variable
- **Prismatic joint:** d_i is variable

Rules for Coordinate Assignment



- Z_i is an axis about which the rotation is considered or along which the translation takes place
- If Z_{i-1} and Z_i axes are parallel to each other, X axis will be directed from Z_{i-1} to Z_i along their common normal

- If Z_{i-1} and Z_i axes intersect each other, X axis can be selected along either of two remaining directions
- If Z_{i-1} and Z_i axes act along a straight line, X axis can be selected anywhere in a plane perpendicular to them
- Y axis is decided as $Y = Z \times X$

We have

$$\begin{aligned} {}^{i-1}_i T &= {}^{i-1}_A T {}^A_B T {}^B_C T {}^C_i T \\ &= \text{ROT}(Z, \theta_i) \text{TRANS}(Z, d_i) \text{ROT}(X, \alpha_i) \text{TRANS}(X, a_i) \\ &= \text{Screw}_Z \text{Screw}_X \end{aligned}$$

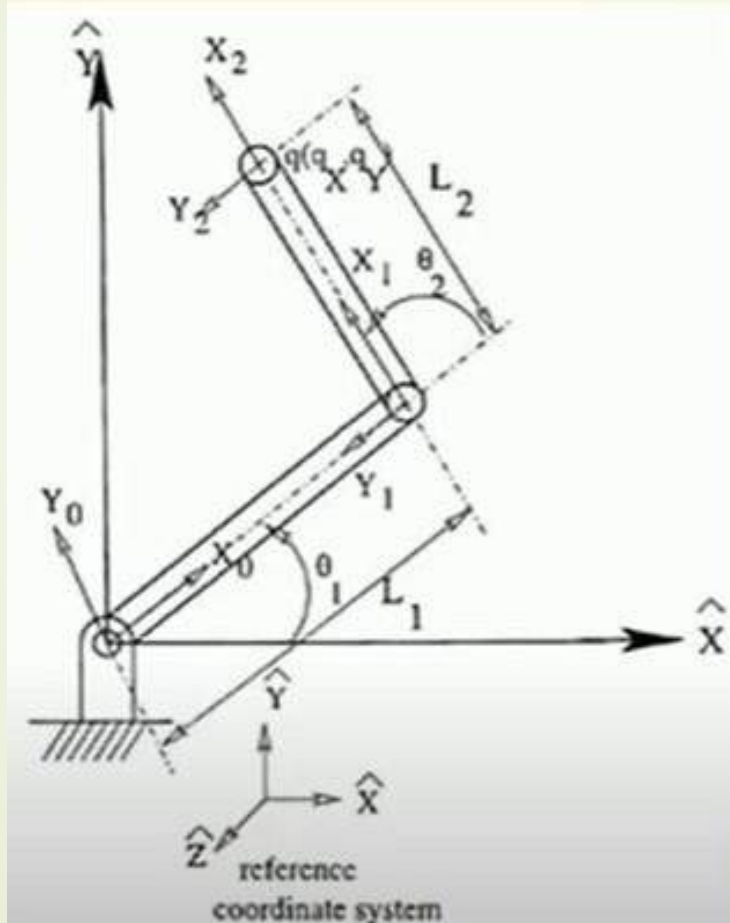


$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now, ${}^{i-1}_i T = [{}^{i-1}_iT]^{-1}$

$$= \begin{bmatrix} c\theta_i & s\theta_i & 0 & -a_i \\ -s\theta_i c\alpha_i & c\theta_i c\alpha_i & s\alpha_i & -d_i s\alpha_i \\ s\theta_i s\alpha_i & -c\theta_i s\alpha_i & c\alpha_i & -d_i c\alpha_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example 1

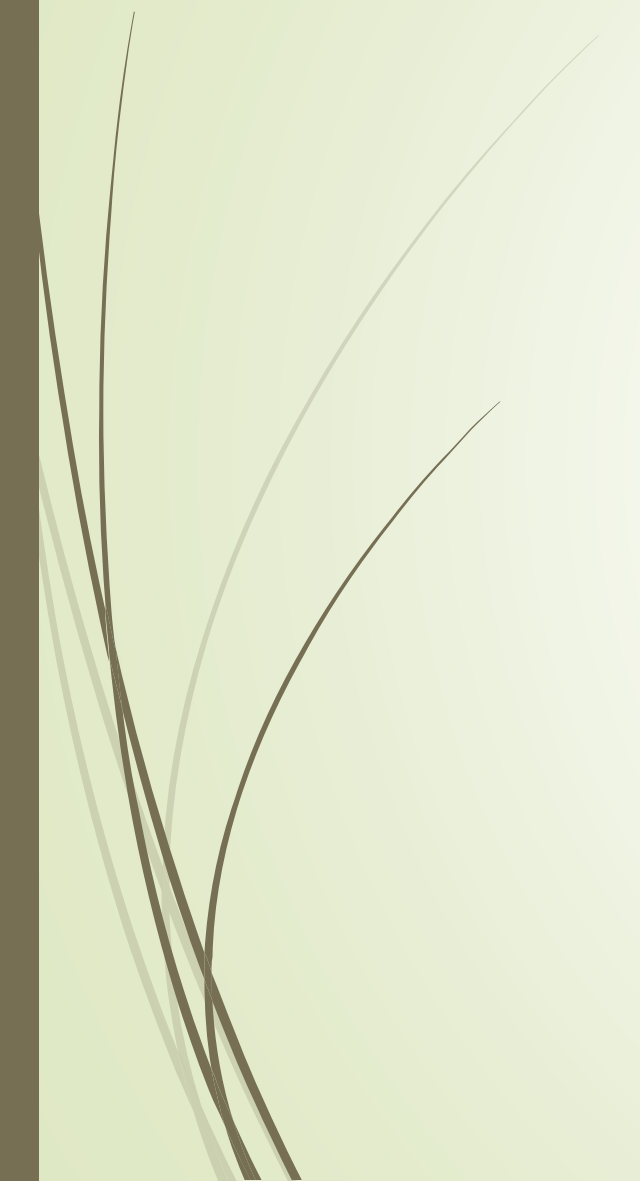


Frame	θ_i	d_i	α_i	a_i
1	θ_1	0	0	L_1
2	θ_2	0	0	L_2

Forward Kinematics

$${}^{\text{Base}}T_2 = {}^{\text{Base}}T_1^1 T_2^1$$

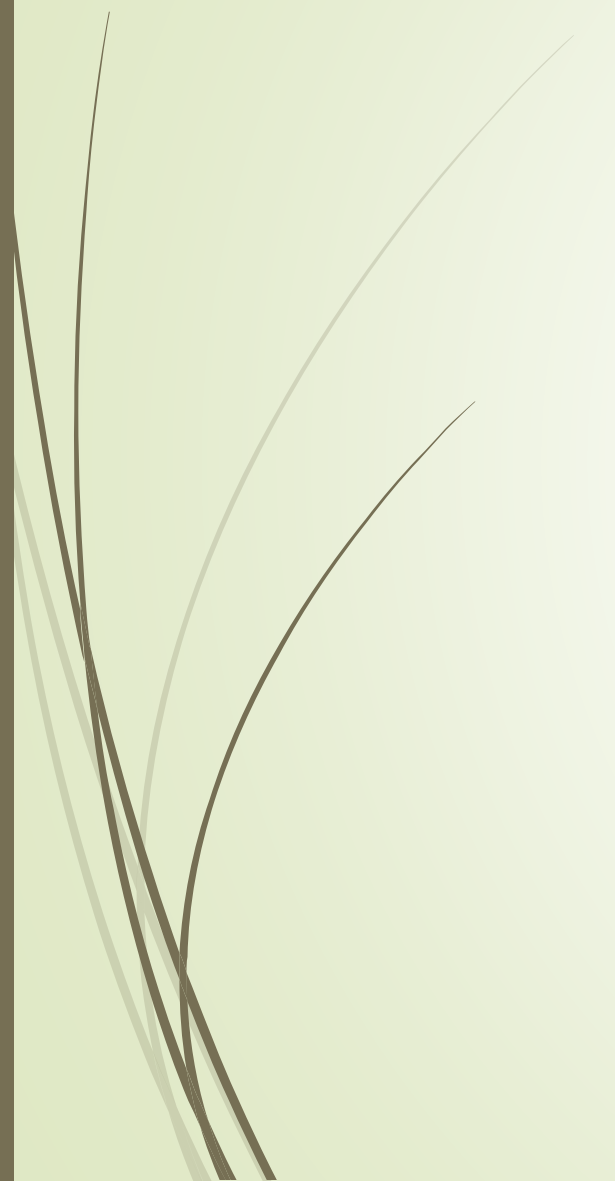
$$\begin{aligned} {}^{\text{Base}}T_1 &= \text{ROT}(\hat{Z}, \theta_1) \text{TRANS}(\hat{X}, L_1) \\ &= \begin{bmatrix} c_1 & -s_1 & 0 & L_1 c_1 \\ s_1 & c_1 & 0 & L_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$


$$\begin{aligned} {}^1_2T^a &= ROT(\hat{Z}, \theta_2) TRANS(\hat{X}, L_2) \\ &= \begin{bmatrix} c_2 & -s_2 & 0 & L_2 c_2 \\ s_2 & c_2 & 0 & L_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
 {}_2^{Base}T &= {}_1^{Base}T_2^1 T \\
 &= \begin{bmatrix} c_{12} & -s_{12} & 0 & L_1 c_1 + L_2 c_{12} \\ s_{12} & c_{12} & 0 & L_1 s_1 + L_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

Inverse Kinematic

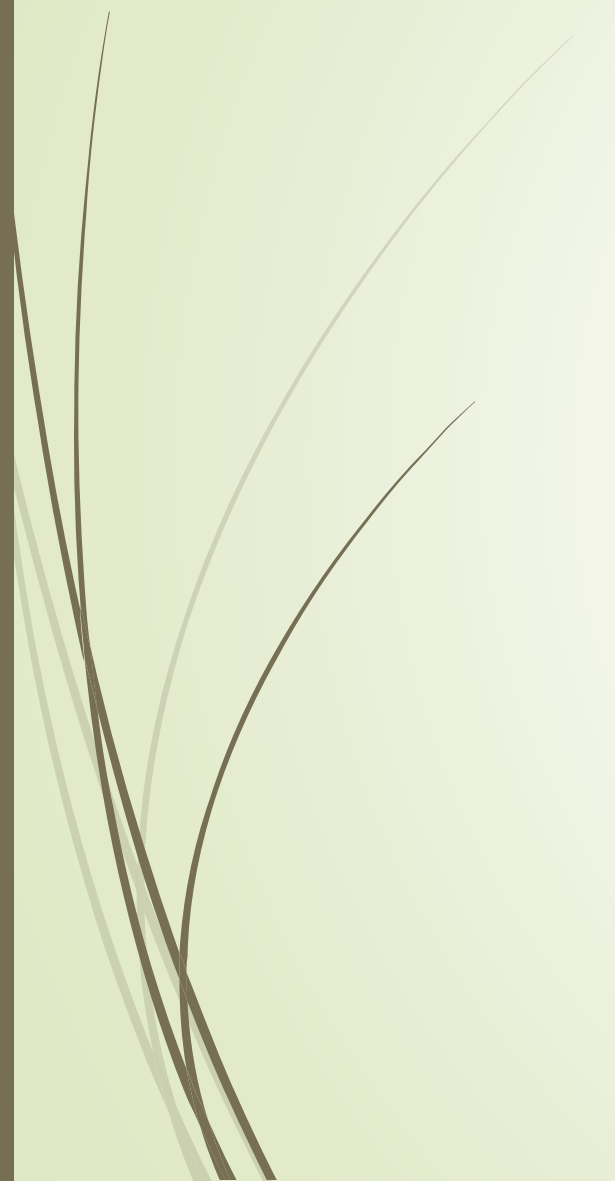
$${}^{\text{Base}}_2T = \begin{bmatrix} c\phi & -s\phi & 0 & q_x \\ s\phi & c\phi & 0 & q_y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$


$$q_x = L_1 c_1 + L_2 c_{12}$$

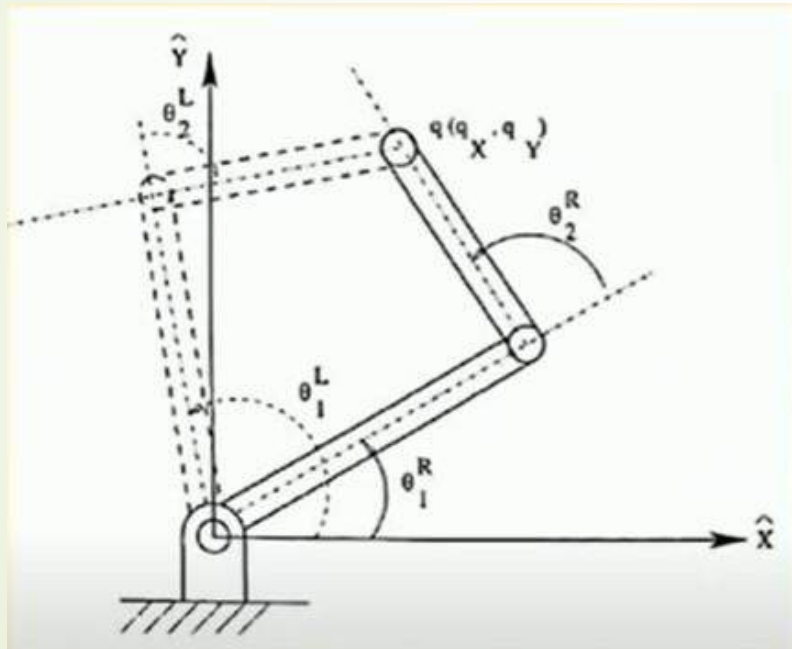
$$q_y = L_1 s_1 + L_2 s_{12}$$

$$q_x^2 + q_y^2 = L_1^2 + L_2^2 + 2L_1 L_2 c_{12} c_1 + 2L_1 L_2 s_{12} s_1,$$

$$c_2 = \frac{q_x^2 + q_y^2 - L_1^2 - L_2^2}{2L_1 L_2}$$


$$\theta_2 = \arccos\left(\frac{q_x^2 + q_y^2 - L_1^2 - L_2^2}{2L_1L_2}\right).$$

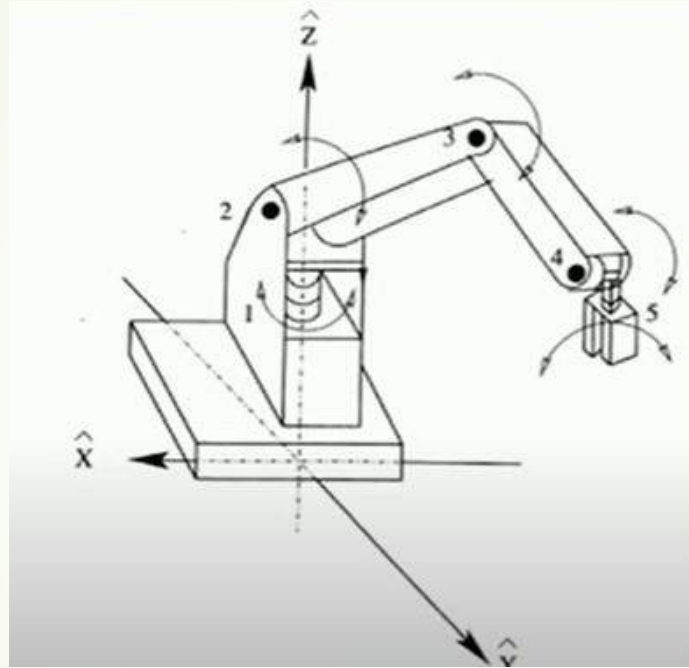
$$\begin{aligned} q_x &= L_1c_1 + L_2c_1c_2 - L_2s_1s_2 \\ &= c_1(L_1 + L_2c_2) - s_1(L_2s_2) \end{aligned}$$

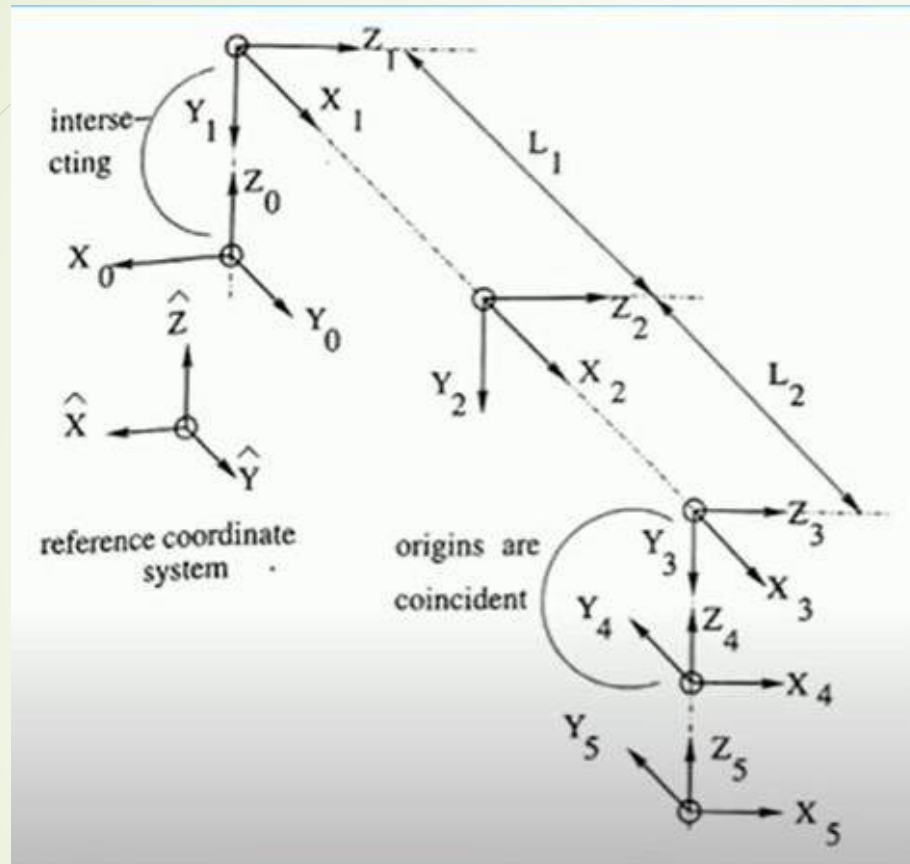


$$q_y = \rho \cos(\psi - \theta_1)$$

$$\theta_1 = \psi - \arctan\left(\frac{q_x}{q_y}\right)$$

Example 2





Frame	θ_i	d_i	α_i	a_i
1	θ_1	0	-90	0
2	θ_2	0	0	L_1
3	θ_3	0	0	L_2
4	θ_4	0	90	0
5	θ_5	0	0	0

Forward Kinematics

$${}^0_5T = {}^0_1T {}^1_2T {}^2_3T {}^3_4T {}^4_5T$$

$$\begin{aligned} {}^0_1T &= Rot(\hat{Z}, \theta_1) Rot(\hat{X}, -90) \\ &= \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Here, c_1 and s_1 denote $c\theta_1$ (or, $\cos\theta_1$) and $s\theta_1$ (or, $\sin\theta_1$), respectively.

$${}^0_1T^{-1} = \begin{bmatrix} c_1 & s_1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -s_1 & c_1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = Rot(\hat{Z}, \theta_2) Trans(\hat{X}, L_1)$$

$$= \begin{bmatrix} c_2 & -s_2 & 0 & L_1 c_2 \\ s_2 & c_2 & 0 & L_1 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} {}^2_3T &= Rot(\hat{Z}, \theta_3) Trans(\hat{X}, L_2) \\ &= \begin{bmatrix} c_3 & -s_3 & 0 & L_2 c_3 \\ s_3 & c_3 & 0 & L_2 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$



$${}^3_4T = Rot(\hat{Z}, \theta_4) Rot(\hat{X}, 90)$$

$$= \begin{bmatrix} c_4 & 0 & s_4 & 0 \\ s_4 & 0 & -c_4 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^4_5T = Rot(\hat{Z}, \theta_5)$$

$$= \begin{bmatrix} c_5 & -s_5 & 0 & 0 \\ s_5 & c_5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned}
 {}^0_5T &= {}^0_1T {}^1_2T {}^2_3T {}^3_4T {}^4_5T \\
 &= \begin{bmatrix} v_{11} & v_{12} & v_{13} & p_x \\ v_{21} & v_{22} & v_{23} & p_y \\ v_{31} & v_{32} & v_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 v_{11} &= c_1 c_{234} c_5 - s_1 s_5 \\
 v_{12} &= -c_1 c_{234} s_5 - s_1 c_5 \\
 v_{13} &= c_1 s_{234} \\
 v_{21} &= s_1 c_{234} c_5 + c_1 s_5 \\
 v_{22} &= -s_1 c_{234} s_5 + c_1 c_5 \\
 v_{23} &= s_1 s_{234} \\
 v_{31} &= -s_{234} c_5 \\
 v_{32} &= s_{234} s_5 \\
 v_{33} &= c_{234} \\
 p_x &= c_1 (L_1 c_2 + L_2 c_{23}) \\
 p_y &= s_1 (L_1 c_2 + L_2 c_{23}) \\
 p_z &= -L_1 s_2 - L_2 s_{23}
 \end{aligned}$$

Inverse Kinematics

$${}^0_5T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & q_x \\ r_{21} & r_{22} & r_{23} & q_y \\ r_{31} & r_{32} & r_{33} & q_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} {}^0_5T &= {}^0_1T {}^1_2T {}^2_3T {}^3_4T {}^4_5T \\ \Rightarrow {}^0_1T^{-1}({}^0_5T) &= {}^1_2T {}^2_3T {}^3_4T {}^4_5T \end{aligned}$$

$$\Rightarrow \begin{bmatrix} c_1 & s_1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -s_1 & c_1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & q_x \\ r_{21} & r_{22} & r_{23} & q_y \\ r_{31} & r_{32} & r_{33} & q_z \\ 0 & 0 & 0 & 1 \end{bmatrix} =$$

$$\Rightarrow \begin{bmatrix} r_{11}c_1 + r_{21}s_1 & r_{12}c_1 + r_{22}s_1 & r_{13}c_1 + r_{23}s_1 & q_xc_1 + q_ys_1 \\ -r_{31} & -r_{32} & -r_{33} & -q_z \\ -r_{11}s_1 + r_{21}c_1 & -r_{12}s_1 + r_{22}c_1 & -r_{13}s_1 + r_{23}c_1 & -q_xs_1 + q_yc_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} c_{234}c_5 & -c_{234}s_5 & s_{234} & L_1c_2 + L_2c_{23} \\ s_{234}c_5 & -s_{234}s_5 & -c_{234} & L_1s_2 + L_2s_{23} \\ s_5 & c_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$q_x c_1 + q_y s_1 = L_1 c_2 + L_2 c_{23}$$

$$-q_x s_1 + q_y c_1 = 0$$

$$q_z = -L_1 s_2 - L_2 s_{23}$$

$$s_{234} = r_{13} c_1 + r_{23} s_1$$

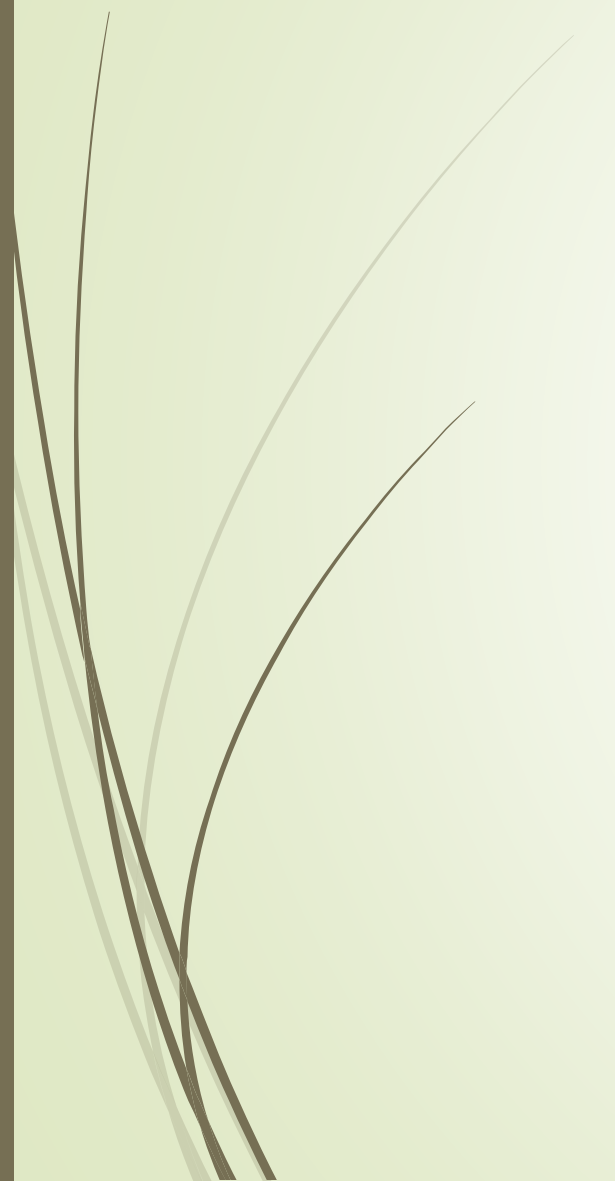
$$c_{234} = r_{33}$$

$$-r_{11} s_1 + r_{21} c_1 = s_5$$

$$-r_{12} s_1 + r_{22} c_1 = c_5$$

$$-q_x s_1 + q_y c_1 = 0$$

$$\Rightarrow \theta_1 = \arctan\left(\frac{q_y}{q_x}\right)$$


$$\rho \sin \alpha c_2 - \rho \cos \alpha s_2 = q_x c_1 + q_y s_1$$

$$\rho \sin(\alpha - \theta_2) = q_x c_1 + q_y s_1$$

$$\rho \cos(\alpha - \theta_2) = -q_z$$

$$\begin{aligned}
 q_x^2 + q_y^2 + q_z^2 &= L_1^2 + L_2^2 + 2L_1L_2c_3 \\
 \Rightarrow \theta_3 &= \arccos\left(\frac{q_x^2 + q_y^2 + q_z^2 - L_1^2 - L_2^2}{2L_1L_2}\right)
 \end{aligned}$$

$$\begin{aligned}
 L_1c_2 + L_2c_{23} &= q_xc_1 + q_ys_1 \\
 \Rightarrow (L_1 + L_2c_3)c_2 - (L_2s_3)s_2 &= q_xc_1 + q_ys_1
 \end{aligned}$$

Let us assume $L_1 + L_2c_3 = \rho \sin \alpha$ and $L_2s_3 = \rho \cos \alpha$,
 where $\rho \neq 0$ and $\rho = \sqrt{(L_1 + L_2c_3)^2 + (L_2s_3)^2}$; $\alpha = \arctan\left(\frac{L_1 + L_2c_3}{L_2s_3}\right)$.
 Thus, the above expression can be written as follows:



$$\tan(\alpha - \theta_2) = \frac{q_x c_1 + q_y s_1}{-q_z}$$

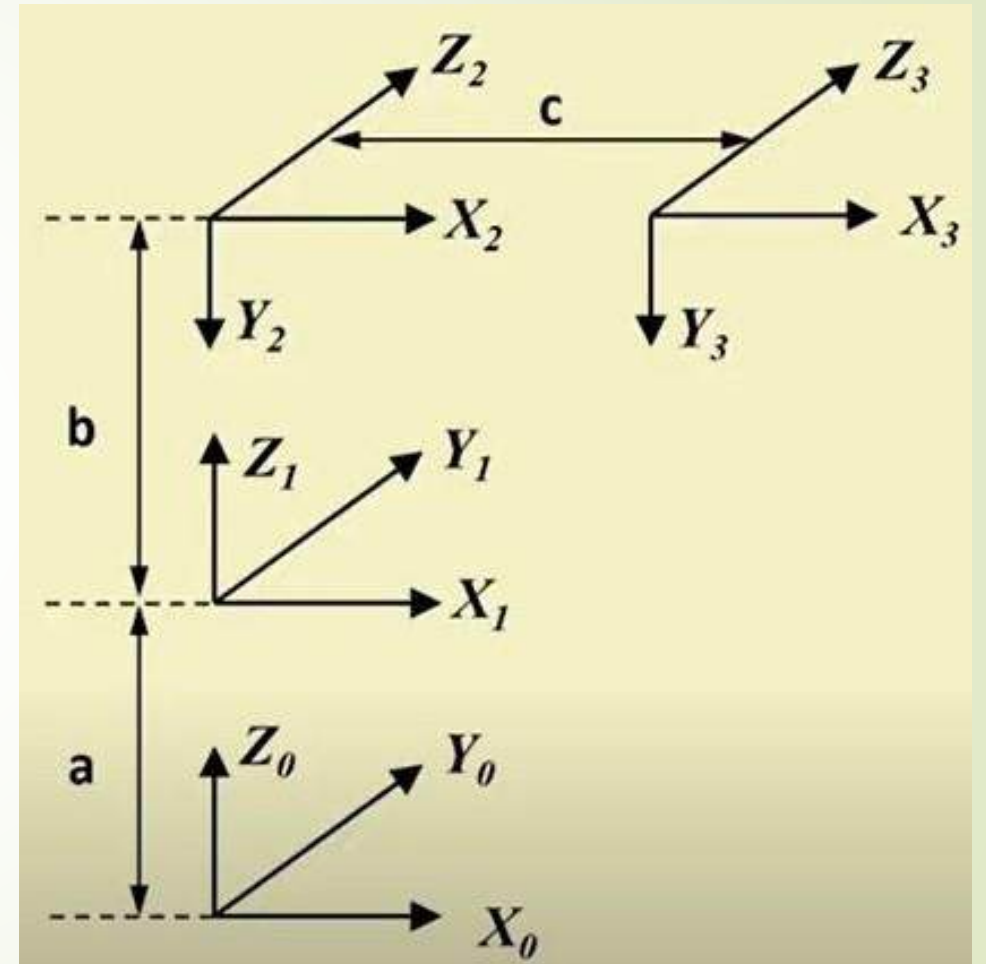
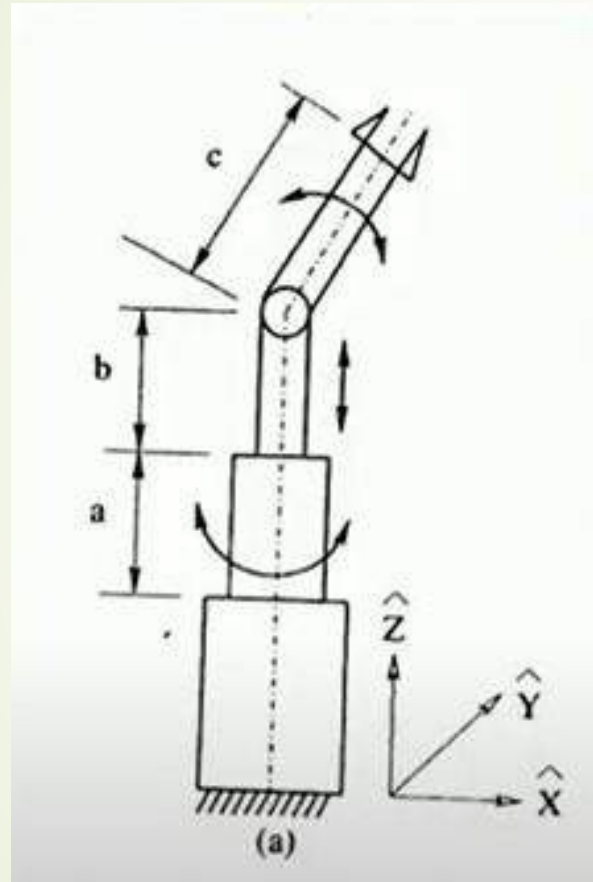
$$\Rightarrow \theta_2 = \alpha - \arctan\left(\frac{q_x c_1 + q_y s_1}{-q_z}\right)$$

$$\theta_2 + \theta_3 + \theta_4 = \arctan\left(\frac{r_{13}c_1 + r_{23}s_1}{r_{33}}\right)$$

$$\Rightarrow \theta_4 = \arctan\left(\frac{r_{13}c_1 + r_{23}s_1}{r_{33}}\right) - \theta_2 - \theta_3$$

$$\theta_5 = \arctan\left(\frac{-r_{11}s_1 + r_{21}c_1}{-r_{12}s_1 + r_{22}c_1}\right)$$

An Example

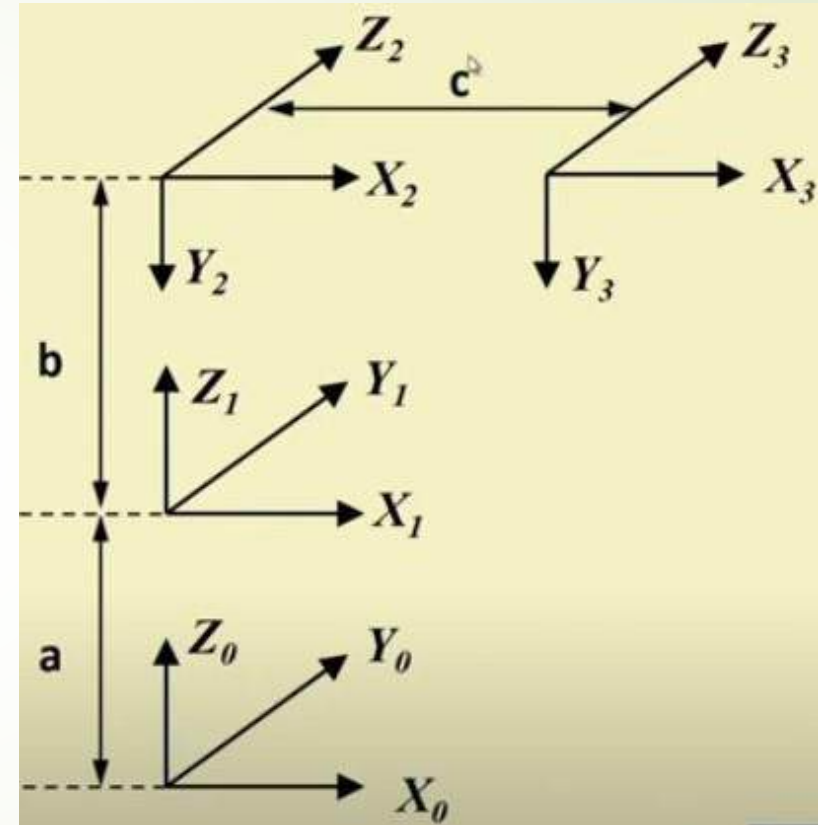
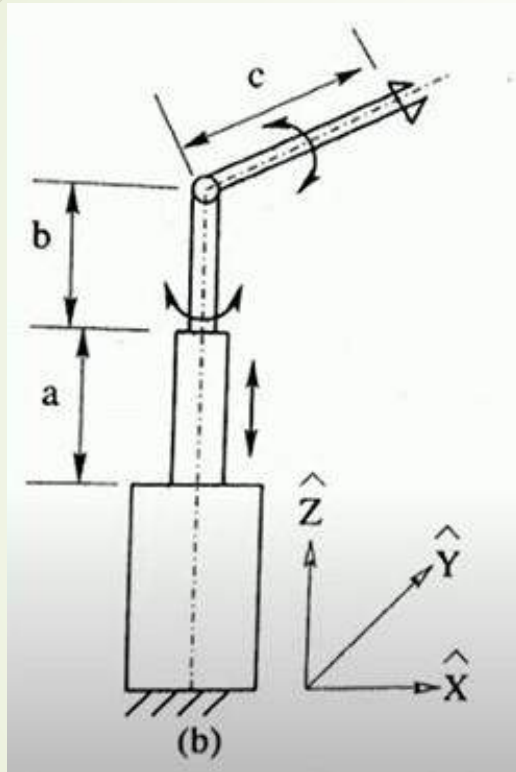


D-H parameter's table:

Frame	θ_i	d_i	α_i	a_i
1	θ_1	a^*	0	0
2	0	b	-90	0
3	θ_3	0	0	c

a^* : fixed offset

Another Example

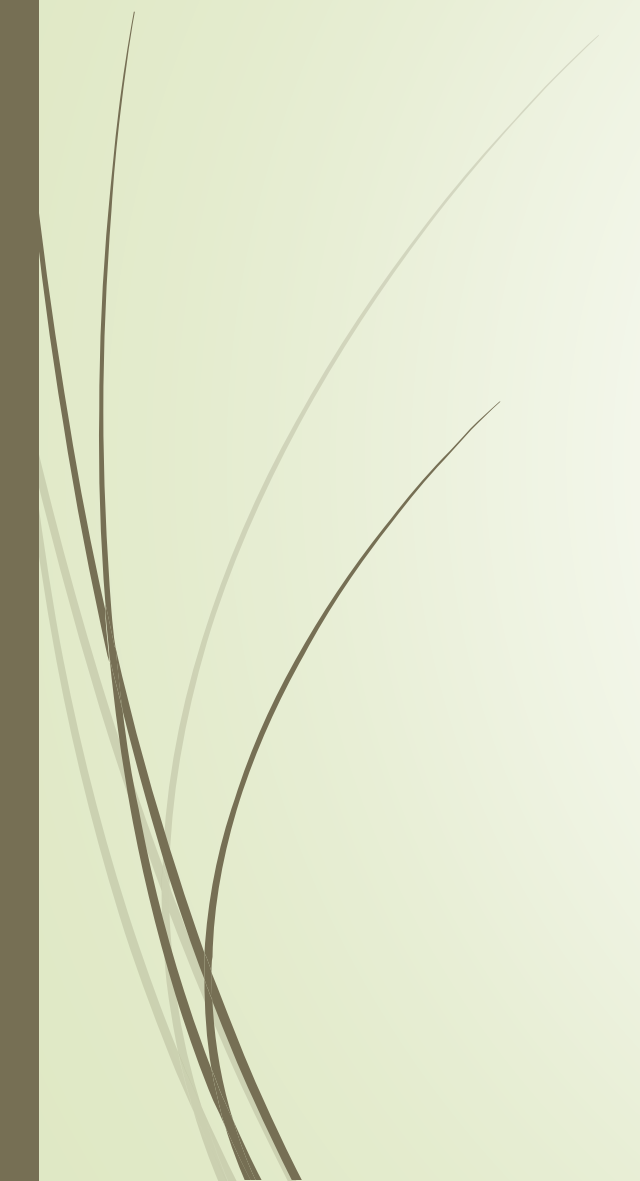




D-H parameter's table:

Frame	θ_i	d_i	α_i	a_i
1	0	a	0	0
2	θ_2	b*	-90	0
3	θ_3	0	0	c

b* : fixed offset



Example 1

Forward Kinematics

Problem Statement

A 2-link robotic arm has the following parameters:

- Link 1 length: $l_1 = 3$ m
- Link 2 length: $l_2 = 2$ m
- Joint 1 angle: $\theta_1 = 30^\circ$
- Joint 2 angle: $\theta_2 = 45^\circ$

Find the (x, y) position of the end-effector using Forward Kinematics (FK).

Step 1: Forward Kinematics Equations

The position of the end-effector is given by:

$$x = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2)$$

$$y = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2)$$

Example 1

Forward Kinematics

Step 2: Plug in Values

$$x = (3 \cos 30^\circ) + (2 \cos(30^\circ + 45^\circ))$$

$$y = (3 \sin 30^\circ) + (2 \sin(30^\circ + 45^\circ))$$

Using trigonometric values:

- $\cos 30^\circ = 0.866$, $\sin 30^\circ = 0.5$
- $\cos 75^\circ = 0.2588$, $\sin 75^\circ = 0.9659$

$$x = (3 \times 0.866) + (2 \times 0.2588)$$

$$y = (3 \times 0.5) + (2 \times 0.9659)$$

Step 3: Solve for (x, y)

$$x = 2.598 + 0.5176 = 3.1156$$

$$y = 1.5 + 1.9318 = 3.4318$$

Thus, the end-effector position is:

$$(x, y) = (3.12, 3.43) \text{ m}$$

The end-effector is located at (3.12 m, 3.43 m).

Example 2

Inverse Kinematics

Problem Statement

A robot's end-effector is at $(x, y) = (3.12, 3.43)$ m. The robot arm has links of length $l_1 = 3$ m and $l_2 = 2$ m. Find the joint angles θ_1 and θ_2 .

Step 1: Use the Inverse Kinematics Equations

$$\cos \theta_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}$$

$$\theta_2 = \cos^{-1} \left(\frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2} \right)$$

$$\theta_1 = \tan^{-1} \left(\frac{y}{x} \right) - \tan^{-1} \left(\frac{l_2 \sin \theta_2}{l_1 + l_2 \cos \theta_2} \right)$$

Step 2: Plug in Values

$$\cos \theta_2 = \frac{(3.12)^2 + (3.43)^2 - (3)^2 - (2)^2}{2(3)(2)}$$

After computation: $\theta_2 = 45^\circ$

The joint angles are $\theta_1 = 29^\circ, \theta_2 = 45^\circ$

Given Parameters	Forward Kinematics (Find x, y)	Inverse Kinematics (Find θ_1, θ_2)
$l_1 = 3 \text{ m}, l_2 = 2 \text{ m}$	$x = 3.12 \text{ m}, y = 3.43 \text{ m}$	$\theta_1 = 29^\circ, \theta_2 = 45^\circ$
$\theta_1 = 30^\circ, \theta_2 = 45^\circ$	$(3.12, 3.43) \text{ m}$	$29^\circ, 45^\circ$

Question

- Consider a 2-link robotic arm with three reference frames: Frame 0 (base frame), Frame 1 (attached to the first link), and Frame 2 (attached to the second link).
- The robotic arm operates in a 2D plane. The following parameters are provided:
- Link 1: Length $L1 = 4 \text{ units}$
- Rotation $\theta_1 = 45^\circ$ about the Z-axis Translation along the X-axis from Frame 0 to Frame 1
- Link 2: Length $L2 = 3 \text{ units}$
- Rotation $\theta_2 = 43^\circ$ about the Z-axis Translation along the X-axis from Frame 1 to Frame 2
- **a) Calculate the overall transformation matrix of the robotic arm**
- **b) Describe the position and orientation based on the overall transformation matrix.**
- **C) Sketch a graph to visualize the robotic arm**

1. Transformation from Frame 0 to Frame 1 (0T_1)

$${}^0T_1 = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & L_1 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Substituting $L_1 = 4$ and $\theta_1 = 45^\circ$:

$${}^0T_1 = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 & 4 \\ \sin 45^\circ & \cos 45^\circ & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Since $\cos 45^\circ = \sin 45^\circ = \frac{\sqrt{2}}{2}$:

$${}^0T_1 = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & 4 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2. Transformation from Frame 1 to Frame 2 (1T_2)

$${}^1T_2 = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 & L_2 \\ \sin \theta_2 & \cos \theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Substituting $L_2 = 3$ and $\theta_2 = 43^\circ$:

$${}^1T_2 = \begin{bmatrix} \cos 43^\circ & -\sin 43^\circ & 0 & 3 \\ \sin 43^\circ & \cos 43^\circ & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Approximating values:

$$\cos 43^\circ \approx 0.7314, \quad \sin 43^\circ \approx 0.6819$$

$${}^1T_2 = \begin{bmatrix} 0.7314 & -0.6819 & 0 & 3 \\ 0.6819 & 0.7314 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Compute the Overall Transformation Matrix

$(T_{TOTAL} = T1 \cdot T2)$

Multiplying:

$${}^0T_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & 4 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0.7314 & -0.6819 & 0 & 3 \\ 0.6819 & 0.7314 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Multiplication Step-by-Step

Computing rotation components:

$$R = \begin{bmatrix} \frac{\sqrt{2}}{2} \cdot 0.7314 + (-\frac{\sqrt{2}}{2} \cdot 0.6819) & \frac{\sqrt{2}}{2} \cdot (-0.6819) + (-\frac{\sqrt{2}}{2} \cdot 0.7314) & 0 \\ \frac{\sqrt{2}}{2} \cdot 0.7314 + \frac{\sqrt{2}}{2} \cdot 0.6819 & \frac{\sqrt{2}}{2} \cdot (-0.6819) + \frac{\sqrt{2}}{2} \cdot 0.7314 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Approximating values:

$$R = \begin{bmatrix} 0.035 & -0.9993 & 0 \\ 0.9993 & 0.035 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Computing position components:

$$P = \begin{bmatrix} \frac{\sqrt{2}}{2} \cdot 3 + 4 \\ \frac{\sqrt{2}}{2} \cdot 3 + 0 \\ 0 \end{bmatrix}$$

Approximating values:

$$P = \begin{bmatrix} 4 + 2.121 \\ 2.121 \\ 0 \end{bmatrix} = \begin{bmatrix} 6.121 \\ 2.121 \\ 0 \end{bmatrix}$$

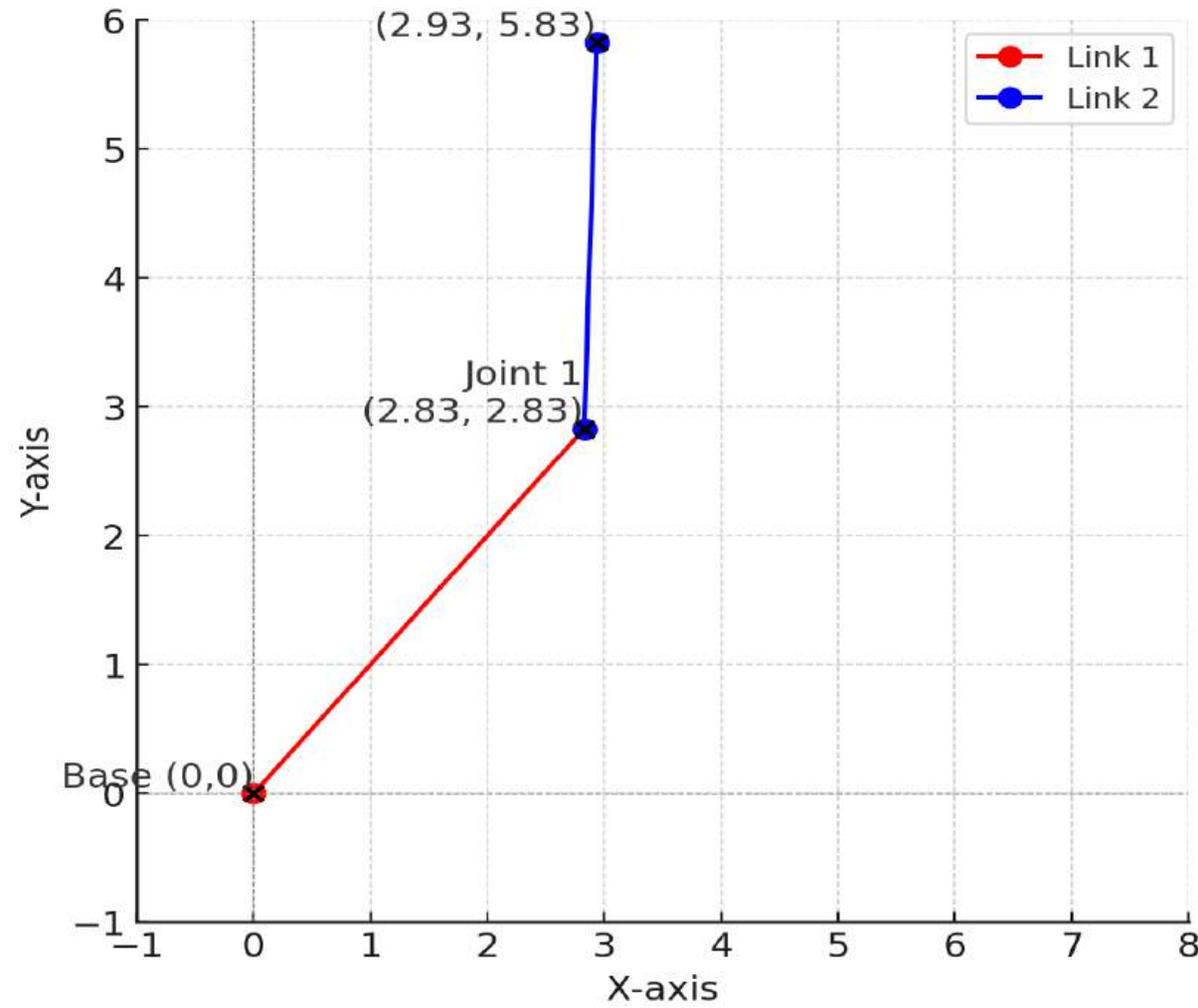
Thus, the overall transformation matrix:

$${}^0T_2 = \begin{bmatrix} 0.035 & -0.9993 & 0 & 6.121 \\ 0.9993 & 0.035 & 0 & 2.121 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Position: The end-effector (Frame 2) is located at (6.121, 2.121, 0) in Frame 0.

Orientation: The rotation matrix shows that Frame 2 is rotated by approximately 88° (sum of 45° and 43°) relative to Frame 0.

A graph that visualize the robotic arm in 2D space



Robotics

Topic 3 : Trajectory planning



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Definition

Trajectory planning in robotics involves determining a feasible and optimal path for a robot to follow from a starting position to a goal while considering the robot's kinematic and dynamic constraints, as well as obstacles in the environment.

This is a crucial component for autonomous robots, including industrial robots, mobile robots, and humanoid robots.

Concepts in Robotics Trajectory Planning

Configuration Space (C-Space):

- Represents all possible positions and orientations of the robot.
- Each point in C-space corresponds to a unique configuration of the robot.

Kinematic and Dynamic Constraints:

- Kinematic Constraints: Limits on the robot's movements based on its mechanical structure (e.g., joint limits, maximum velocities).
- Dynamic Constraints: Limits based on the forces and torques that the robot's actuators can generate (e.g., maximum accelerations, inertia).

Concepts in Robotics Trajectory Planning

Path Planning vs. Trajectory Planning:

- Path Planning: Focuses on finding a collision-free path in C-space.
- Trajectory Planning: Involves determining a time-parameterized path that the robot can follow, considering its kinematic and dynamic constraints.

Polynomial trajectory

Polynomial trajectory planning is a popular method for generating smooth, continuous paths for a robot to follow.

It involves using polynomial equations to describe the trajectory, ensuring that the path is smooth and meets certain boundary conditions and constraints.

This method is particularly useful to describe the position, velocity, and acceleration of the robot over time.

Types of Polynomial Trajectories

- Linear Polynomial: Simplest form, representing a straight-line motion.
- Quadratic Polynomial: represents parabolic trajectories, useful for simple curved paths.
- Cubic Polynomial: generates smooth paths with specified initial and final positions and velocities.
- Quintic Polynomial: specifying initial and final positions, velocities, and accelerations, providing a higher level of smoothness and control.

Formulation of Polynomial Trajectories

- A general polynomial trajectory can be expressed as:

$$p(t) = a_0 + a_1t + a_2t^2 + \dots + a_nt^n$$

- For a cubic polynomial:

$$p(t) = a_0 + a_1t + a_2t^2 + a_3t^3$$

- To determine the coefficients (a^0, a^1, a^2, a^3) boundary conditions are used:

- Initial position: $p(0) = p_0$
- Final position: $p(T) = p_f$
- Initial velocity: $\dot{p}(0) = v_0$
- Final velocity: $\dot{p}(T) = v_f$

These conditions form a system of equations that can be solved for the coefficients.



Cubic Polynomial Trajectory Planning

A Sample Calculation and Implementation



Problem Statement

- Given the following boundary conditions for a cubic polynomial trajectory:
 - Initial position: $p(0) = 0$
 - Final position: $p(T) = 1$
 - Initial velocity: $\dot{p}(0) = 0$
 - Final velocity: $\dot{p}(T) = 0$

where T is the total duration of the trajectory.



Cubic Polynomial Form

- The cubic polynomial can be written as:

$$p(t) = a_0 + a_1t + a_2t^2 + a_3t^3$$

Boundary Conditions

- From the given boundary conditions:
- 1. $p(0) = \alpha_0 = 0$
- 2. $p(T) = 1$
- 3. $p'(0) = \alpha_1 = 0$
- 4. $p'(T) = \alpha_1 + 2 \alpha_2 T + 3 \alpha_3 T^2 = 0$

Solving for Coefficients

► Solving these equations, we get:

► $a_0 = 0$

► $a_1 = 0$

► $a_2 = \frac{3}{T^2}$

► $a_3 = -\frac{2}{T^3}$

1. $p(0) = 0$

$$0 = a_0$$

$$\text{So, } a_0 = 0$$

2. $p(T) = 1$

$$1 = a_0 + a_1T + a_2T^2 + a_3T^3$$

Since $a_0 = 0$, this simplifies to:

$$1 = a_1T + a_2T^2 + a_3T^3$$

3. $\dot{p}(0) = 0$

$$\dot{p}(t) = a_1 + 2a_2t + 3a_3t^2$$

$$0 = a_1$$

$$\text{So, } a_1 = 0$$

4. $\dot{p}(T) = 0$

$$0 = a_1 + 2a_2T + 3a_3T^2$$

Since $a_1 = 0$, this simplifies to:

$$0 = 2a_2T + 3a_3T^2$$

$$2a_2T = -3a_3T^2$$

$$a_2 = -\frac{3a_3T}{2}$$

We now have two equations:

$$1. \quad 1 = a_2T^2 + a_3T^3$$

$$2. \quad a_2 = -\frac{3a_3T}{2}$$

Substitute a_2 in the first equation:

$$1 = -\frac{3a_3T}{2}T^2 + a_3T^3$$

$$1 = -\frac{3a_3T^3}{2} + a_3T^3$$

$$1 = a_3T^3 - \frac{3a_3T^3}{2}$$

$$1 = \frac{2a_3T^3 - 3a_3T^3}{2}$$

$$1 = -\frac{a_3T^3}{2}$$

$$a_3T^3 = -2$$

$$a_3 = -\frac{2}{T^3}$$

Substitute a_3 back into a_2 :

$$a_2 = -\frac{3a_3T}{2}$$

$$a_2 = -\frac{3}{2} \left(-\frac{2}{T^3} \right) T$$

$$a_2 = \frac{3}{T^2}$$

Final Coefficients

The coefficients of the cubic polynomial are:

$$a_0 = 0$$

$$a_1 = 0$$

$$a_2 = \frac{3}{T^2}$$

$$a_3 = -\frac{2}{T^3}$$



Final Polynomial

- The cubic polynomial trajectory is:
- $p(t) = \frac{3}{T^2}t^2 - \frac{2}{T^3}t^3$

Verification

- Verification of the boundary conditions:
- 1. Initial position $p(0) = 0$
- 2. Final position $p(T) = 1$
- 3. Initial velocity $p'(0) = 0$
- 4. Final velocity $p'(T) = 0$

To verify, we check the boundary conditions:

1. Initial position $p(0)$:

$$p(0) = \frac{3}{T^2}(0)^2 - \frac{2}{T^3}(0)^3 = 0$$

2. Final position $p(T)$:

$$p(T) = \frac{3}{T^2}T^2 - \frac{2}{T^3}T^3 = 3 - 2 = 1$$

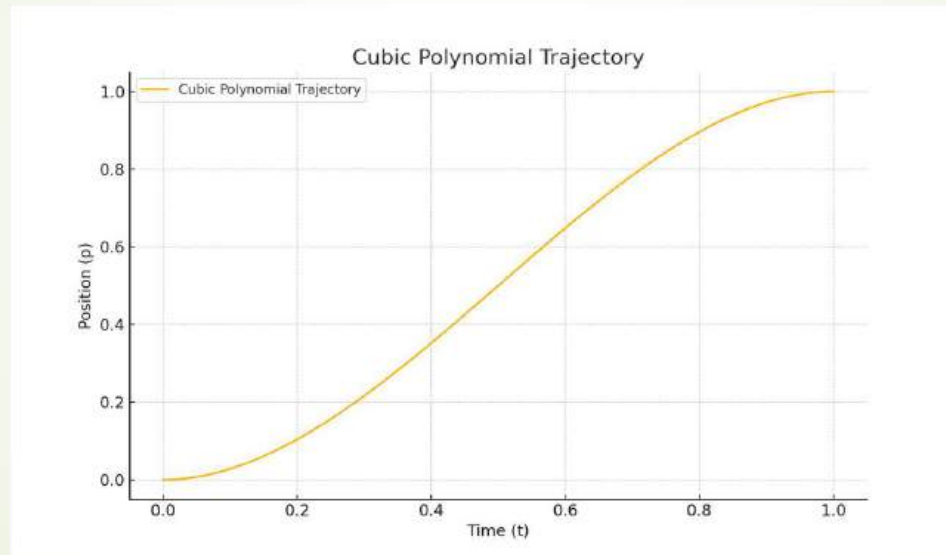
3. Initial velocity $\dot{p}(0)$:

$$\begin{aligned}\dot{p}(t) &= \frac{6}{T^2}t - \frac{6}{T^3}t^2 \\ \dot{p}(0) &= \frac{6}{T^2}(0) - \frac{6}{T^3}(0)^2 = 0\end{aligned}$$

4. Final velocity $\dot{p}(T)$:

$$\dot{p}(T) = \frac{6}{T^2}T - \frac{6}{T^3}T^2 = 6/T - 6/T = 0$$

Graphical Representation of the Trajectory



Common Schemes in Trajectory planning

Trajectory planning

```
graph TD; A[Trajectory planning] --> B[Cartesian Scheme]; A --> C[Joint-space Scheme];
```

Cartesian Scheme

Computationally expensive,
as inverse kinematics problem
has to be solved on-line

Joint-space Scheme

Comparism of Schemes

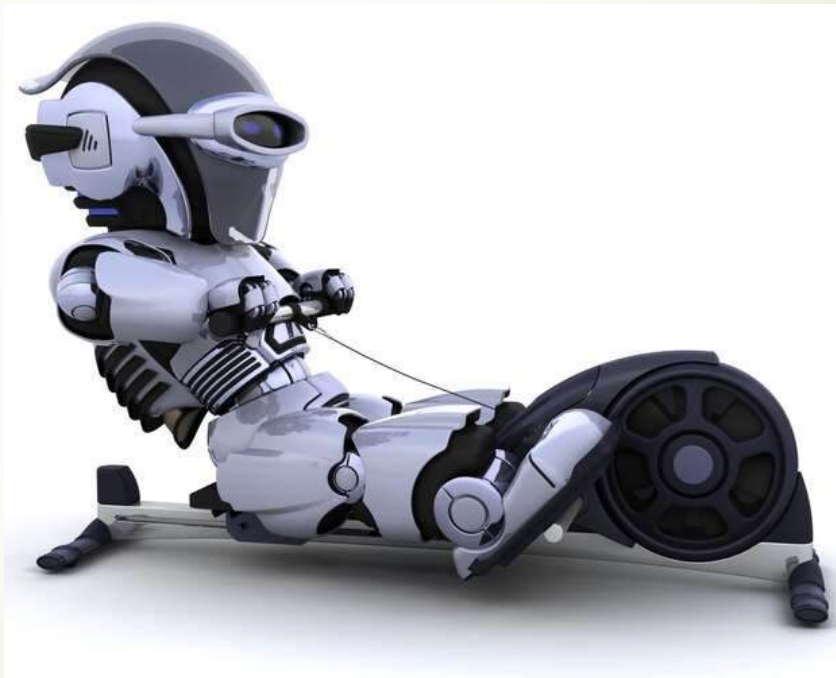
Feature	Cartesian Scheme	Joint-space Scheme
Definition	The trajectory is planned in Cartesian space (X, Y, Z) and then converted into joint-space movements using inverse kinematics .	The trajectory is planned directly in joint space ($\theta_1, \theta_2, \dots, \theta_n$) without converting from Cartesian coordinates.
Complexity	High: Requires solving inverse kinematics at each time step, making it computationally expensive.	Lower: No need for inverse kinematics, making it computationally efficient .
Accuracy	Ensures that the end-effector follows an exact Cartesian path .	The end-effector's path is not explicitly controlled in Cartesian space, but joint movements are smooth.
Computational Cost	High, due to solving inverse kinematics online.	Lower, as joint angles are directly interpolated.
Path Smoothness	Cartesian trajectories may cause jerky joint motions if inverse kinematics solutions change rapidly.	Naturally smooth in joint space, as motion follows simple interpolation between joint angles.
Suitability	Best when precise end-effector motion is needed, such as in welding, painting, or surgical robotics .	Best when smooth joint motions are required, such as in pick-and-place tasks or general robotic arm control .



Any Question?

ROBOTICS

Topic 4: Robot Dynamics



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Robotic dynamics

- ▶ Robotic dynamics deals with the forces and torques that act on a robotic system as it moves.
 - ▶ Unlike kinematics (which focuses only on positions and velocities), dynamics considers mass, inertia, forces, and torques, making it essential for real-world applications like control, simulation, and stability analysis.
- 

Concepts in Robot Dynamics

- Forward Dynamics: Given torques (τ), compute resulting motion.
- Inverse Dynamics: Given motion ($\theta, \dot{\theta}, \ddot{\theta}$), find joint torques (τ).
- Robotic Joint Torque Components:
 - Inertia Term: Depends on mass distribution (moment of inertia tensor).
 - Coriolis & Centrifugal Forces: Affect motion in dynamic conditions.
 - Gravity Term: Depends on mass and gravity.

Forward Dynamics

- Forward dynamics is the process of computing the resulting motion of a robotic system given the applied joint torques or forces. It is used in robot simulations, control design, and motion prediction.
- Given the joint torques (τ), compute the joint accelerations ($\ddot{\theta}$) and then integrate over time to find velocities ($\dot{\theta}$) and positions (θ).
- General Equation

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + G(\theta) = \tau$$

Solving for joint accelerations ($\ddot{\theta}$):

$$\ddot{\theta} = M^{-1}(\tau - C(\theta, \dot{\theta})\dot{\theta} - G(\theta))$$

Where:

- $M(\theta) \rightarrow$ Mass (inertia) matrix
- $C(\theta, \dot{\theta}) \rightarrow$ Coriolis & Centrifugal forces
- $G(\theta) \rightarrow$ Gravity forces
- $\tau \rightarrow$ Applied joint torques
- Once we find $\ddot{\theta}$, we integrate over time to get velocity and position:

$$\dot{\theta}(t) = \int \ddot{\theta} dt$$

$$\theta(t) = \int \dot{\theta} dt$$

Inverse Dynamics

- Inverse dynamics is the process of computing the required joint torques or forces to achieve a desired motion of a robotic system. It is widely used in robot control, motion planning, and force-based interactions.
- Given joint positions (θ), velocities ($\dot{\theta}$), and accelerations ($\ddot{\theta}$), find the required torques (τ) to achieve this motion.
- Uses the robot's dynamic equations to compute forces that compensate for inertia, Coriolis, centrifugal, and gravity effects.
- General Formulation

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + G(\theta) = \tau$$

Where:

- $M(\theta) \rightarrow$ Mass (inertia) matrix
- $C(\theta, \dot{\theta}) \rightarrow$ Coriolis & Centrifugal forces
- $G(\theta) \rightarrow$ Gravity forces
- $\tau \rightarrow$ General Formulation



Equations of Motion: Newton-Euler vs. Lagrangian Methods

► Newton-Euler Equations

- Uses forces and torques directly.
- Based on Newton's Second Law ($\mathbf{F} = m\mathbf{a}$) and Euler's rotational dynamics ($\boldsymbol{\tau} = \mathbf{I}\boldsymbol{\alpha}$)

► Recursive algorithm:

Forward Pass: Compute linear/angular velocities.

Backward Pass: Compute forces/torques.

Faster and used in real-time controllers.

Example

- Two-Link Planar Manipulator
- For a 2-DOF robotic arm, given:
 - Applied joint torques: τ_1, τ_2
 - Initial joint positions: θ_1, θ_2
 - Initial joint velocities: $\dot{\theta}_1, \dot{\theta}_2$

- We compute joint accelerations using:

$$\ddot{\theta} = M^{-1}(\tau - C\dot{\theta} - G)$$

Then integrate to find velocity and position over time

Question

➤ A 2-DOF planar robotic arm has two rotary joints and is affected by gravity. Given the robot's mass properties, inertia, and applied joint torques, determine how the robot moves using forward dynamics.

➤ Given Data:

- Link 1: Length $l_1=1\text{m}$, Mass $m_1=2\text{kg}$, Inertia $I_1=0.5\text{kg.m}^2$
- Link 2: Length $l_2=0.8\text{m}$, Mass $m_2=1.5\text{kg}$, Inertia $I_2=0.3\text{kg.m}^2$
- Initial Conditions:
 - $\theta_1(0) = 30^\circ, \theta_2(0) = 45^\circ$
 - $\dot{\theta}_1(0) = 0, \dot{\theta}_2(0) = 0$

➤ We compute joint accelerations using:

$$\ddot{\theta} = M^{-1}(\tau - C\dot{\theta} - G)$$

Then integrate to find velocity and position over time

Understanding the Forward Dynamics Equation

- The equation governing motion is:

$$\ddot{\theta} = M^{-1}(\tau - C\dot{\theta} - G)$$

- where:

- M is the mass (inertia) matrix.
- C represents Coriolis and centrifugal forces.
- G is the gravity torque vector.
- τ is the applied joint torque vector.
- $\theta, \dot{\theta}$ and $\ddot{\theta}$ are joint positions, velocities, and accelerations.

- Since we are considering only the effect of gravity, we assume:

- No applied joint torques: $\tau = [0,0]$.
- No initial velocity: $\dot{\theta}(0) = [0,0]$.

- Thus, the equation simplifies to: $\ddot{\theta} = M^{-1}(-G)$

Compute the Inertia Matrix M

- The **inertia**
- **matrix** for a 2-DOF planar robotic arm is:

$$M(\theta) = \begin{bmatrix} I_1 + I_2 + m_2 l_1^2 + 2m_2 l_1 l_2 \cos \theta_2 & I_2 + m_2 l_1 l_2 \cos \theta_2 \\ I_2 + m_2 l_1 l_2 \cos \theta_2 & I_2 \end{bmatrix}$$

- Substituting given values:

$$I_1 = 0.5, I_2 = 0.3$$

$$m_1 = 2, m_2 = 1.5$$

$$l_1 = 1.0, l_2 = 0.8$$

$$\theta_2 = 45^\circ \rightarrow \cos(45^\circ) \approx 0.707$$

$$M(\theta) = \begin{bmatrix} 3.996 & 1.148 \\ 1.148 & 0.3 \end{bmatrix}$$

Coriolis and Centrifugal Forces $C(\theta, \dot{\theta})$

- The Coriolis and centrifugal matrix is:

$$C(\theta, \dot{\theta}) = \begin{bmatrix} -m_2 l_1 l_2 \sin(\theta_2) (\dot{\theta}_2) & -m_2 l_1 l_2 \sin(\theta_2) (\dot{\theta}_1 + \dot{\theta}_2) \\ m_2 l_1 l_2 \sin(\theta_2) \dot{\theta}_1 & 0 \end{bmatrix}$$

- Since both joint velocities are initially zero:

$$C(\theta, \dot{\theta}) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Compute the Gravity Vector G

- The **gravity vector** is given by:

$$G(\theta) = \begin{bmatrix} m_1 g l_1 \cos(\theta_1) + m_2 g (l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)) \\ m_2 g l_2 \cos(\theta_1 + \theta_2) \end{bmatrix}$$

- where

$g = 9.81 \text{ m/s}^2$ (acceleration due to gravity),

m_1, m_2 = masses of links 1 and 2,

l_1 = full length of link 1,

θ_1, θ_2 = joint angles.

- Substituting values:

$$G(\theta) = \begin{bmatrix} 2 \times 9.81 \times 1 \times \cos(30^\circ) + 1.5 \times 9.81 \times (1 \times \cos(30^\circ) + 0.8 \times \cos(30^\circ + 45^\circ)) \\ 1.5 \times 9.81 \times 0.8 \times \cos(30^\circ + 45^\circ) \end{bmatrix}$$

$$G(\theta) = \begin{bmatrix} 17.0 + 15.8 \\ 3.05 \end{bmatrix} = \begin{bmatrix} 32.8 \\ 3.05 \end{bmatrix}$$

Computing Joint Accelerations $\ddot{\theta}$

- Now we compute the joint accelerations $\ddot{\theta}_1$ and $\ddot{\theta}_2$ using the forward dynamics equation:

$$\ddot{\theta} = M^{-1}(\tau - C\dot{\theta} - G)$$

- Since the applied torques are zero and the Coriolis matrix is also zero:

$$\ddot{\theta} = M^{-1}(-G)$$

- Computing the M^{-1}
- Multiplying $-G$
- Thus, the **joint accelerations**

Lagrangian Method

- Uses energy (kinetic & potential) equations.
- Based on the Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \tau_i$$

where

q represents the coordinates (joint angles, positions)

$\dot{q}$ is the velocity

τ is the force/torque

- $L = K - P$ (*Lagrangian = Kinetic Energy - Potential Energy*)
- This method leads to explicit equations of motion in the form of:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau$$

where

$M(q)$ is the mass/inertia matrix.

$C(q, \dot{q})\dot{q}$ represents Coriolis and centrifugal forces.

$G(q)$ is the gravitational force vector.

- More computationally expensive but useful for symbolic analysis.

Lagrange-Euler Method for Computing Joint Torques

- 1. Define the Generalized Coordinates
- For an n – *link* robotic manipulator:
 - Rotary joints: $q_i = \theta_i$ (joint angles).
 - Prismatic joints: $q_i = d_i$ (linear displacements)

➤ The Lagrangian function is defined as:

➤ $L = K - P$

where:

K = Kinetic Energy

P = Potential Energy

Lagrange-Euler Method for Computing Joint Torques

- Compute the Kinetic and Potential Energy
- Kinetic Energy (K)
- For each link i :

$$K_i = \frac{1}{2}m_i v_i^2 + \frac{1}{2}\omega_i^T I_i \omega_i$$

- where:
 - m_i = mass of link i
 - v_i = linear velocity of link i
 - ω_i = angular velocity of link i
 - I_i = inertia tensor of link i
- The total kinetic energy of the manipulator is:

$$K = \sum_{i=1}^n K_i$$

Lagrange-Euler Method for Computing Joint Torques

- **Potential Energy (P)**

- For each link i , potential energy is due to gravity:

$$P_i = m_i g h_i$$

- where h_i is the height of the center of mass of link i
- The total potential energy of the manipulator is:

$$P = \sum_{i=1}^n P_i$$

Lagrange-Euler Method for Computing Joint Torques

- Compute the Equations of Motion
- Using the Lagrange-Euler equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \tau_i$$

- where τ_i is the torque or force required at joint i

1. Compute $\frac{\partial L}{\partial \dot{q}_i}$ (momentum term).
2. Differentiate with respect to time to obtain $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right)$
3. Compute $\frac{\partial L}{\partial q_i}$: (position-dependent forces).
4. Solve for τ_i using the Lagrange-Euler equation.

Example

► Two-Link Planar Manipulator

For a **2-DOF robotic arm**, given:

- Joint positions: θ_1, θ_2
- Joint velocities: $\dot{\theta}_1, \dot{\theta}_2$
- Joint accelerations: $\ddot{\theta}_1, \ddot{\theta}_2$
- We compute required torques τ_1, τ_2

$$\tau_1 = M_{11}\ddot{\theta}_1 + M_{12}\ddot{\theta}_2 + C_1 + G_1$$

$$\tau_2 = M_{21}\ddot{\theta}_1 + M_{22}\ddot{\theta}_2 + C_2 + G_2$$

Where:

- $M_{ij} \rightarrow$ Mass matrix elements
- $C_i \rightarrow$ Coriolis & Centrifugal forces
- $G_j \rightarrow$ Gravity forces

Question

- ▶ Consider a 2-DOF planar manipulator with rotary joints at θ_1, θ_2 .
- Compute kinetic energy and potential energy for both links.
- Apply the Lagrange-Euler equation to find torques τ_1, τ_2
- ▶ Given:
 - Link 1: Length l_1 , Mass m_1 , Inertia I_1 .
 - Link 2: Length l_2 , Mass m_2 , Inertia I_2 .
 - Joint Variables: θ_1, θ_2 (joint angles).
 - Gravitational acceleration: g
- ▶ Each link has rotational motion, so we use generalized coordinates $q_1 = \theta_1, q_2 = \theta_2$

Compute Kinetic Energy (K)

- ▶ The kinetic energy of each link consists of translational and rotational energy:

$$K = \sum_{i=1}^2 \left(\frac{1}{2} m_i v_i^2 + \frac{1}{2} I_i \omega_i^2 \right)$$

- ▶ Velocity of the Links
 - The center of mass (COM) of each link is located at its midpoint:
 - Link 1 COM Position:

- ▶
$$x_1 = \frac{l_1}{2} \cos \theta_1, \quad y_1 = \frac{l_1}{2} \sin \theta_1$$

➤ Link 2 COM Position:

$$x_2 = l_1 \cos \theta_1 + \frac{l_2}{2} \cos(\theta_1 + \theta_2)$$
$$y_2 = l_1 \sin \theta_1 + \frac{l_2}{2} \sin(\theta_1 + \theta_2)$$

➤ Differentiate to obtain velocities:

$$v_1^2 = \left(\frac{dx_1}{dt} \right)^2 + \left(\frac{dy_1}{dt} \right)^2$$

$$v_2^2 = \left(\frac{dx_2}{dt} \right)^2 + \left(\frac{dy_2}{dt} \right)^2$$

➤ Using differentiation:

$$v_1^2 = \left(\frac{l_1}{2} \dot{\theta}_1 \sin \theta_1 \right)^2 + \left(\frac{l_1}{2} \dot{\theta}_1 \cos \theta_1 \right)^2$$

$$v_2^2 = \left(-l_1 \dot{\theta}_1 \sin \theta_1 - \frac{l_2}{2} (\dot{\theta}_1 + \dot{\theta}_2) \sin(\theta_1 + \theta_2) \right)^2 + \left(l_1 \dot{\theta}_1 \cos \theta_1 + \frac{l_2}{2} (\dot{\theta}_1 + \dot{\theta}_2) \cos(\theta_1 + \theta_2) \right)^2$$

- Thus, the kinetic energy is:

$$K = \frac{1}{2}m_1 \left(\frac{l_1}{2} \dot{\theta}_1 \right)^2 + \frac{1}{2}I_1 \dot{\theta}_1^2 \\ + \frac{1}{2}m_2 (v_2^2) + \frac{1}{2}I_2 (\dot{\theta}_1 + \dot{\theta}_2)^2$$

- **Compute Potential Energy (P)**
- Each link's potential energy is based on height:

$$P = \sum_{i=1}^2 m_i g h_i$$

$$P_1 = m_1 g \frac{l_1}{2} \sin \theta_1$$

$$P_2 = m_2 g \left(l_1 \sin \theta_1 + \frac{l_2}{2} \sin(\theta_1 + \theta_2) \right)$$

$$P = m_1 g \frac{l_1}{2} \sin \theta_1 + m_2 g \left(l_1 \sin \theta_1 + \frac{l_2}{2} \sin(\theta_1 + \theta_2) \right)$$

- **Apply Lagrange-Euler Equation**
- The equation of motion for each joint

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_i} \right) - \frac{\partial L}{\partial \theta_i} = \tau_i$$

where $L = K - P$.

1. Compute $\frac{\partial L}{\partial \dot{\theta}_1}$ and $\frac{\partial L}{\partial \dot{\theta}_2}$.
2. Differentiate w.r.t time to get $\frac{d}{dt}$.
3. Compute $\frac{\partial L}{\partial \theta_1}$ and $\frac{\partial L}{\partial \theta_2}$.
4. Solve for τ_1 and τ_2 .

► **Solve for Joint Torques τ_1, τ_2**

► After substitution, the final torques are:

$$\tau_1 = M_{11}\ddot{\theta}_1 + M_{12}\ddot{\theta}_2 + C_1 + G_1$$

$$\tau_2 = M_{21}\ddot{\theta}_1 + M_{22}\ddot{\theta}_2 + C_2 + G_2$$

where:

- M_{ij} are elements of the mass matrix,
- C_i represents Coriolis & centrifugal effects,
- G_i are gravity torques.

Question

- Consider a 2-DOF planar manipulator with rotary joints at θ_1, θ_2 . The system consists of two links with the following given parameters:

- Given Data

- Link 1:

- Length $L_1 = 1 \text{ m}$
- Mass $m_1 = 2 \text{ kg}$
- Moment of Inertia $I_1 = 0.5 \text{ kgm}^2$

- Link 2:

- Length $L_2 = 0.8 \text{ m}$
- Mass $m_2 = 1.5 \text{ kg}$
- Moment of Inertia $I_2 = 0.2 \text{ kg} \cdot \text{m}^2$

- Assume joint positions and velocities: $\theta_1 = 30^\circ, \theta_2 = 45^\circ$

$$\dot{\theta}_1 = 1.2 \text{ rad/s}, \dot{\theta}_2 = -0.8 \text{ rad/s}$$

Gravitational Acceleration: $g = 9.81 \text{ m/s}^2$

Compute Kinetic Energy (K)

Given the formula

$$K = \sum_{i=1}^2 \left(\frac{1}{2} m_i v_i^2 + \frac{1}{2} I_i \omega_i^2 \right)$$

where:

- v_1, v_2 are the velocities of the centers of mass of Link 1 and Link 2.
- ω_1, ω_2 are the angular velocities.

Velocity of Center of Mass

For Link 1, the center of mass is at $\frac{l_1}{2}$, so: $v_1 = \frac{l_1}{2} \dot{\theta}_1$

For Link 2, using kinematics: $v_2^2 = l_1^2 \dot{\theta}_1^2 + \frac{l_2^2}{4} (\dot{\theta}_1 + \dot{\theta}_2)^2 + l_1 l_2 \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2) \cos \theta_2$

Since both links rotate, $\omega_1 = \dot{\theta}_1$, $\omega_2 = \dot{\theta}_1 + \dot{\theta}_2$

Therefore, KE expression $K = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} I_1 \dot{\theta}_1^2 + \frac{1}{2} m_2 v_2^2 + \frac{1}{2} I_2 (\dot{\theta}_1 + \dot{\theta}_2)^2$

$$K = \frac{1}{2}(2) \left(\frac{1}{2}(1)(1.2)^2 \right) + \frac{1}{2}(0.5)(1.2)^2 + \frac{1}{2}(1.5)(1^2(1.2)^2 + \frac{0.8^2}{4}(1.2 + 0.8)^2 + 1(0.8)(1.2)(1.2 + 0.8) \cos 45^\circ) + \frac{1}{2}(0.2)(1.2 + 0.8)^2$$

$$K = 3.70 \text{ J}$$

Compute Potential Energy (P)

From the given formula:

$$P = \sum_{i=1}^2 m_i g h_i$$

where:

- **For Link 1:**

$$h_1 = \frac{l_1}{2} \sin \theta_1$$

$$P_1 = m_1 g h_1 = m_1 g \frac{l_1}{2} \sin \theta_1$$

- **For Link 2:**

$$h_2 = l_1 \sin \theta_1 + \frac{l_2}{2} \sin(\theta_1 + \theta_2)$$

$$P_2 = m_2 g h_2 = m_2 g \left(l_1 \sin \theta_1 + \frac{l_2}{2} \sin(\theta_1 + \theta_2) \right)$$

- Thus,

$$P = m_1 g h_1 + m_2 g h_2$$

$$P = (2)(9.81) \frac{1}{2} \sin(30^\circ) + (1.5)(9.81) \left(1 \sin(30^\circ) + \frac{0.8}{2} \sin(30^\circ + 45^\circ) \right)$$

$$P = 17.95 \text{ J}$$



Compute Lagrangian (L)

▀ $L = K - P$

▀ $L = 3.70 - 17.95 = -14.25$

Apply Lagrange-Euler Equations

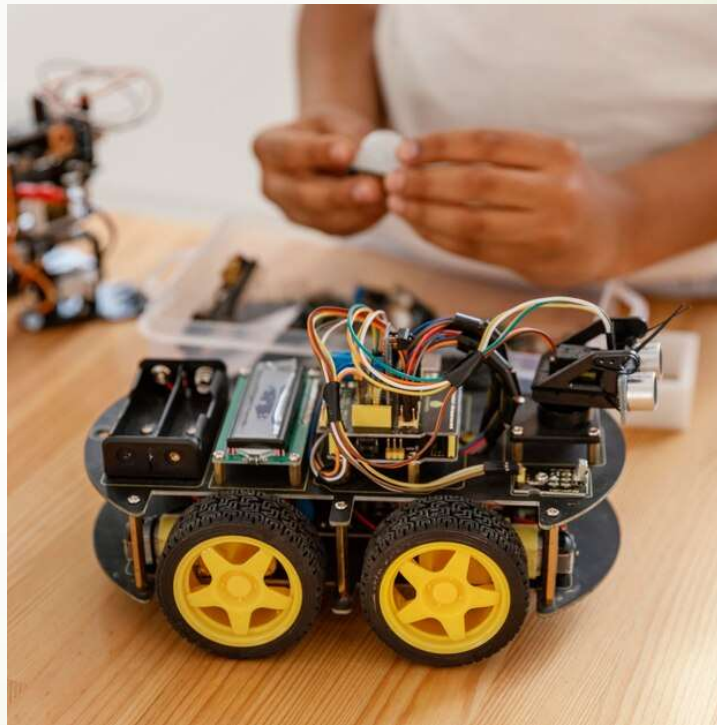
- ▶ The Lagrange equation is:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_i} \right) - \frac{\partial L}{\partial \theta_i} = \tau_i$$

- ▶ For each joint $i(\theta_1, \theta_2)$, we compute:
- ▶ **Compute Mass Matrix M**
- ▶ The mass matrix elements:
- ▶ Compute Coriolis and Centrifugal Terms
- ▶ Compute Gravity Terms
- ▶ Compute Torques

ROBOTICS

Topic 5: Control Scheme



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Outline

- Control Schemes
- Control of Motor
- Mathematical Representation of Joint Torque
- Partition Control Scheme
- Partition Control Scheme in Action
- Control Architecture in Robotics



Introduction

- Robots are controlled using various schemes to ensure precise motion and stability.
- Control schemes refer to the methods or strategies used to regulate and manage the movement and operation of a robot.
- Control schemes encompass various methods for programming and controlling a robot's movements and behaviour.
- They are designed to handle variables like position, velocity, force, and torque

Control of Motor



A DC motor is often used at each joint, where torque is proportional to the armature current.



Control is necessary to manage joint torque effectively.

$$\tau_m \propto I_A$$
$$\tau = K_m \cdot I_A$$

Mathematical Representation of Joint Torque

Joint torque t can be represented as follows:

$$t = D(\theta)\ddot{\theta} + h(\theta, \dot{\theta}) + C(\theta)$$

where

$D(\theta)$ = Inertia terms (related to mass and motion resistance)

$h(\theta, \dot{\theta})$ = Coriolis and centrifugal terms (effects due to motion)

$C(\theta)$ = Gravity terms (gravitational forces acting on the robot)

Partitioned Control Scheme

- To control a robotic joint, a Partitioned Control Scheme is considered
- It involves separating different components of torque for better efficiency.

- General Control Equation

$$\tau = \alpha \tau' + \beta$$

- This equation defines the total torque (τ) applied to a robot joint, which consists of two key components:
- $\alpha \tau' \rightarrow$ Represents the effect of the control signal (τ') scaled by the inertia matrix (α).
- $\beta \rightarrow$ Represents the combined effects of Coriolis, centrifugal, gravity, and friction forces.

Partitioned Control Scheme

■ Definition of Components

- $\alpha = D(\theta)$ represents the inertia term.
- $D(\theta)$ is the inertia matrix, which depends on the joint position (θ).
- It accounts for the resistance to acceleration due to the mass and shape of the robotic links.

$$\beta = h(\theta, \dot{\theta}) + C(\theta) + F(\theta, \dot{\theta})$$

■ β represents all the dynamic effects acting on the joint:

■ $h(\theta, \dot{\theta}) \rightarrow$ Coriolis and centrifugal forces

These forces arise due to the motion of the robotic arm.

■ $C(\theta) \rightarrow$ Gravity force

The effect of gravity on the robotic joint.

■ $F(\theta, \dot{\theta}) \rightarrow$ Friction and other resistive forces

Includes joint friction and external resistances.

Understanding the Partitioned Control Scheme

- The partitioning method separates the dynamic effects for easier control:
- The controller first computes a control signal τ' based on feedback.
- The inertia scaling (α) adjusts the control torque.
- The disturbance compensation (β) accounts for non-linear effects.

Partitioned Control Scheme in Action

■ Given Data

- Desired joint position: $\theta_d = 60^\circ$
- Current joint position: $\theta = 45^\circ$
- Desired joint velocity: $\dot{\theta}_d = 10^\circ/s$
- Current joint velocity: $\dot{\theta} = 5^\circ/s$
- Proportional gain: $K_p = 3$
- Derivative gain: $K_D = 2$
- Inertia matrix: $D(\theta) = 5$
- Coriolis and centrifugal forces: $h(\theta, \dot{\theta}) = 4$
- Gravity effect: $C(\theta) = 2$
- Friction forces: $F(\theta, \dot{\theta}) = 3$

Partitioned Control Scheme in Action

- Compute the Control Signal τ'
- The PD control law is used:

$$\tau' = K_P \cdot E + K_D \cdot \dot{E}$$

- Where:

$$E = \theta_d - \theta = 60^\circ - 45^\circ = 15^\circ$$

$$\dot{E} = \dot{\theta}_d - \dot{\theta} = 10^\circ/s - 5^\circ/s = 5^\circ/s$$

- Substituting values:

$$\tau' = (3 \times 15) + (2 \times 5)$$

$$\tau' = 45 + 10 = 55$$

Partitioned Control Scheme in Action

- Compute α and β
- From the given formulas:

$$\alpha = D(\theta) = 5$$

$$\beta = h(\theta, \dot{\theta}) + C(\theta) + F(\theta, \dot{\theta})$$

$$\beta = 4 + 2 + 3 = 9$$

- Compute the Final Torque τ
- Using the formula:

$$\tau = \alpha\tau' + \beta$$

$$\tau = (5 \times 55) + 9$$

- Substituting values:

$$\tau = 275 + 9 = 284$$



Partitioned Control Scheme in Action

- ▶ Apply the Torque
- ▶ The computed torque $\tau = 284$ is applied to the robotic joint.
- ▶ This moves the joint closer to the desired position $\theta_d = 60^0$
- ▶ The control loop repeats continuously, updating values to minimize the error.

Control Architecture in Robotics

Figure 1 represents a robot joint control scheme using a partitioned control approach.

It combines feedback control with the system dynamics to regulate joint motion accurately.

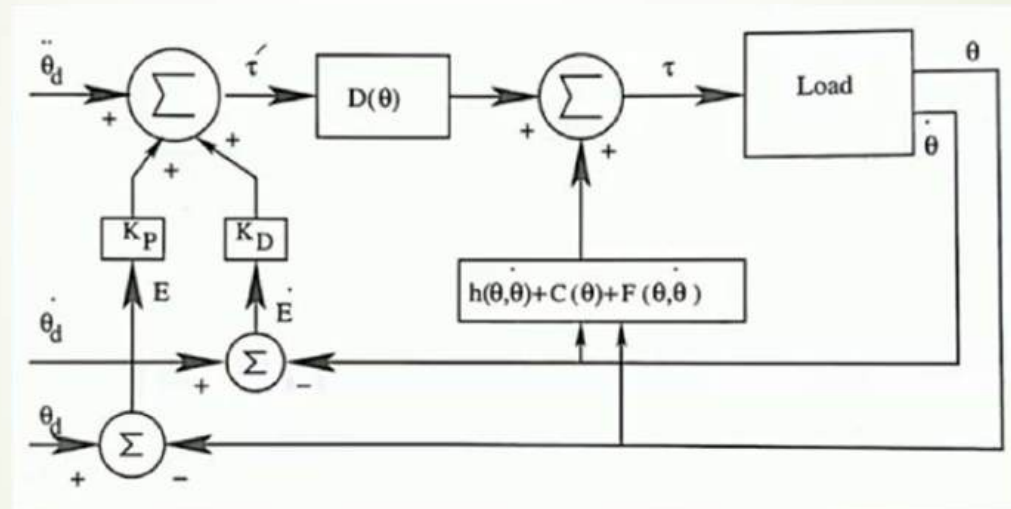


Figure 1: Partition Control

Control Architecture

► Desired Motion Input (θ , $\dot{\theta}_d$)

- θ_d = Desired joint position.
- $\dot{\theta}_d$ = Desired joint velocity.
- $\ddot{\theta}_d$ = Desired joint acceleration.
 - These inputs represent the reference trajectory that the controller will track.

► Error Calculation (E)

- The error (E) is calculated as: $E = \theta_d - \theta$
- This represents the difference between the desired and actual joint position.

► Proportional-Derivative (PD) Control

- The controller consists of Proportional (K_P) and Derivative (K_D) gains.
- Proportional term (K_P): Helps correct position errors.
- Derivative term (K_D): Helps stabilize the system by considering velocity error.
- The sum of these terms generates a control signal.

Control Architecture

- Dynamics Partitioning
- The system dynamics are divided into:
 - $D(\theta)$: Inertia matrix – Represents how the joint resists motion due to its mass.
 - $h(\theta, \dot{\theta})$: Coriolis and centrifugal forces – Accounts for motion-induced forces.
 - $C(\theta)$: Gravity compensation – Adjusts for gravitational effects.
 - $F(\theta, \dot{\theta})$: Other friction and forces – Models additional resistances.
- **Torque Computation (τ)**
 - The control torque τ is computed considering
$$\tau = D(\theta) \cdot (\text{Control signal}) + (h(\theta, \dot{\theta}) + C(\theta) + F(\theta, \dot{\theta}))$$
 - This ensures the applied torque accounts for all system effects.
- **Load (Joint Motion)**
 - The torque τ is applied to the robot joint/load, causing it to move.
 - The actual joint position θ is fed back to the controller to adjust future torque.

PUMA

Robots use different control architectures to manage motion and ensure precision.

A well-known robotic manipulator, PUMA (Programmable Universal Machine for Assembly), follows such control principles.

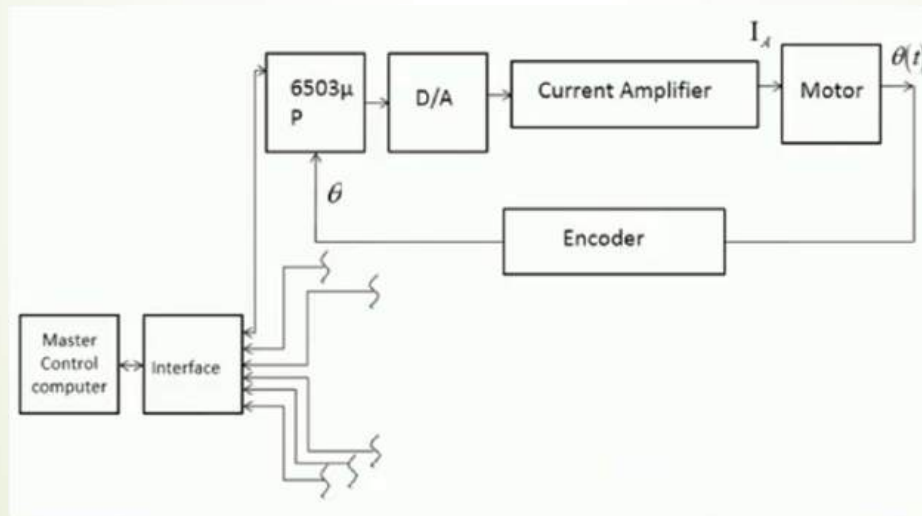


Figure 2: PUMA Architecture



Importance of Control Schemes

- Ensure precise movement of robotic joints
 - Improve efficiency and accuracy
 - Help robots adapt to changing environments
 - Minimize errors and prevent collisions.
- 

ROBOTICS

Topic 6: Sensors



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SENSORS

- ❖ Human-beings collect information of the surroundings using their sensors, namely eyes, ears, nose, skin etc. , in order to perform various tasks.
 - ❖ A sensor is used to take measurement of physical variable.
 - ❖ A sensor requires **calibration**
 - ❖ Sensors are used to build intelligent robots
- 

Classification of sensors

1.

SENSORS

Interna Sensors

(used to operate the drive units)

Ex. Position sensors,
velocity sensors,
Acceleration Sensors,
Force / Moment sensors

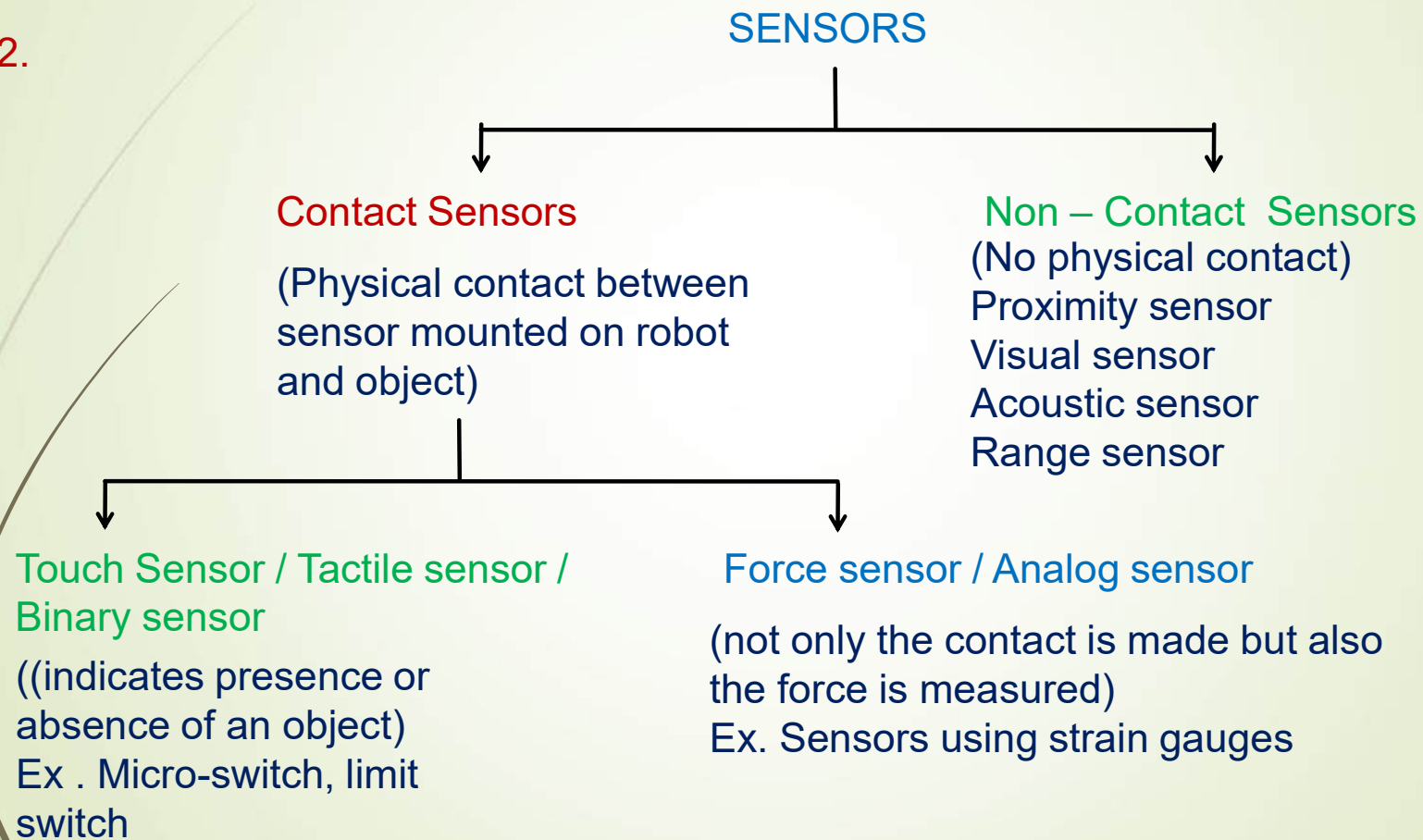
External Sensors

(used to collect information of the environment)

Ex. Temperature sensors,
Visual sensors,
Proximity Sensors,
Acoustic sensor

Classification of sensors (cont.)

2.





Characteristics of Sensors

- ❖ **Range** : Difference between the maximum and minimum values of the input that can be measured.
- ❖ **Response** : should be capable of responding to the changes in minimum time.
- ❖ **Accuracy** : deviation from exact quantity
- ❖ **Sensitivity** : change in output / change in input
- ❖ **Linearity**: constant sensitivity
- ❖ **Repeatability**: deviation from reading to reading , when these are taken for several times under identical condtions
- ❖ **Resolution**



Touch sensor

- ❖ Used to indicate whether contact has been made between two objects
 - ❖ Does not determine the magnitude of contact force
 - ❖ Ec. : Micro-switch , Limit switch
- 

Connected to robot's wrist

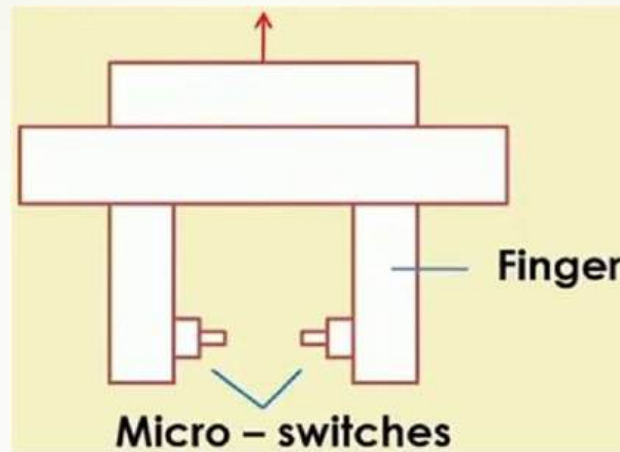


Figure: Micro – switches placed on two fingers of a robotic hand

Position sensor

1. Potentiometer

Linear Potentiometer

Angular Potentiometer

Angular Potentiometer

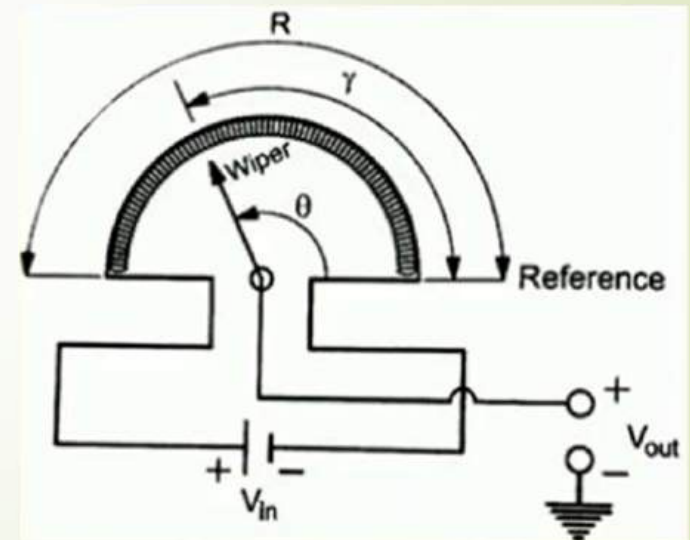
Θ : Angular displacement of the wiper with respect to the reference

R: Total resistance

r : Resistance of the coil between the wiper and the reference

V_{in} : Input voltage

V_{out} : Output voltage



Angular Potentiometer (contd.)

$$\frac{V_{out}}{r} = \frac{V_{in}}{R}$$
$$\Rightarrow V_{out} = \frac{r}{R} V_{in}$$

For the known values of V_{in} , R ; $V_{out} = f(r)$

By measuring V_{out} , r can be determined and hence, angular displacement Θ

Demerit

Resistance of the wire is temperature dependent

Potentiometer is temperature sensitive

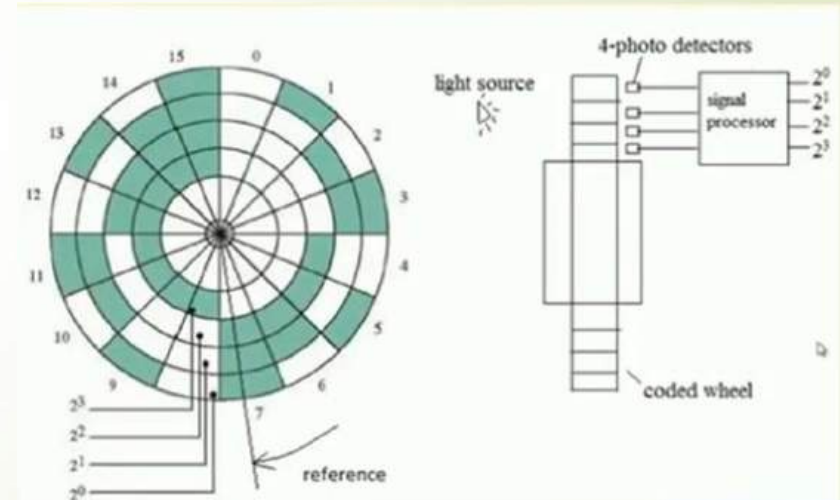
2. Optical Encoder

Absolute Optical Encoder

Incremental Optical Encoder

Absolute Optical Encoder

- ❖ It is mounted on the shaft a rotary device
- ❖ To generate digital word identifying actual position of the shaft measured from zero position.



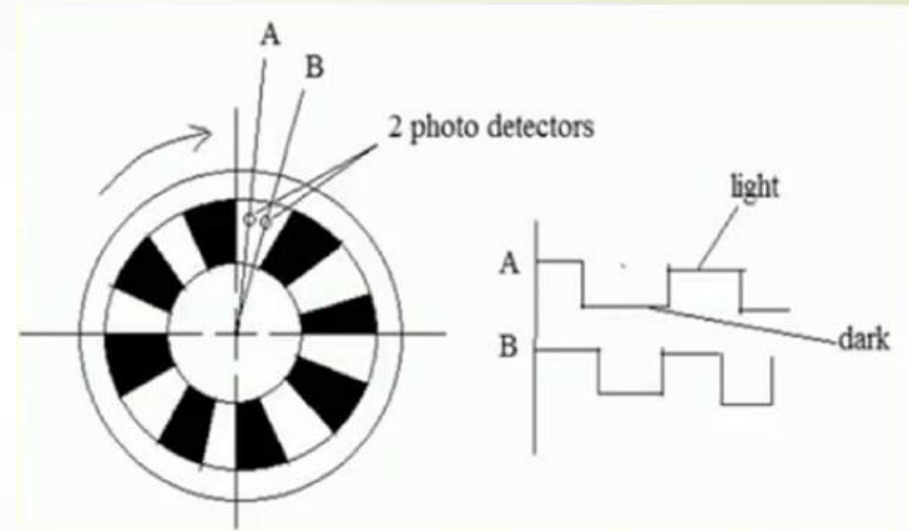
Absolute Optical Encoder (contd.)

Resolution : 1 part in 2^n , where n : number of concentric rings (tracks)

2^3	2^2	2^1	2^0	DV
0	0	0	0	→ 0
0	0	0	1	→ 1
0	0	1	0	→ 2
	:	:		
	:	:		
1	1	1	0	→ 14
1	1	1	1	→ 15

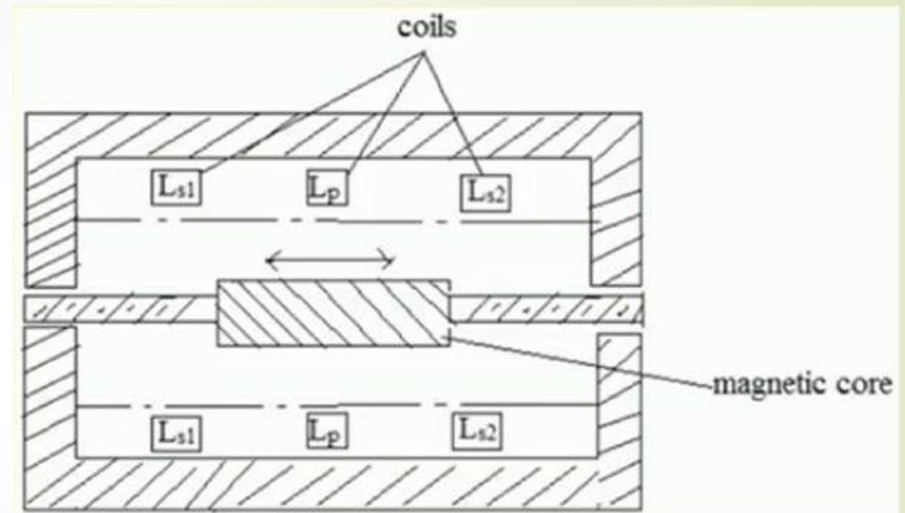
Incremental optical encoder

- ❖ Consists of one coded disc and two photo –detectors
- ❖ By counting the number of light and dark zones, angular displacement can be measured with respect to known starting position.
- ❖ It can determine the direction of rotation also
- ❖ It is construction – wise – simpler, less accurate and less expensive



Linear Variable Differential Transformer (LVDT)

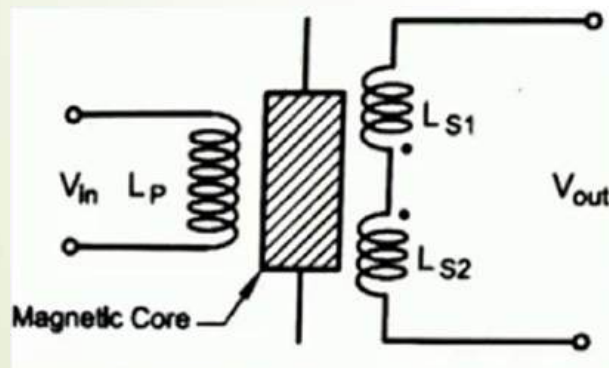
- ❖ It consists of two parts: fixed casing and moving magnetic core
- ❖ In-between the fixed casing and magnetic core, there are one primary (L_p) and two secondary (L_{s1} , L_{s2}) coils
- ❖ Produced voltage output is proportional to the displacement of moving part relative to the fixed one.



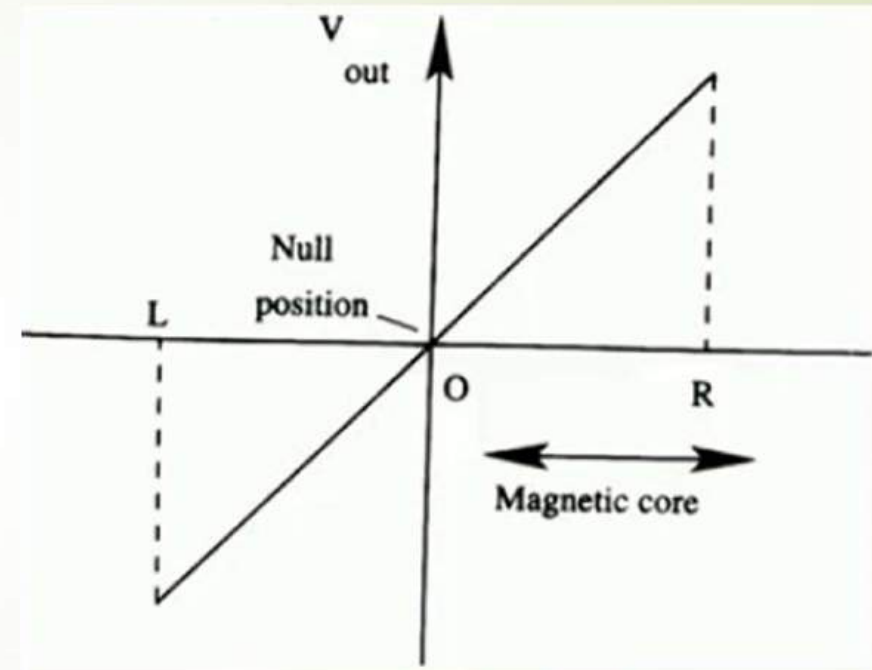
LVDT (contd.)

- ❖ AC voltage is applied to L_p
- ❖ L_{s1} and L_{s2} are connected in series

$$V_{out} = V_{L_{s2}} - V_{L_{s1}}$$

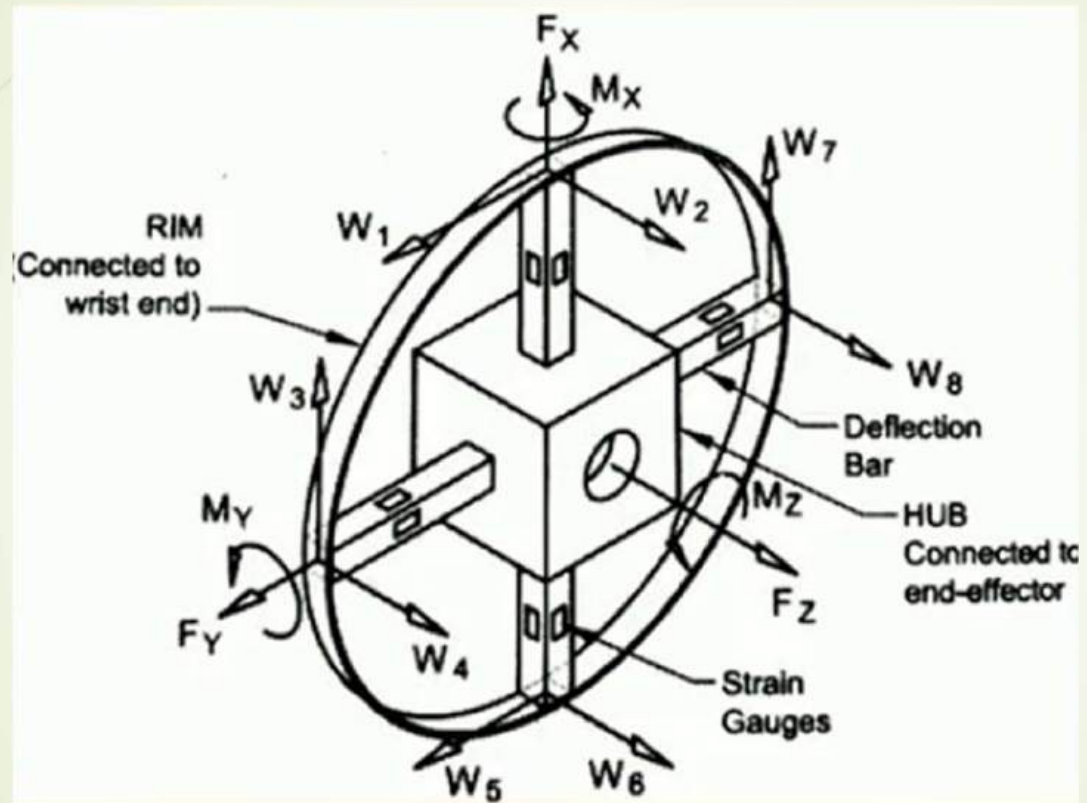


LVDT: equivalent electrical circuit



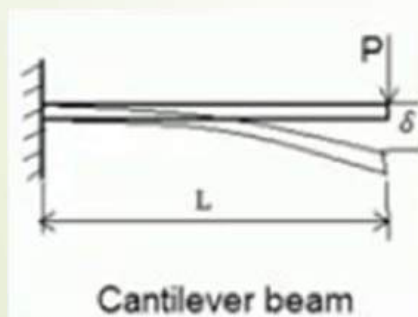
Calibration curve

Force / Moment sensor



Force / Moment sensor (contd.)

- ❖ It is placed between the wrist and end-effector and
- ❖ It consists of 4 deflection bars. Two pairs of strain gauges are mounted on each deflection bar. One end of each deflection bar is rigidly supported by a hub, which is connected to the end-effector end. The other ends of the deflection bars are supported by a common rim, which is connected to the wrist end.
- ❖ External forces cause deflection of the mechanical structure, which are measured using strain gauges.



$$\delta = \frac{PL^3}{3EI}$$

Force / Moment sensor (contd.)

- ❖ Strain gauges is connected to potentiometer circuit, whose output voltage is proportional to the deflection and hence, force.
- ❖ Three components of force (F) and moment (M) each are determined by adding and subtracting the respective components of force. $F = CMW$

$$\begin{array}{l} \text{Forces/} \\ \text{moments} \end{array} \rightarrow \begin{bmatrix} F_x \\ F_y \\ F_z \\ M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & \cdots & -C_{18} \\ C_{21} & C_{22} & C_{23} & \cdots & -C_{28} \\ & & \cdot & & \\ & & \cdot & & \\ & & \cdot & & \\ C_{61} & C_{62} & C_{63} & \cdots & -C_{68} \end{bmatrix} \begin{bmatrix} W_1 \\ W_2 \\ \cdot \\ \cdot \\ \cdot \\ W_8 \end{bmatrix}$$

Calibration matrix
 Readings of the strain gauges


$$F_X = W_3 C_{13} + W_7 C_{17}$$

$$F_Y = W_1 C_{21} + W_5 C_{25}$$

$$F_Z = W_2 C_{32} + W_4 C_{34} + W_6 C_{36} + W_8 C_{38}$$

$$M_X = W_4 C_{44} + W_8 C_{48}$$

$$M_Y = W_2 C_{52} + W_6 C_{56}$$

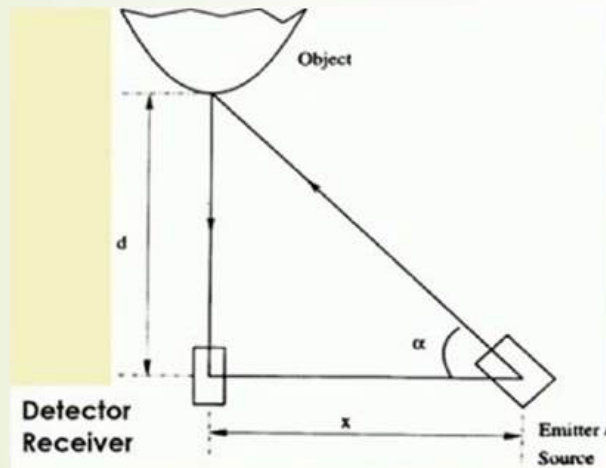
$$M_Z = W_1 C_{61} + W_3 C_{63} + W_5 C_{65} + W_7 C_{67}$$

Precautions

- ❖ Strain gauges are to be properly mounted on the deflection bars
- ❖ Sensor should be operated within the elastic limit of its material (deflection bars) .

Range Sensor

It measures the distance between the sensor (detector) mounted on the robot's body and the object.

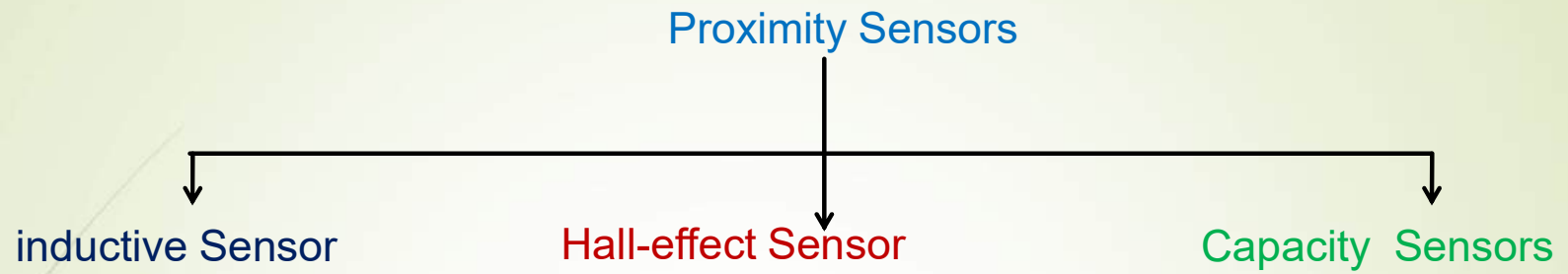


$$d/a = \tan \Theta$$

$$d = a \tan \Theta$$

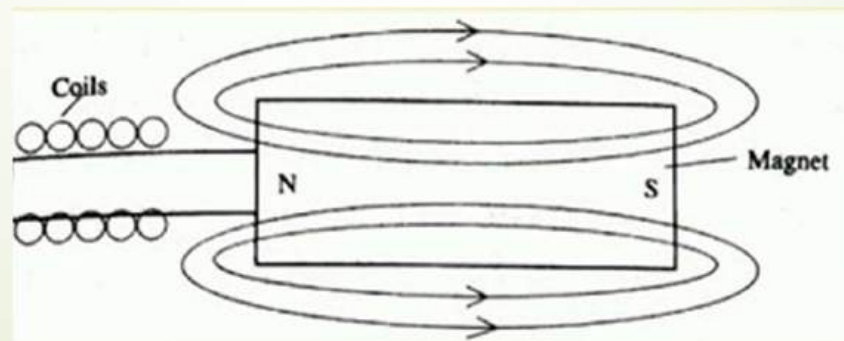
Knowing the values of a and Θ , d can be calculated.

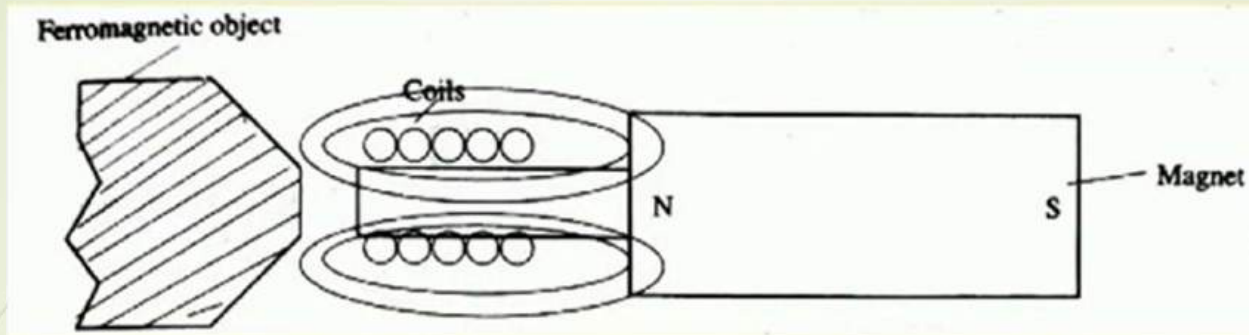
Triangulation method



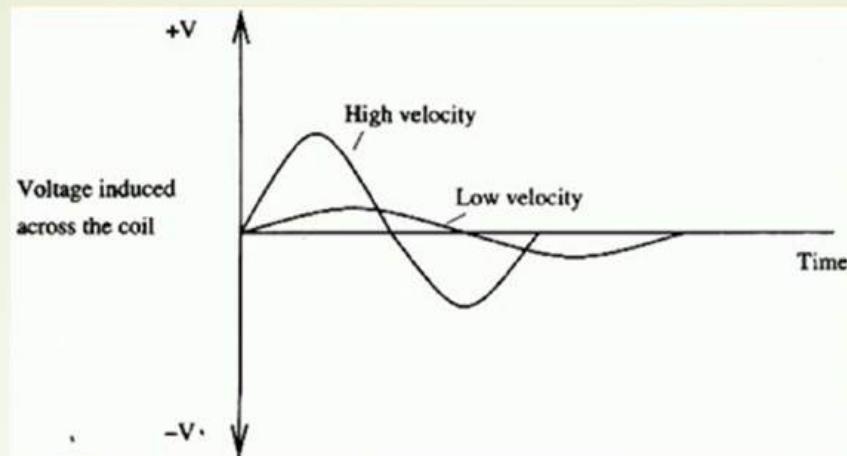
inductive Sensor

- ❖ It consists of a permanent magnet and a wound coil placed next to it



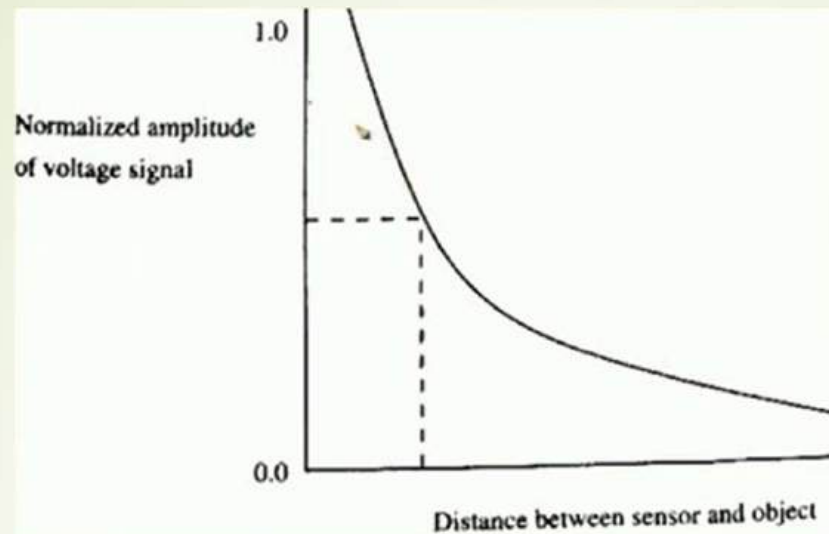


- ❖ The nature of flux lines changes, as the sensor comes closer to a ferromagnetic object.
- ❖ The flux lines change, as the ferromagnetic object either enters or leaves the field of the magnet.
- ❖ Rate of change of the magnetic flux is proportional to induced current (voltage)



Inductive sensor:
Induced voltage vs. time

- ❖ Voltage induced across the coil depends on the speed at which the object either enters or leaves the magnetic field
- ❖ Polarity of the voltage depends on whether the object enters or leaves the field.



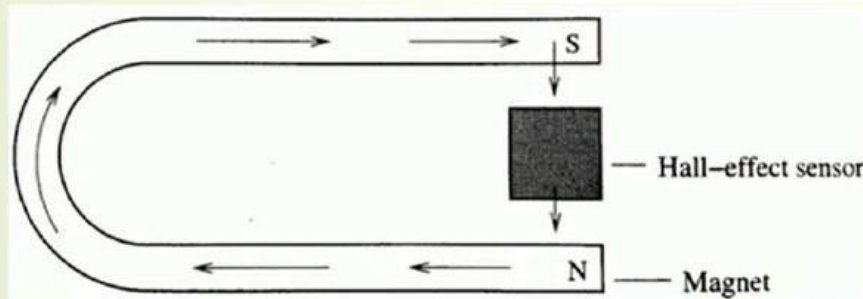
Calibration curve

- ❖ The figure shows the calibration curve corresponding to the measured amplitude of voltage signal, the distance between the sensor and object can be determined.

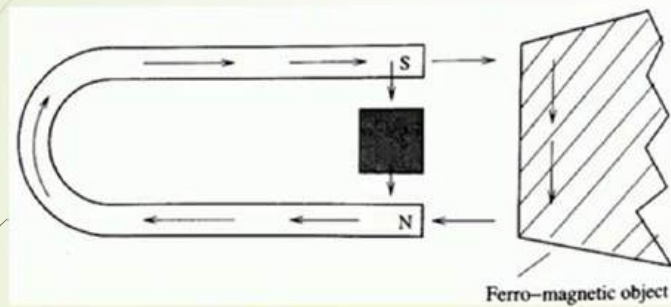
Hall – effect Sensors (for Ferro-magnetic object)

- ❖ It works based on the principle of Lorentz force
- ❖ If a charge of amount q is moving with velocity $\vec{V}$ in a magnetic field of strength $\vec{B}$, then the Lorentz force

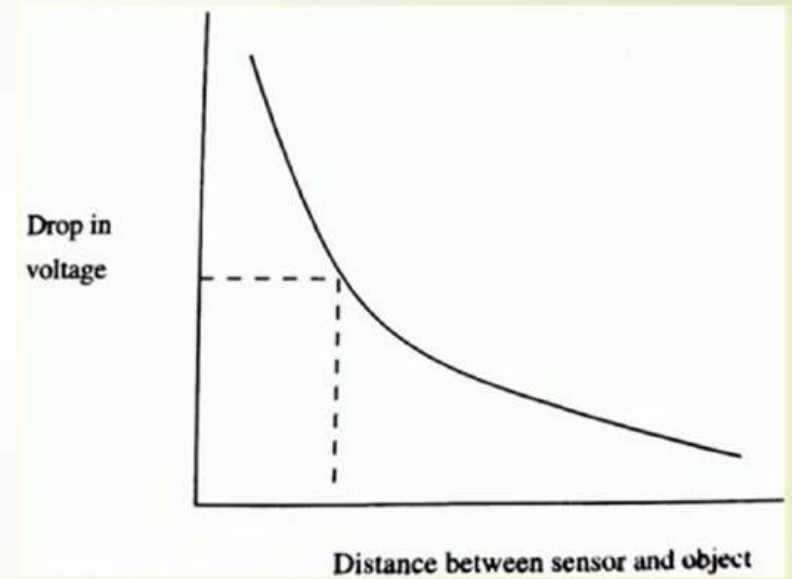
$$\vec{F} = q (\vec{V} \times \vec{B})$$



Hall- effect Sensors (contd.)



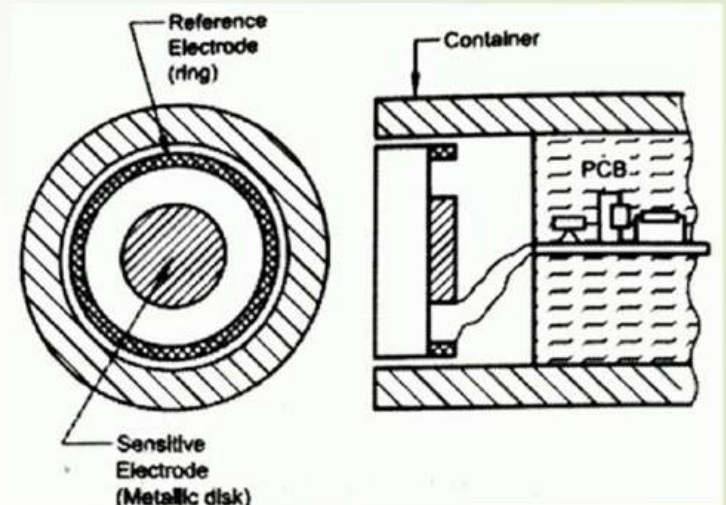
- ❖ Voltage across the semiconductor will be reduced



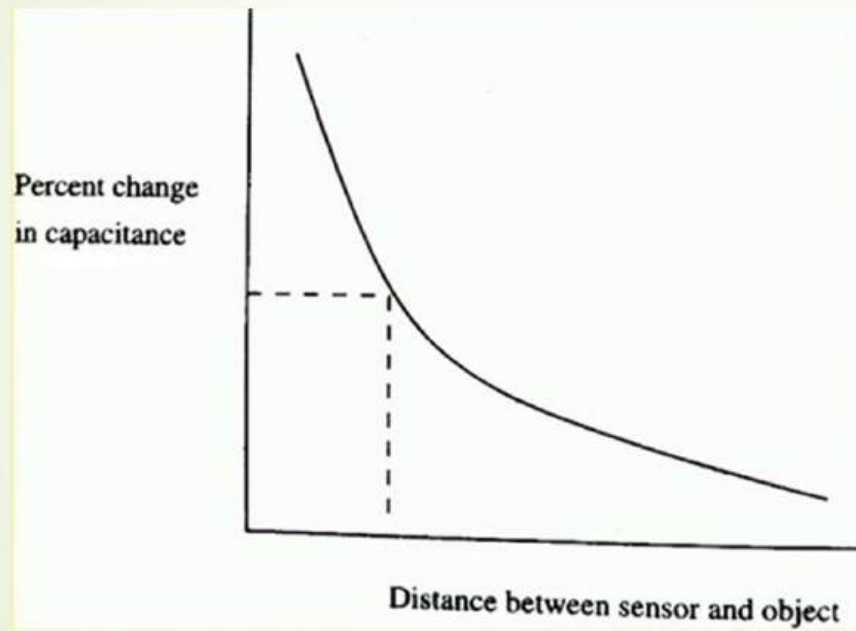
Calibration curve for Hall- effect sensor

Capacitive Sensor (suitable for any material)

- ❖ When an object is brought near to the sensitive electrode, there will be accumulation of charge and consequently, its capacitance changes.
- ❖ When the capacitance of the sensor exceeds a predefined threshold value, oscillation starts.
- ❖ Oscillations are converted into output voltage through PCB



Capacitive sensor (contd.)



Calibration curve

ROBOTICS

Topic 7: Robot Vision



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Robot Vision

- ❖ Robot vision is a technology that enables robots to see and interpret their environment by capturing and processing images.
- ❖ It involves extracting, characterizing, and interpreting objects present in an image captured using a camera.

Purpose

- ❖ Object Identification: Recognizing objects within an image.
- ❖ Navigation: Allowing robots to move through an environment by detecting obstacles.
- ❖ Inspection & Quality Control: Used in manufacturing to identify defects.
- ❖ Surveillance & Security: Detecting intruders or monitoring areas.

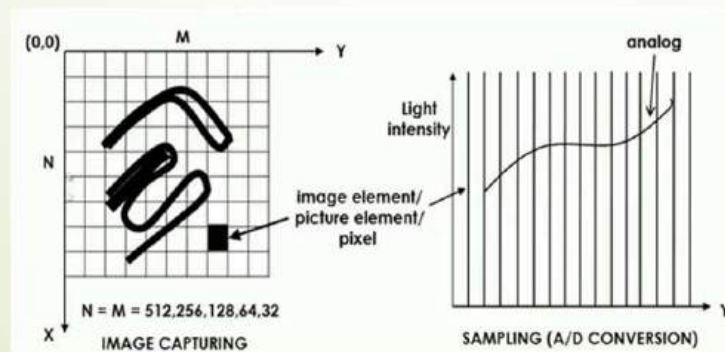
Steps to be Followed

Step 1: Capturing the image

- ❖ The first step in robot vision involves capturing an image of the environment.
- ❖ A Charge-Coupled Device (CCD) camera is used.
- ❖ The camera captures an image, which consists of light intensity variations.

Step 2: Measuring Light Intensity

- ❖ The image consists of variations in brightness.
- ❖ Electron Beam Scanner measures light intensity along a particular direction (e.g., Y-axis).
- ❖ The accumulated charge in photo-sites of the camera sensor is proportional to light intensity.
- ❖ This information is converted from an analog format to digital (A/D conversion).
- ❖ Digitizing the image means converting the brightness levels into numerical values.



Steps to be Followed

Step 3: Storing the Image (Frame Grabbing)

- ❖ The digitized image is stored as an array of pixels.
- ❖ Each pixel has a corresponding light intensity value.
- ❖ This process is known as frame grabbing.
- ❖ Mathematical Representation:

If $F(x,y)$ represents the light intensity at a pixel located at coordinates (x,y) then:

$$F(x,y) = \text{Light intensity at the point } (x, y)$$

Step 4: Preprocessing

- ❖ To enhance the image by removing noise and restoring lost information.
- ❖ Techniques Used in Preprocessing:

1. Masking

A mathematical operation where each pixel's intensity is modified using a mask.

A 3×3 mask is applied, which considers the surrounding pixels to modify the central pixel.

Example:
$$P(x, y) = W_1 f(x-1, y-1) + W_2 f(x-1, y) + \dots + W_9 f(x+1, y+1)$$

where: $W_1, W_2, \dots, W_9$ are weight coefficients and $f(x,y)$ is the intensity of the pixel at (x,y)

Steps to be Followed

2. Neighborhood Averaging

Instead of applying weights, the pixel intensity is averaged with its neighboring pixels.

Formula:

$$P(x, y) = \frac{1}{R} \sum_{(i,j) \in S} f(i, j)$$

where: S is the neighborhood set and R is the total number of pixels in the neighborhood.

3. Median Filtering

A non-linear filtering technique used to reduce noise.

Steps:

1. Collect intensity values of the neighboring pixels (including the current pixel).
2. Sort them in ascending order.
3. Choose the median value to replace the current pixel's intensity.

Steps to be Followed

Step 5: Thresholding

Used to separate objects from the background.

A threshold intensity (T) is chosen.

If the image consists of a black background and a white object:

Pixel = 1 (if intensity $\geq T$) $\rightarrow$ Represents object.

Pixel = 0 (if intensity $< T$) $\rightarrow$ Represents background.

Step 6: Edge Detection

Edge detection helps in finding object boundaries.

Techniques Used for Edge Detection:

1. Gradient Operator

Computes the gradient (rate of change) in intensity.

Uses two masks:

G_x detects edges in the horizontal direction.

G_y detects edges in the vertical direction.

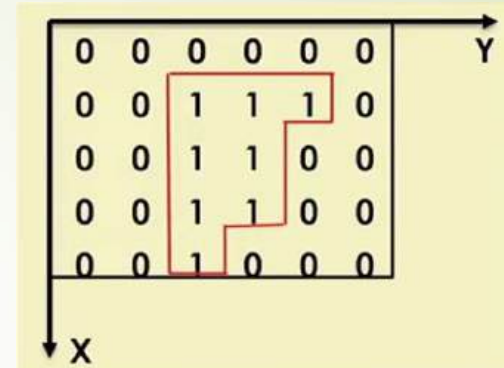
2. Laplace Operator

Measures second-order derivatives to detect edges.

Highlights regions where intensity changes rapidly.

Example:

For a light object on a dark background, edges appear as regions of sharp intensity change.



Steps to be Followed

Step 5: Thresholding Boundary Descriptors

After detecting edges, the next step is to describe object boundaries.

Techniques Used for Boundary Descriptions:

Chain Codes

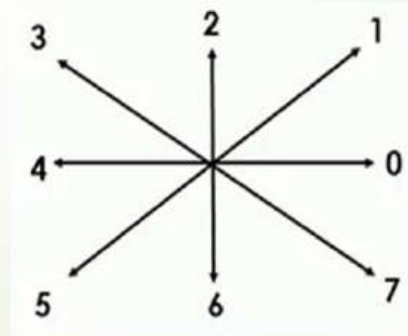
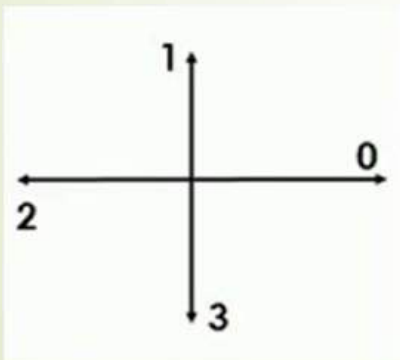
Represents an object's boundary using a sequence of straight-line segments.

Uses a specified length and direction.

Types of Chain Codes:

4-Directional Chain Code: Uses 4 directions (up, down, left, right).

8-Directional Chain Code: Uses 8 directions (includes diagonal movements).



Steps to be Followed

2. Signatures

A 1D representation of the object boundary.

Extracts important features of the shape.

3. Compactness Measurement

Used to distinguish multiple objects in an image.

Formula:

$$Compactness = Perimeter^2 / Area$$

Lower compactness indicates a more circular shape.

Higher compactness indicates a more elongated or irregular shape.

ROBOTICS

Topic 8: Robot Motion Planning



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Dept. of Computer Science and Engineering, UMaT



Definition

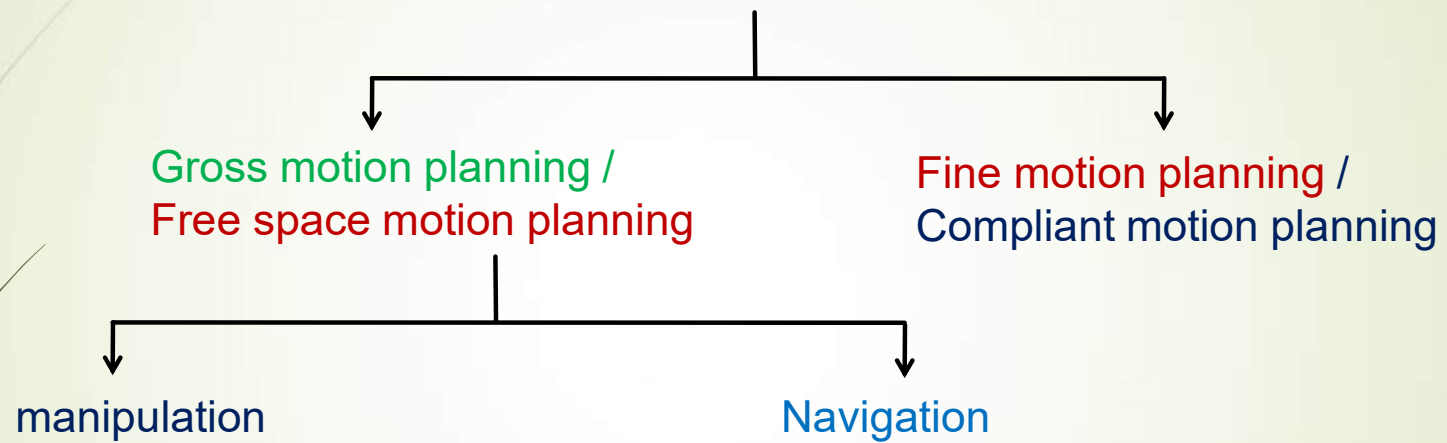
Robot motion planning is the process of determining a collision-free path for a robot to move from an initial position to a target position while navigating through obstacles in an environment.

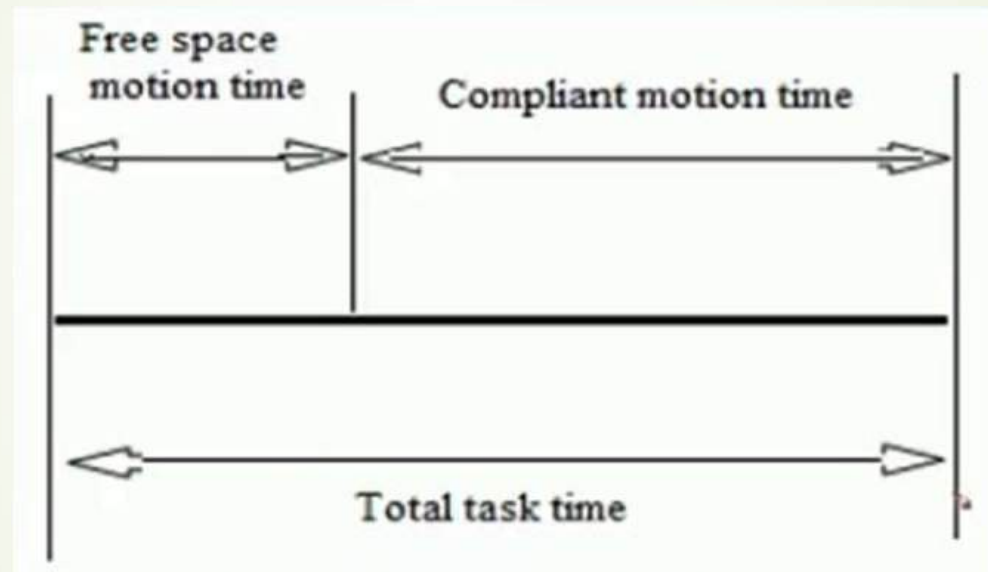
It helps robots to operate autonomously in structured and unstructured environments.

Aim

To determine the course of action / path while moving from an initial position to a final position

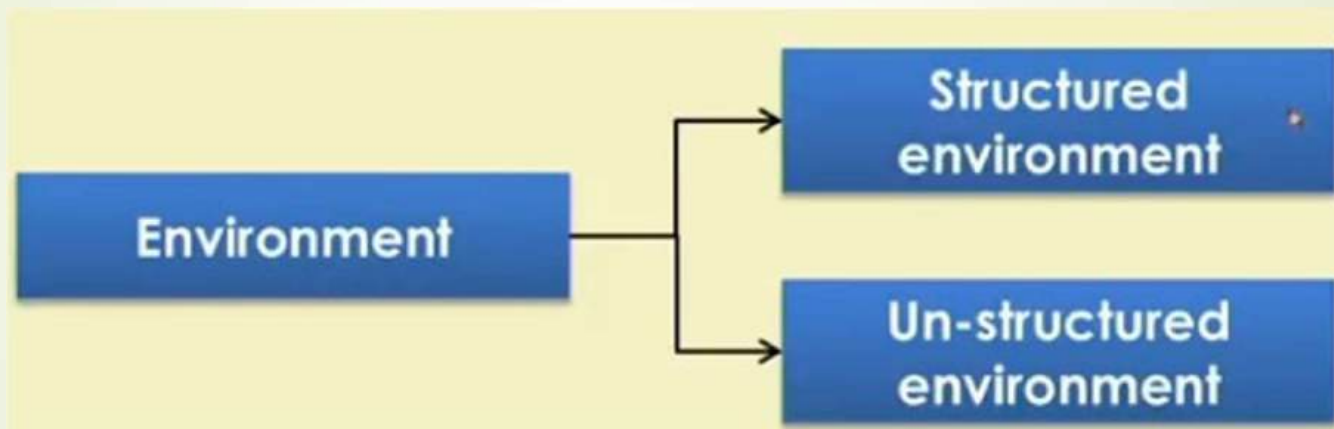
Robot Motion Planning



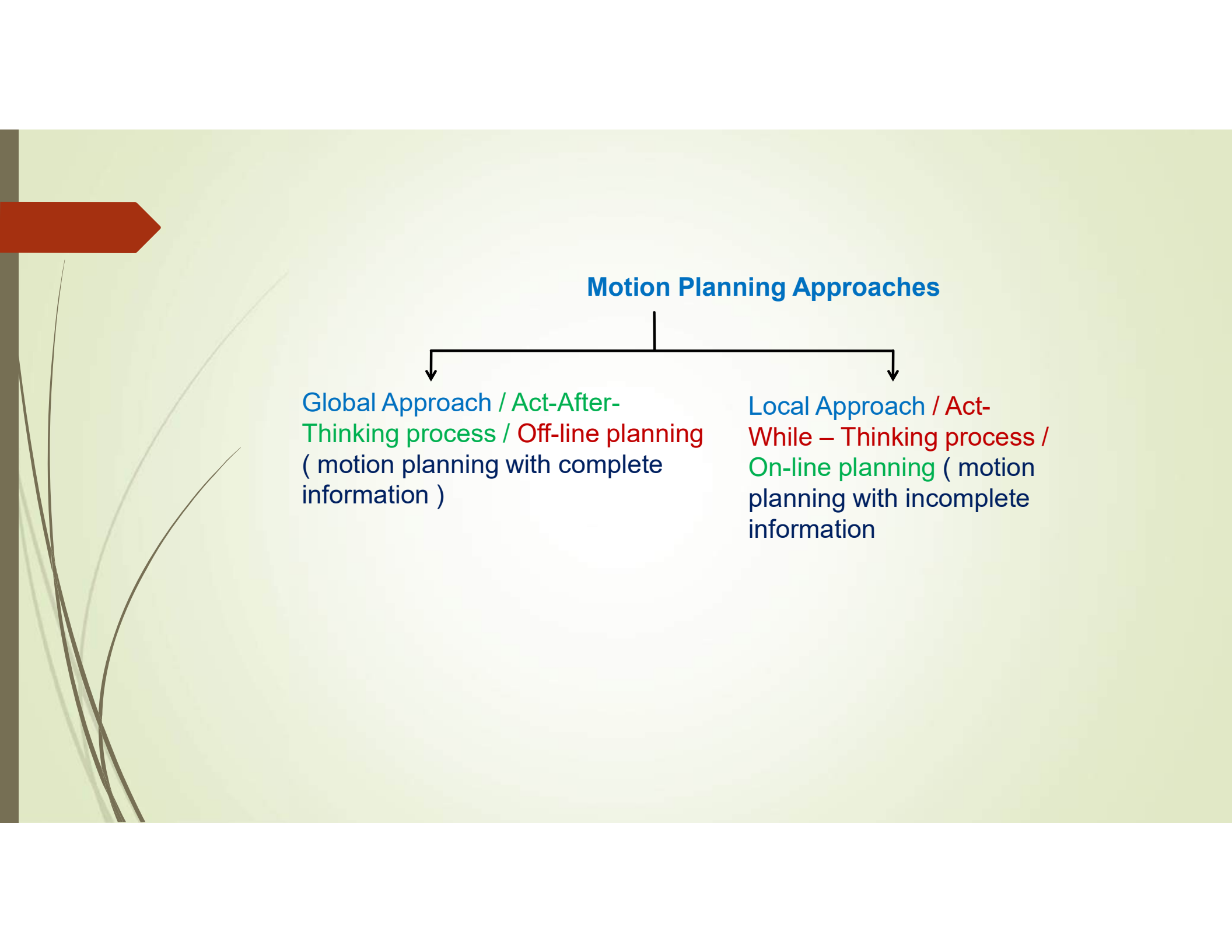


Sequence of Robotic Action





Motion Planning Approaches



```
graph TD; A[Motion Planning Approaches] --> B[Global Approach / Act-After-Thinking process / Off-line planning  
( motion planning with complete information )]; A --> C[Local Approach / Act-While – Thinking process / On-line planning  
( motion planning with incomplete information )];
```

Global Approach / Act-After-Thinking process / Off-line planning
(motion planning with complete information)

Local Approach / Act-While – Thinking process / On-line planning (motion planning with incomplete information)

Motion Planning Schemes

Traditional Schemes / Algorithmic Approaches

Graph-based methods

- (i) Visibility graph
- (ii) Voronoi diagram
- (iii) Cell decomposition
- (iv) Tangent graph
- (v) Accessibility graph

Analytical Approaches

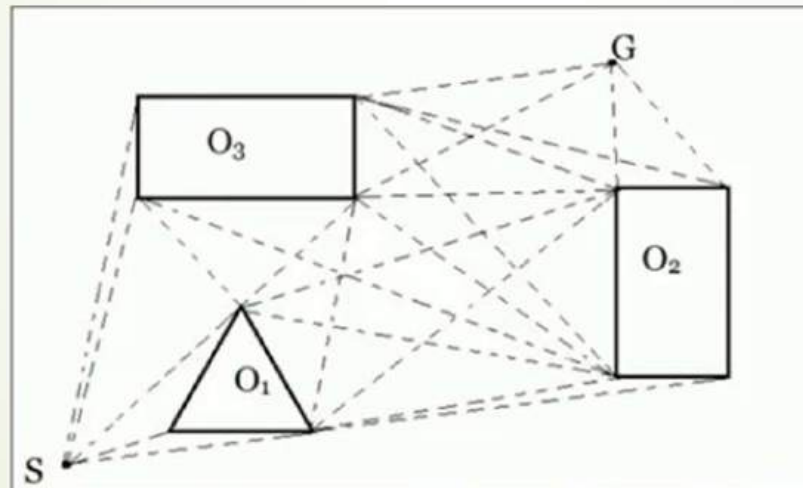
- (i) Potential Field Approach
- (ii) Path velocity decomposition
- (iii) incremental planning
- (iv) Probabilistic Approach
- (v) Relative velocity Approach
- (vi) Reactive Control Strategies (Behavior-Based Robotics)

Non-Traditional schemes (using soft computing)

- (i) Fuzzy logic – based
- (ii) Neural networks –based approaches

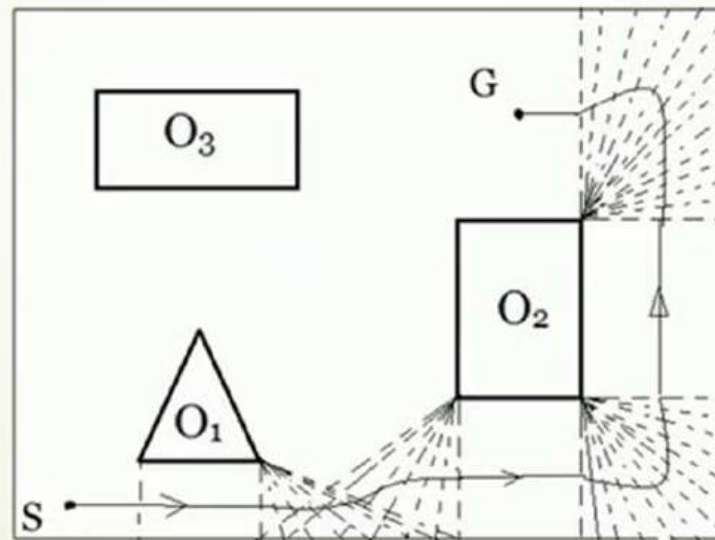
Visibility Graph (Nilsson 1969)

- ❑ It connects those vertices of obstacles, which are visible from one another



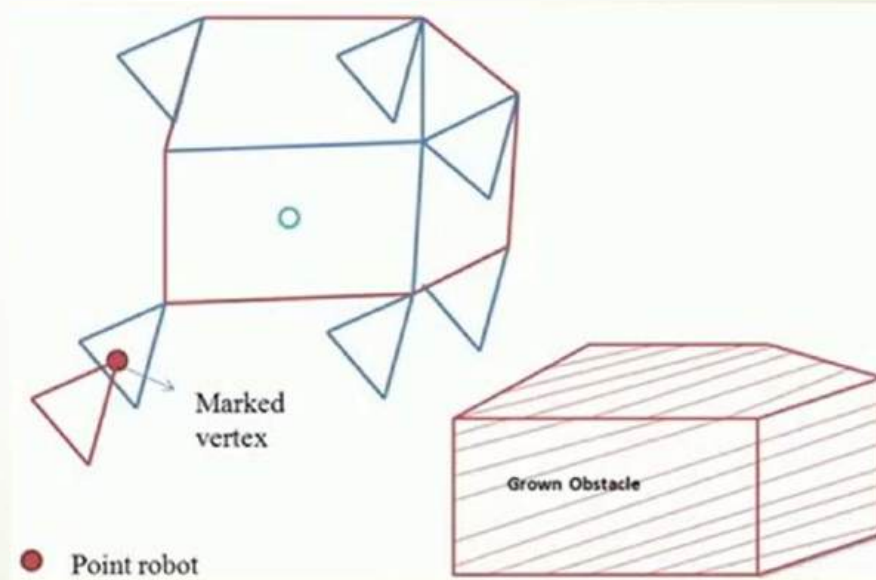
Voronoi Diagram (Dunlaing et al., 1986)

- Represents the locus of points that are equidistant from at least two of the boundaries (obstacle)



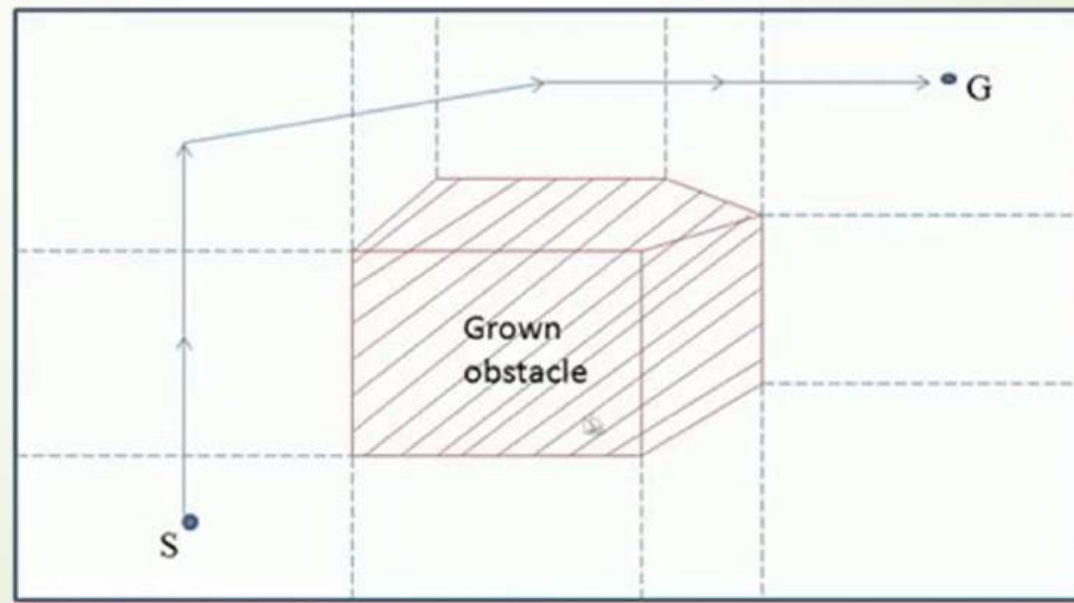
Cell Decomposition (Lozano Perez, 1983)

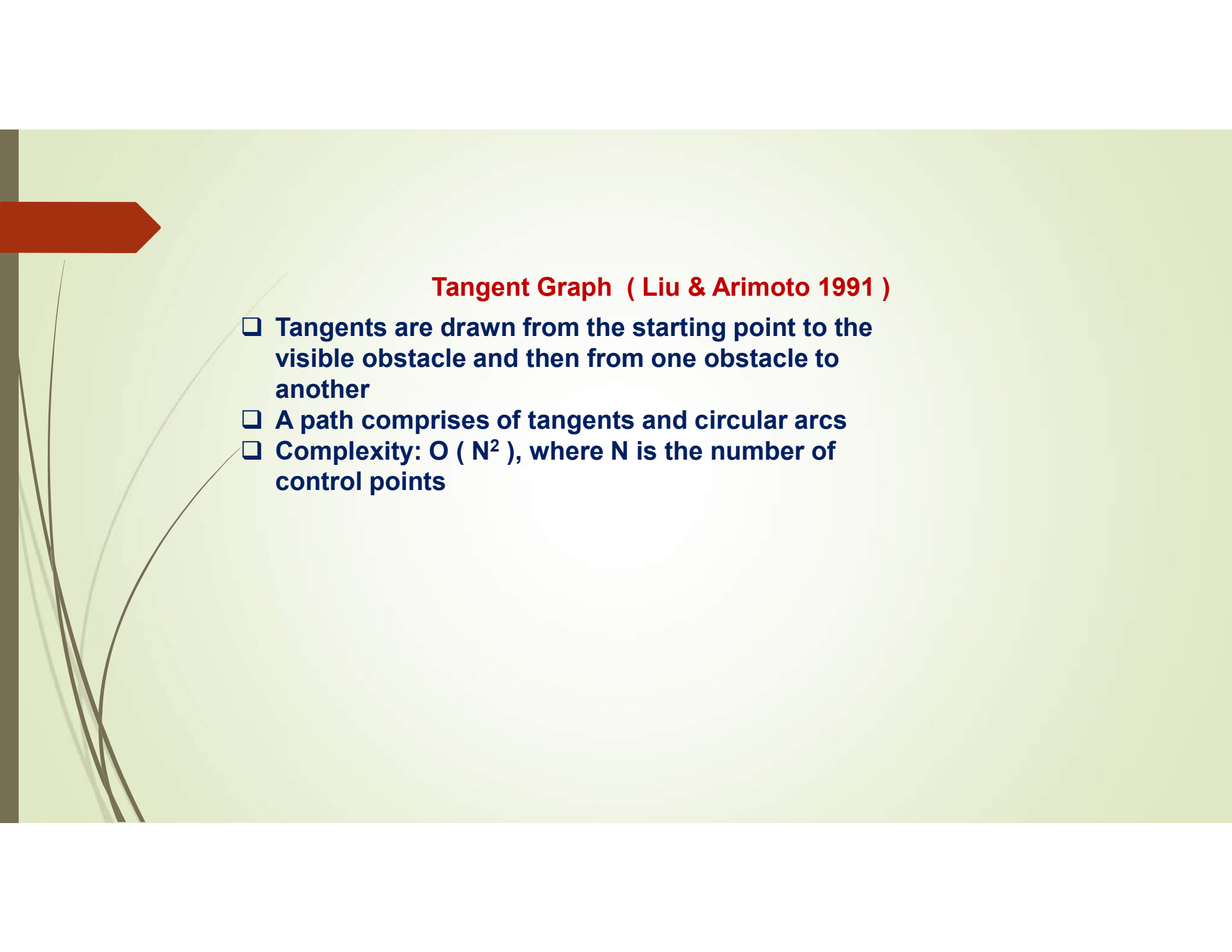
Configuration space (C- space)



Cell Decomposition (contd.)

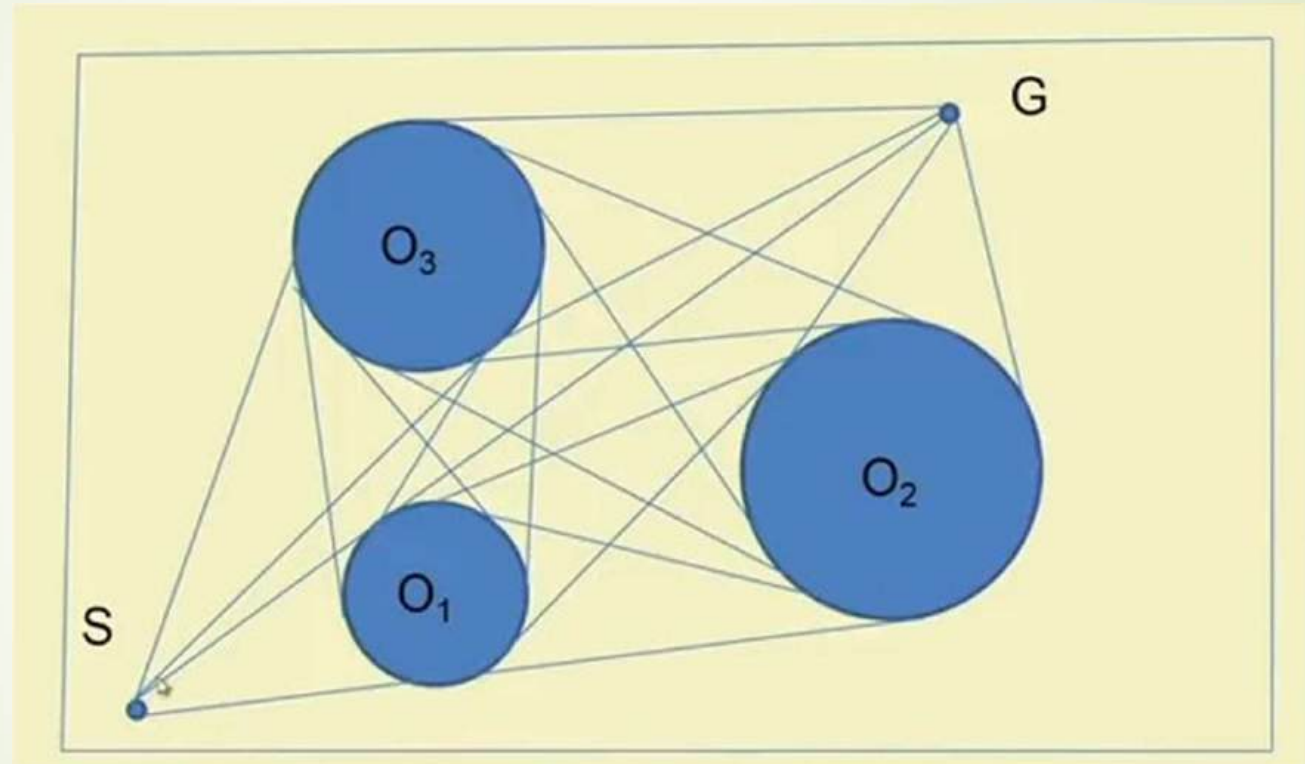
- ❑ Robot's free area is divided into a number of small regions called cells. A connectivity graph is then constructed and searched.





Tangent Graph (Liu & Arimoto 1991)

- ❑ Tangents are drawn from the starting point to the visible obstacle and then from one obstacle to another
- ❑ A path comprises of tangents and circular arcs
- ❑ Complexity: $O(N^2)$, where N is the number of control points



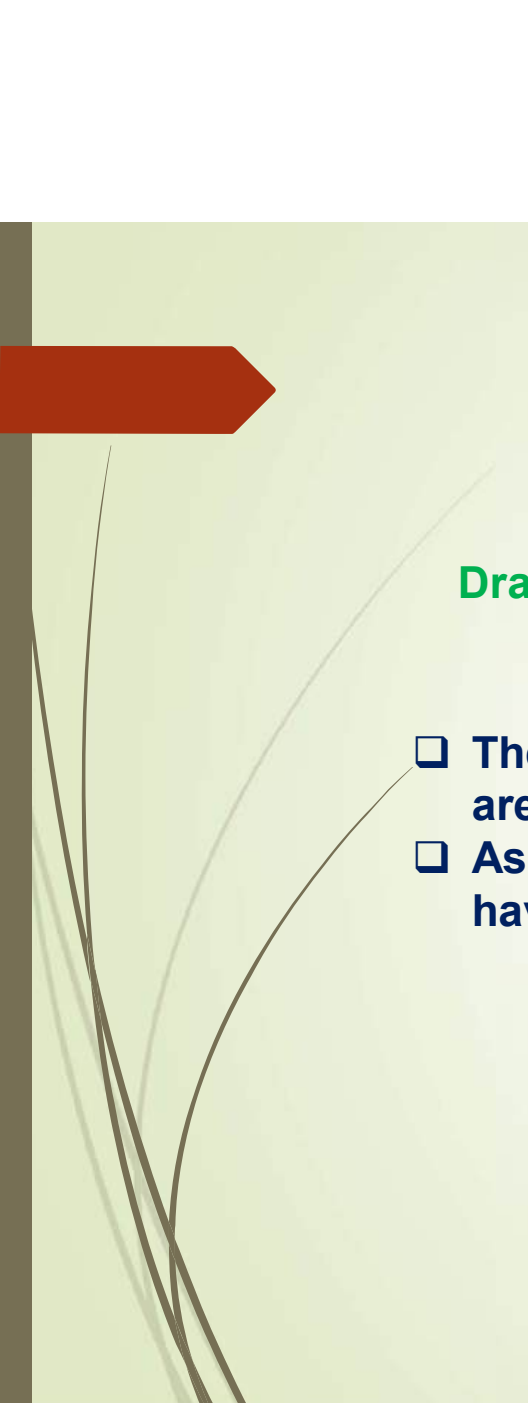


Approaches to solve Moving Obstacle Problems

Path Velocity Decomposition (Kant and Zucker, 1984)

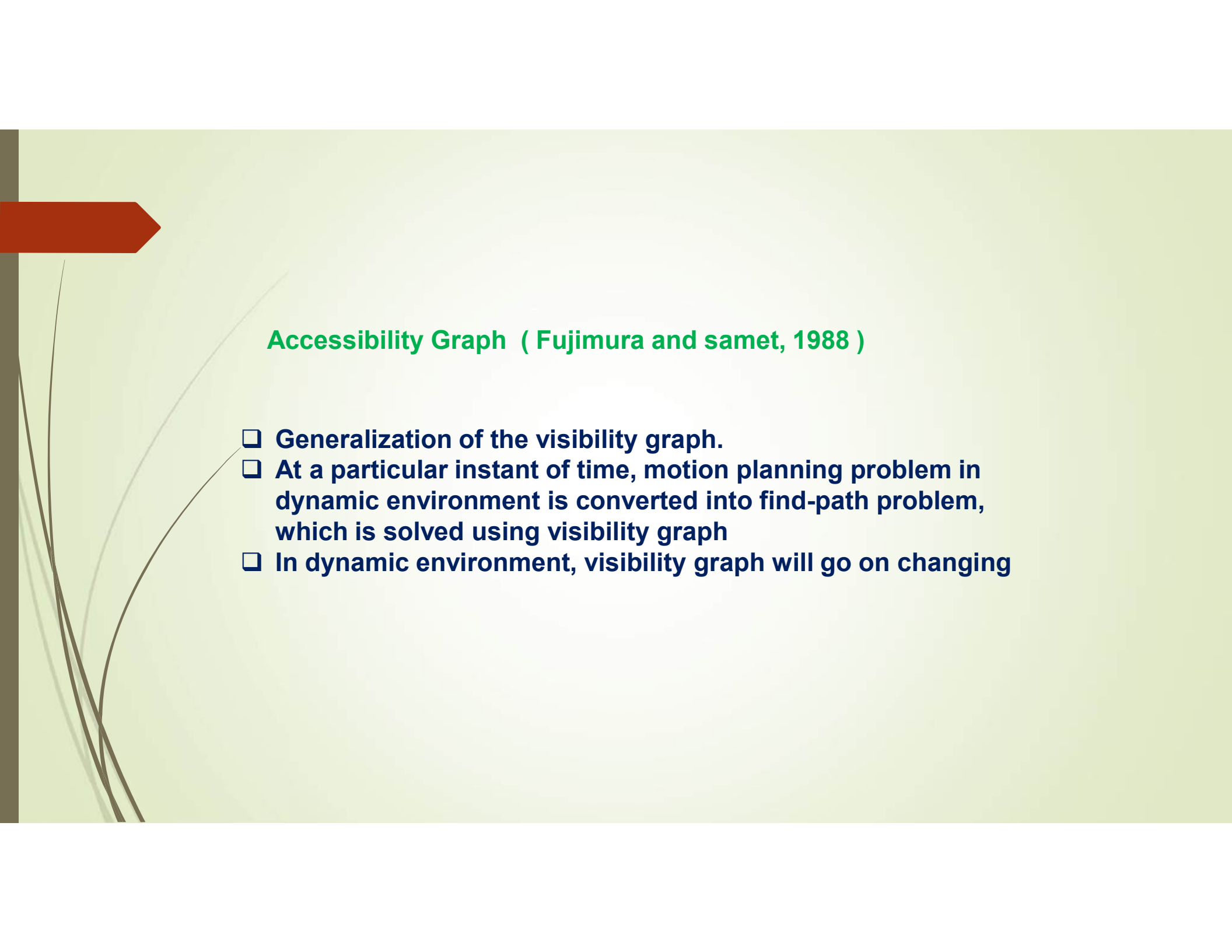
This problem is decomposed into two sub-problems as follows:

- ❑ Path planning problem (PPP) – to plan a path to avoid collision with static obstacles
- ❑ Velocity planning problem (VPP) – to plan the velocity of the robot along the above planned path to avoid collision with moving obstacles.



Drawbacks

- ❑ The path may not be always good, particularly when there are many obstacles
- ❑ As there is a sudden change of velocity of the robot, it will have jerky motion, which is not desirable.



Accessibility Graph (Fujimura and samet, 1988)

- ❑ Generalization of the visibility graph.
- ❑ At a particular instant of time, motion planning problem in dynamic environment is converted into find-path problem, which is solved using visibility graph
- ❑ In dynamic environment, visibility graph will go on changing

Incremental Planning Scheme (Slack & Miller, 1987)

Plan a collision – free path based on predicted position of obstacles



Follow the path until it becomes invalid (validity is checked based on the data collected using sensor (s) / camera(s))



Adjust the predictions & plan a new collision – free path

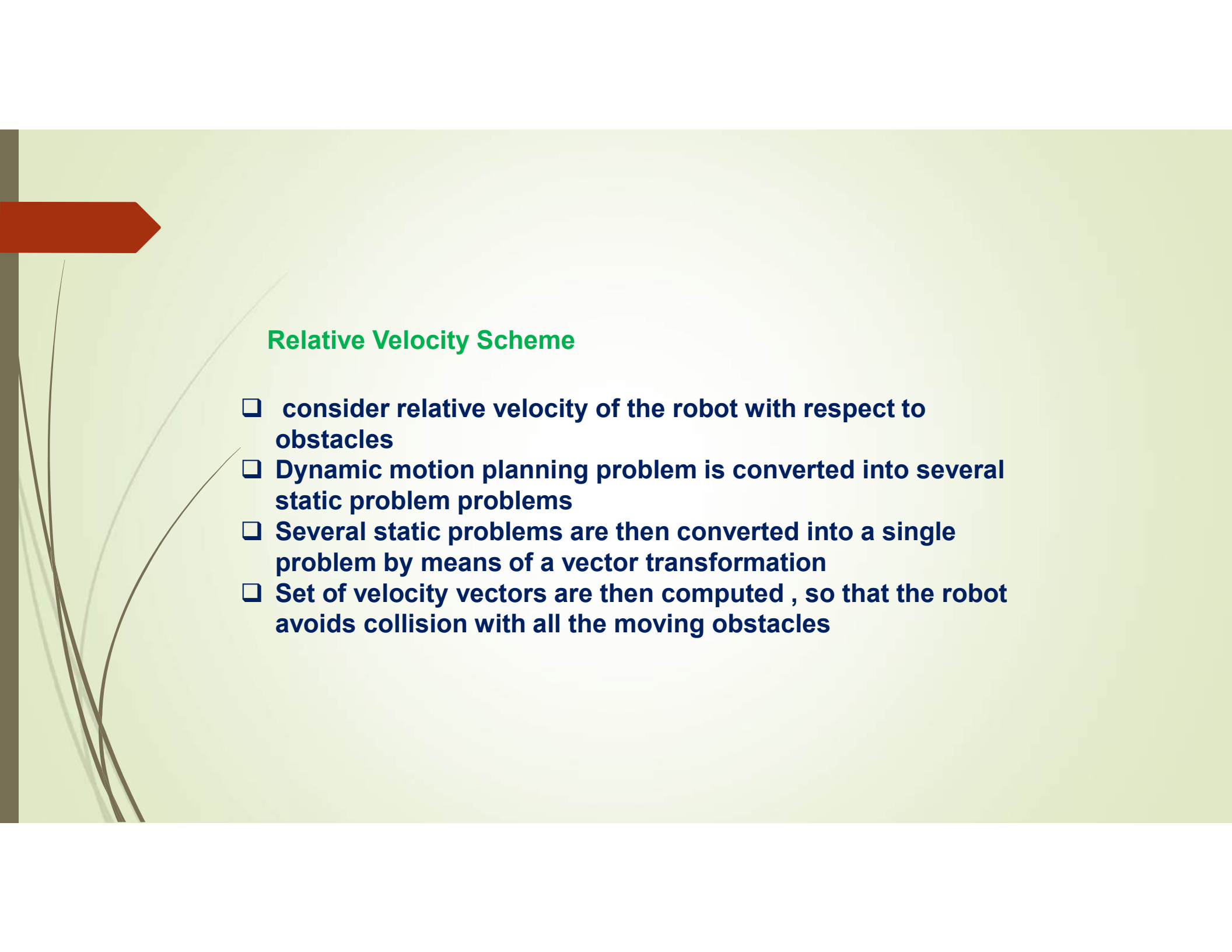




```
graph TD; A[Follow the path until it becomes invalid] --> B[Continue until the robot reaches its destination]; B --> C[ ];
```

**Follow the path until it becomes
invalid**

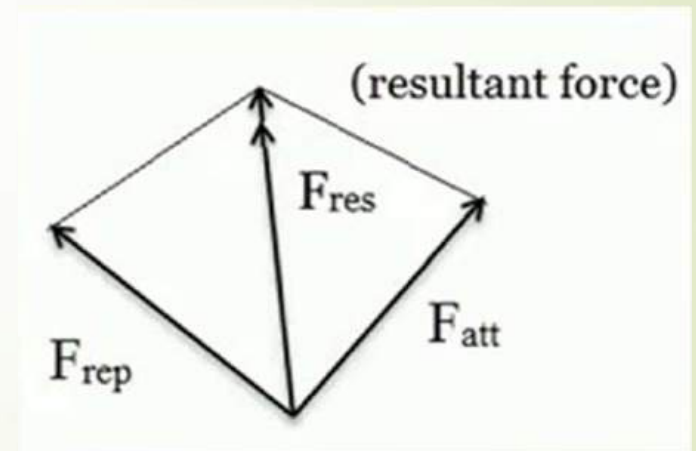
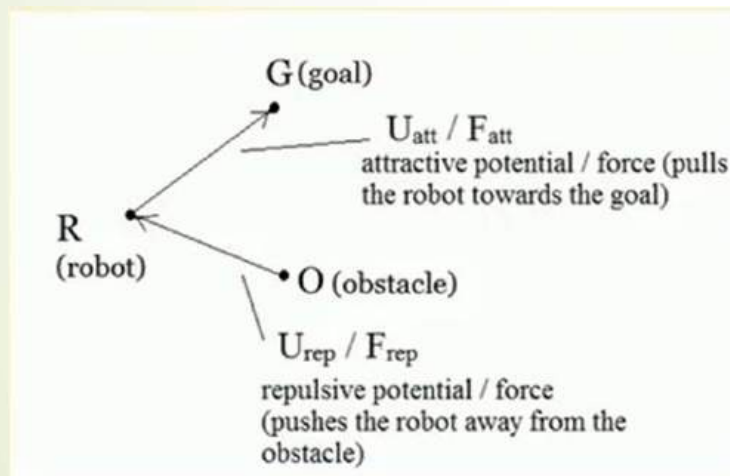
**Continue until the robot reaches its
destination**



Relative Velocity Scheme

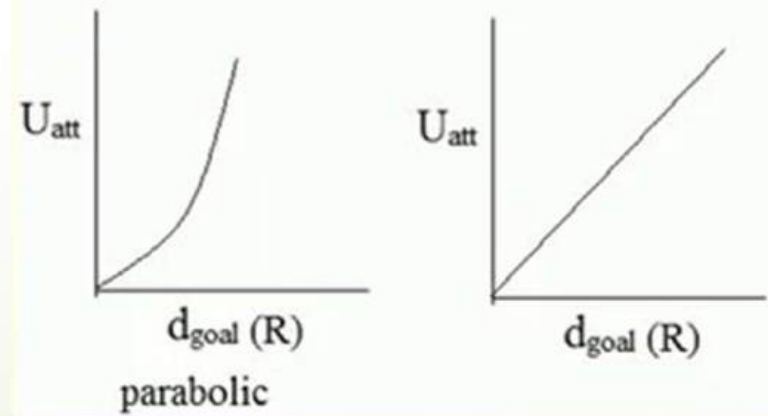
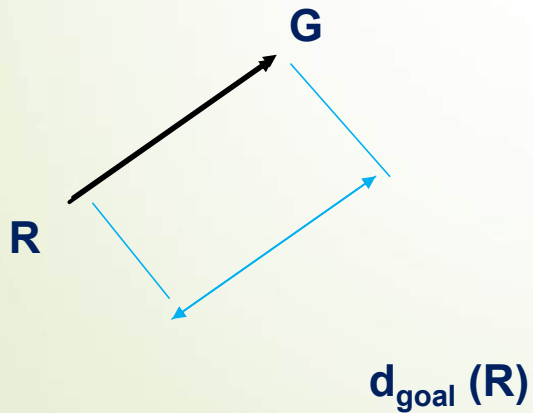
- ❑ consider relative velocity of the robot with respect to obstacles
- ❑ Dynamic motion planning problem is converted into several static problem problems
- ❑ Several static problems are then converted into a single problem by means of a vector transformation
- ❑ Set of velocity vectors are then computed , so that the robot avoids collision with all the moving obstacles

Potential Field Approach (Khatib, 1986)

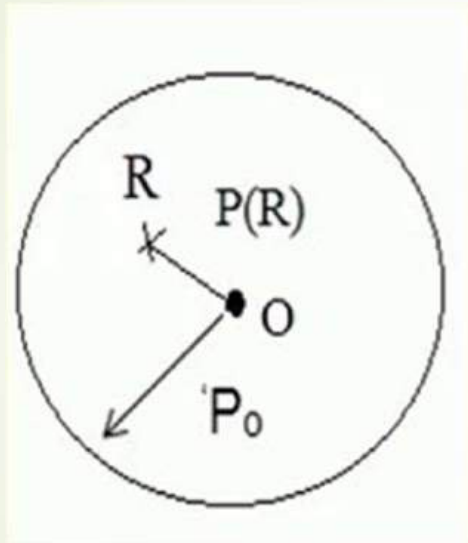


Attractive Potential

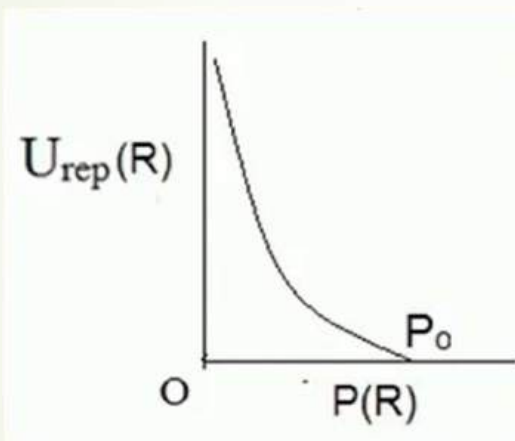
$$U_{att}(R) = \frac{1}{2} \xi d_{goal}^2(R), \text{ if parabolic} \\ = \xi d_{goal}(R), \text{ if conic-well}$$



Repulsive Potential



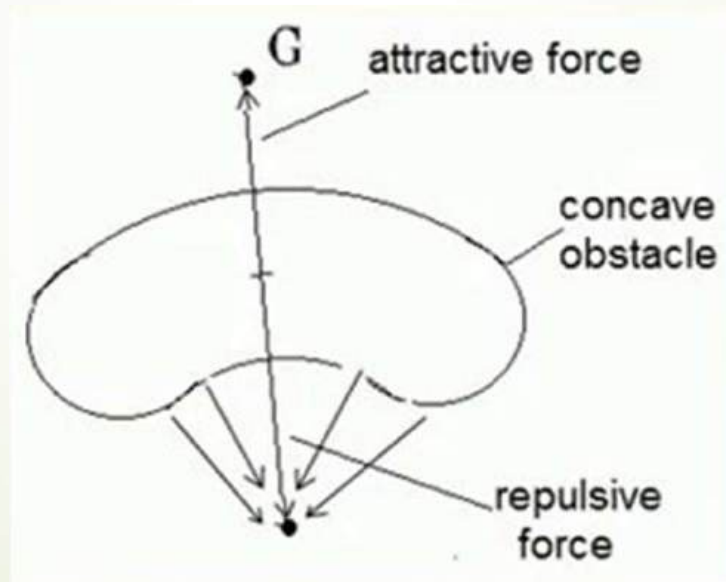
$$U_{\text{rep}}(R) = 0$$

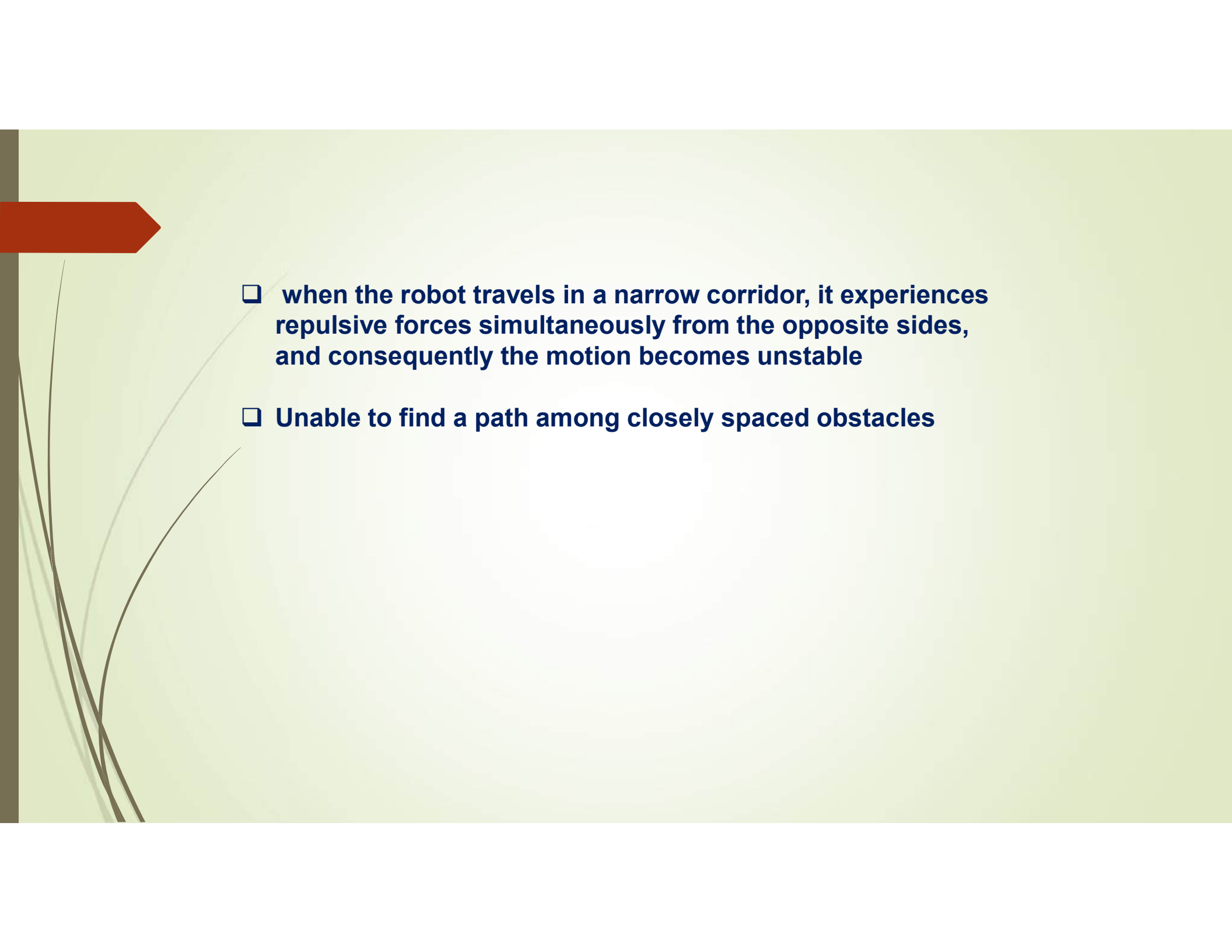


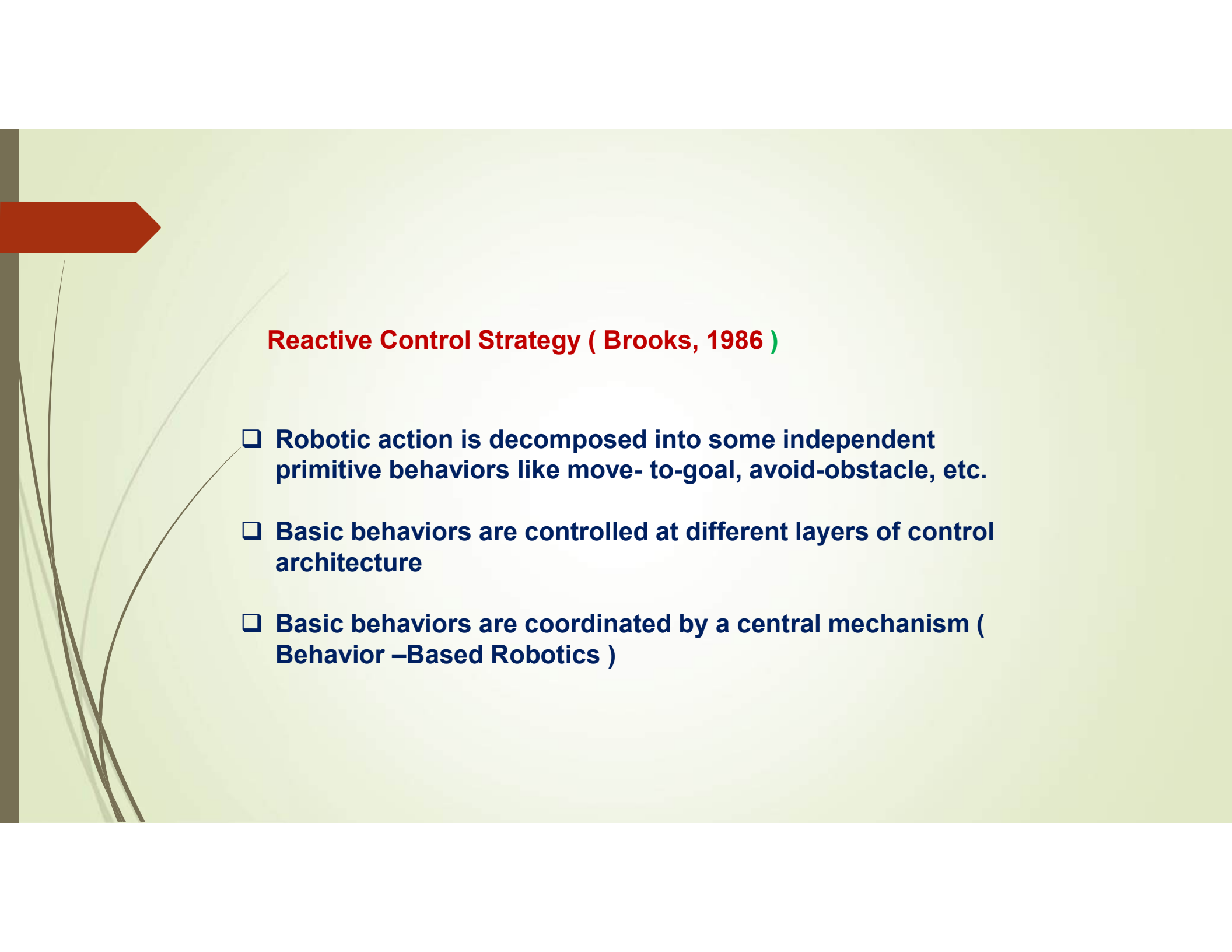
$$U_{rep}(R) = \begin{cases} \frac{1}{2} \eta \left(\frac{1}{P(R)} - \frac{1}{P_o} \right)^2, & \text{if } P(R) < P_o \\ 0, & \text{if } P(R) \geq P_o \end{cases}$$

Drawbacks

- ❑ solution depends on the chosen potential function
- ❑ Chance of local minima problem – when the attractive force is balanced by the repulsive force



- 
- ❑ when the robot travels in a narrow corridor, it experiences repulsive forces simultaneously from the opposite sides, and consequently the motion becomes unstable
 - ❑ Unable to find a path among closely spaced obstacles



Reactive Control Strategy (Brooks, 1986)

- ☐ **Robotic action is decomposed into some independent primitive behaviors like move- to-goal, avoid-obstacle, etc.**
- ☐ **Basic behaviors are controlled at different layers of control architecture**
- ☐ **Basic behaviors are coordinated by a central mechanism (Behavior –Based Robotics)**



Drawbacks

- ❑ The behaviors are hard – wired, thus it is unable to handle behaviors which the programmer did not foresee beforehand
 - ❑ The number of layers increases with the complexity of the problem. It requires a large amount of computer memory.
- 



Computational Complexity

- ❑ **Canny and Reif (1987) -**
Motion planning for a point robot among moving obstacles in 2D plane with bounded velocity is NP – hard.
- 

□ Reif and Sharir (1985) –

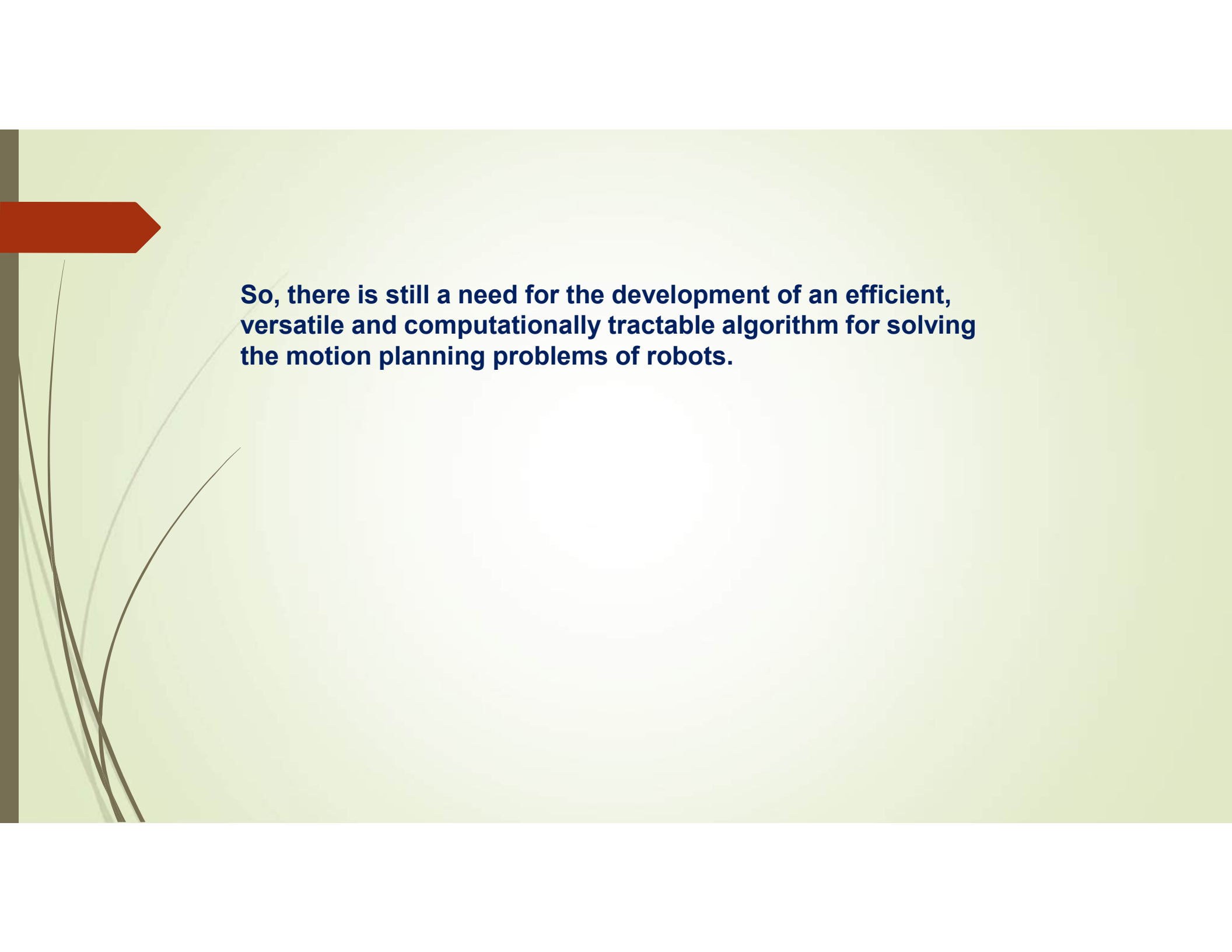
a) Motion planning among moving obstacles in 3-D space without velocity bound is NP –hard

b) Motion planning among moving obstacles in 3-D space with velocity bound is PSPACE -hard



Drawbacks of the Traditional Methods of Motion Planning

- ☐ Traditional methods are computationally expensive even for a simple problem
 - ☐ No versatile algorithm, which is applicable to all the problems
 - ☐ As most of the algorithms do not have an optimization module, the generated path may not be optimal in any sense.
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So, there is still a need for the development of an efficient, versatile and computationally tractable algorithm for solving the motion planning problems of robots.

ROBOTICS

Topic 10: Biped Walking



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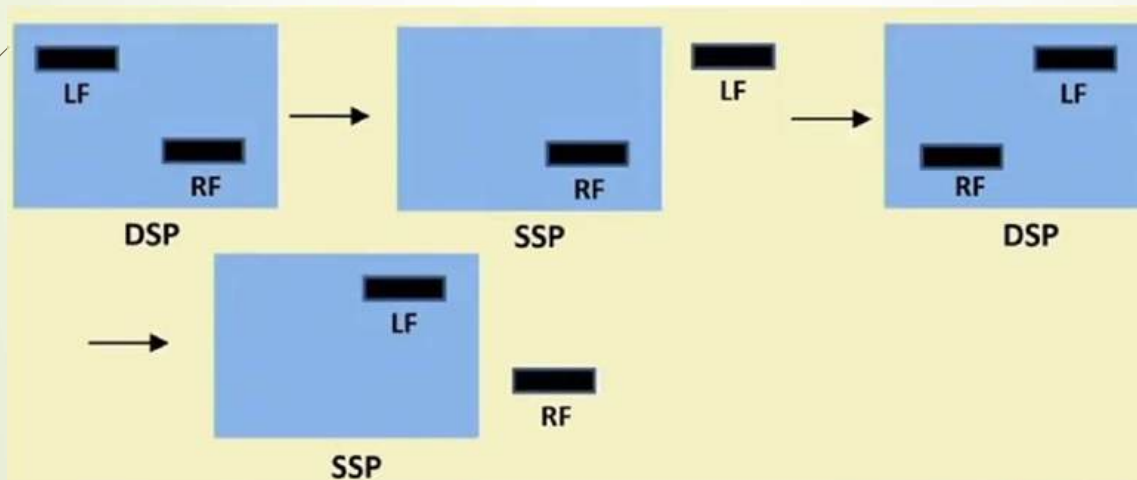


Introduction

- ❑ Biped walking refers to the locomotion of robots that mimic human walking patterns using two legs.
- ❑ It is a complex task due to stability challenges, dynamic balancing, and energy efficiency considerations.
- ❑ A biped robot should be able to walk on plane surface, negotiate staircases, cross ditches, take turn, as the situation demands
- ❑ Challenges in Biped Walking
 - ❑ Stability: Maintaining balance while walking, especially on uneven terrain.
 - ❑ Dynamics: Managing forces and torques during motion.
 - ❑ Energy Efficiency: Reducing power consumption during locomotion.
 - ❑ Control Strategies: Implementing control systems for adaptive movement.

A walking cycle consists of:

- ❑ **Two Single-Support Phases (SSPs) – When only one foot is on the ground.**
- ❑ **Two Double-Support Phases (DSPs) – When both feet touch the ground.**



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- ❑ **Gait** : sequence of leg's movement synchronized with its body movement during walking
 - ❑ **Power consumption** during walking is to be optimized
 - ❑ **Balance** is to be maintained during walking

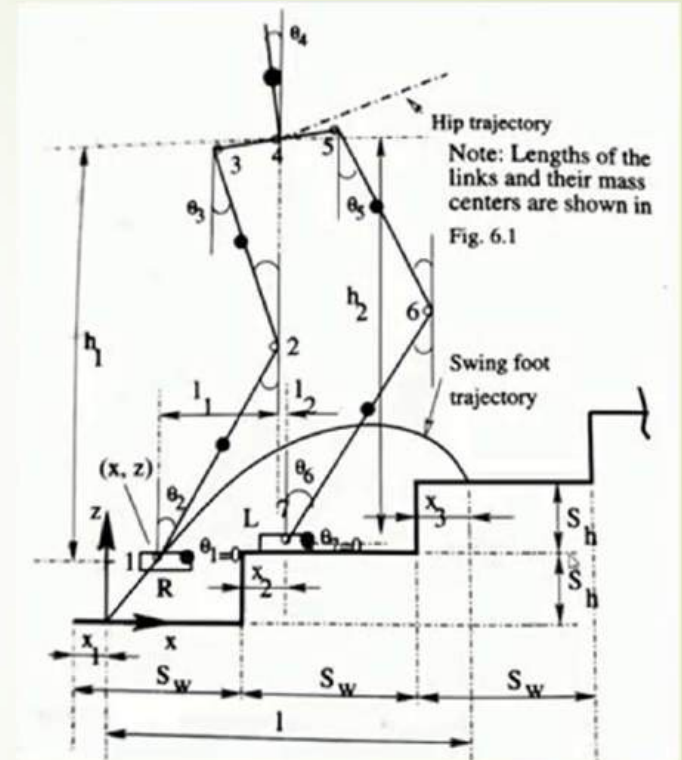
Staircase Ascending(SSP)

- In the present study, movement has been considered in the sagittal plane only.

Key parameters:

$$\text{Step length } l = (2S_w + X_3) - X_1$$

$$\text{Hip height } h = L_2 \cos \theta_2 + L_3 \cos \theta_3$$



Swing foot trajectory generation

The trajectory of the swing foot follows a cubic polynomial equation:

$$Z = C_0 + C_1X + C_2X^2 + C_3X^3$$

Subject to the conditions

$$\begin{aligned} & \text{at } x = 0, z = 0, \\ & \text{at } x = s_w - x_1 - \frac{f_s}{2}, z = s_h + \frac{f_s}{2}, \\ & \text{at } x = 2s_w - x_1 - \frac{f_s}{2}, z = 2s_h + \frac{f_s}{2}, \\ & \text{at } x = 2s_w - x_1 + x_3, z = 2s_h. \end{aligned}$$

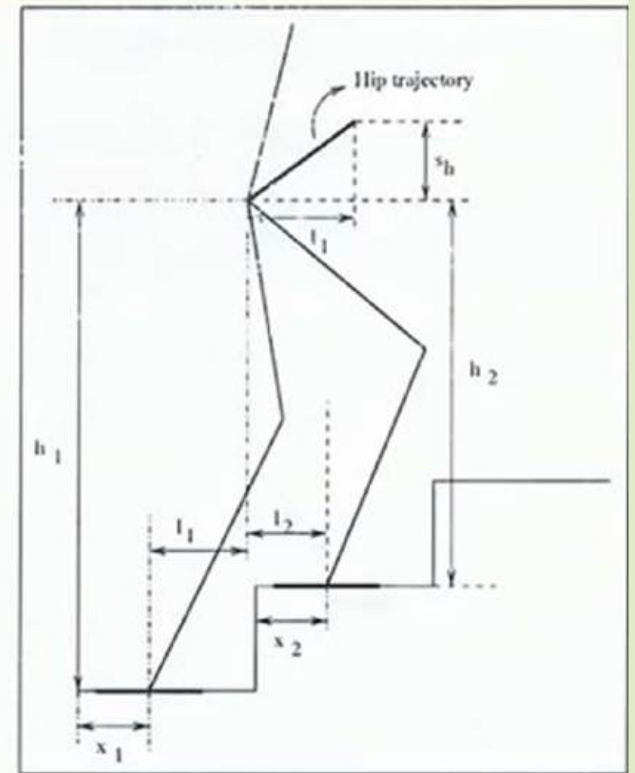
where coefficients are subject to boundary conditions ensuring smooth motion.

Hip joint trajectory generation

Hip trajectory is generated to maintain stability during movement.

h_1 : hip height

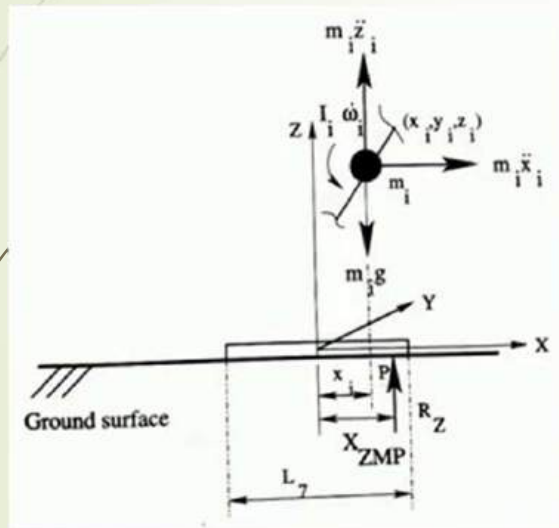
l_1 : distance of the ankle of swing foot from the projection of hip



Dynamic balance analysis

Dynamic balance is analyzed by:

- Calculating torques at different joints.
- Using Denavit-Hartenberg (D-H) parameters for kinematic modeling.



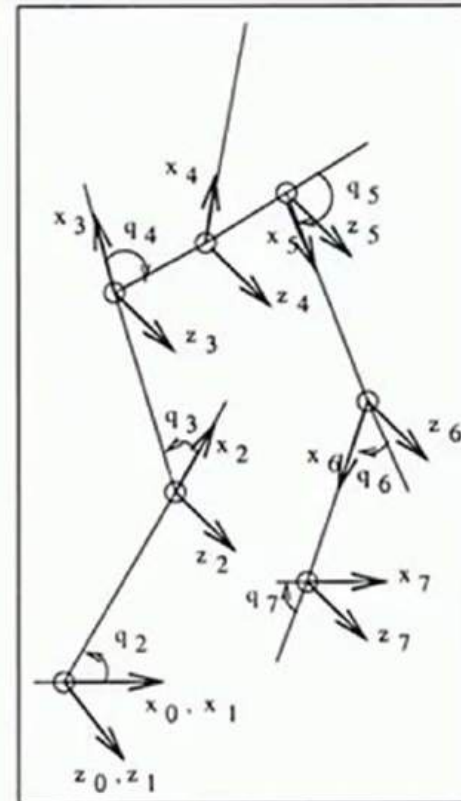
$$\sum_{i=1}^7 m_i (\ddot{z}_i - g) (x_{ZMP} - x_i) + \sum_{i=1}^7 m_i \ddot{x}_i z_i - \sum_{i=1}^7 I_i \dot{\omega}_i = 0$$

$$x_{ZMP} = \frac{\sum_{i=1}^7 (I_i \dot{\omega}_i + m_i x_i (\ddot{z}_i - g) - m_i \ddot{x}_i z_i)}{\sum_{i=1}^7 m_i (\ddot{z}_i - g)}$$

$$x_{DBM} = \left(\frac{L_7}{2} - |x_{ZMP}| \right),$$

Torque calculation

D – H parameter settling



Trajectory for joint angle

$$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5,$$

Joint torques

$$\tau_i = \sum_{k=1}^n D_{ik} \ddot{q}_k + \sum_{k=1}^n \sum_{m=1}^n h_{ikm} \dot{q}_k \dot{q}_m + C_i, \quad i = 1, 2, \dots, n$$

$$\text{where } D_{ik} = \sum_{j=\max(i,k)}^n \text{Tr}(U_{jk} J_j U_{ji}^T) \quad i, k = 1, 2, \dots, n,$$

$$h_{ikm} = \sum_{j=\max(i,k,m)}^n \text{Tr}(U_{jkm} J_j U_{ji}^T) \quad i, k, m = 1, 2, \dots, n,$$

$$C_i = \sum_{j=i}^n (-m_j \bar{g} U_{ji}^j \bar{r}); i = 1, 2, \dots, n$$

Average power consumption

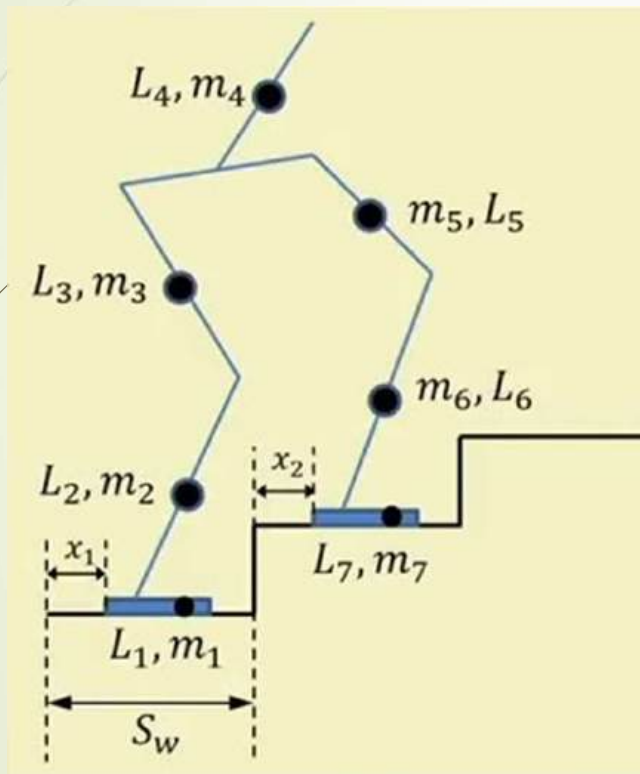
$$P_i = \frac{1}{T} \sum_{i=1}^n \int_0^T (|\tau_i \dot{q}_i| + k\tau_i^2) dt,$$

Mathematical formulation for Double support phase

$$m_{41} = \frac{m_4 X_2}{X_1 + X_2}$$

$$m_{42} = \frac{m_4 X_1}{X_1 + X_2}$$

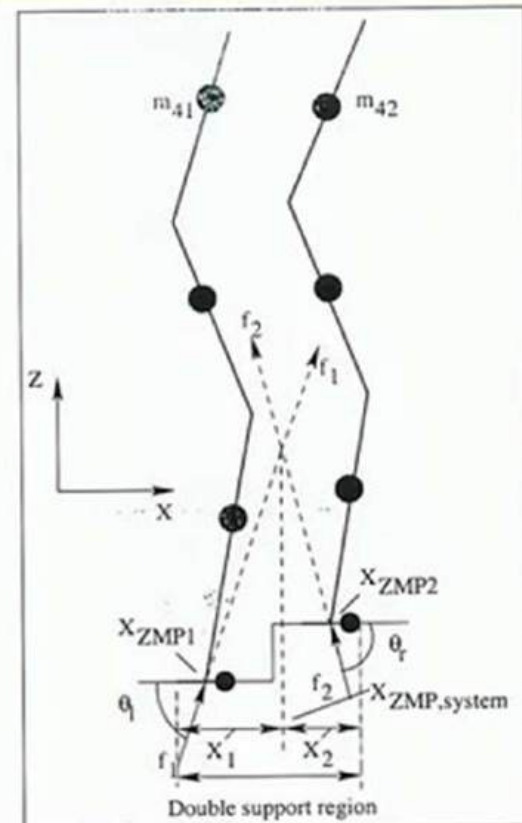
Double support phase



ZMP in double support phase

The ZMP is a key factor in ensuring stability during DSP.

The ZMP is defined as the point where the sum of the moments of forces acting on the robot is zero.

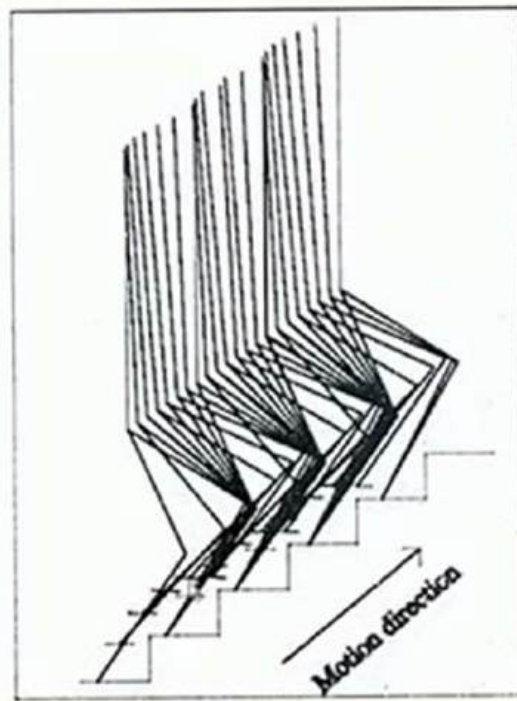




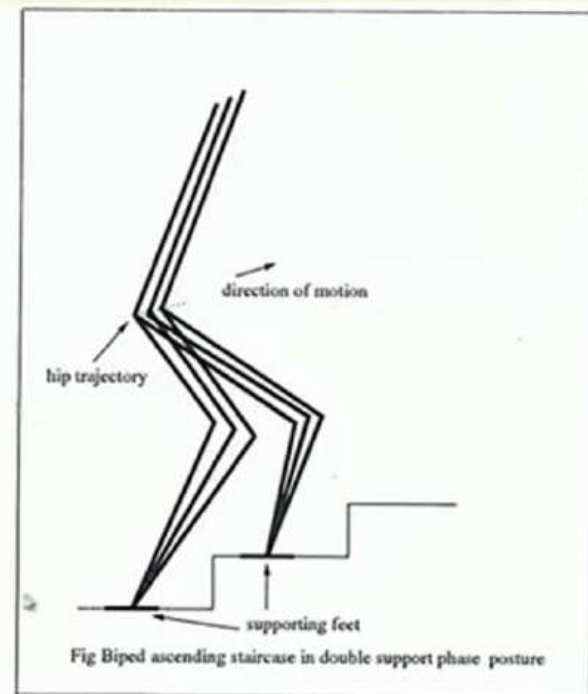
Dynamic balance margin

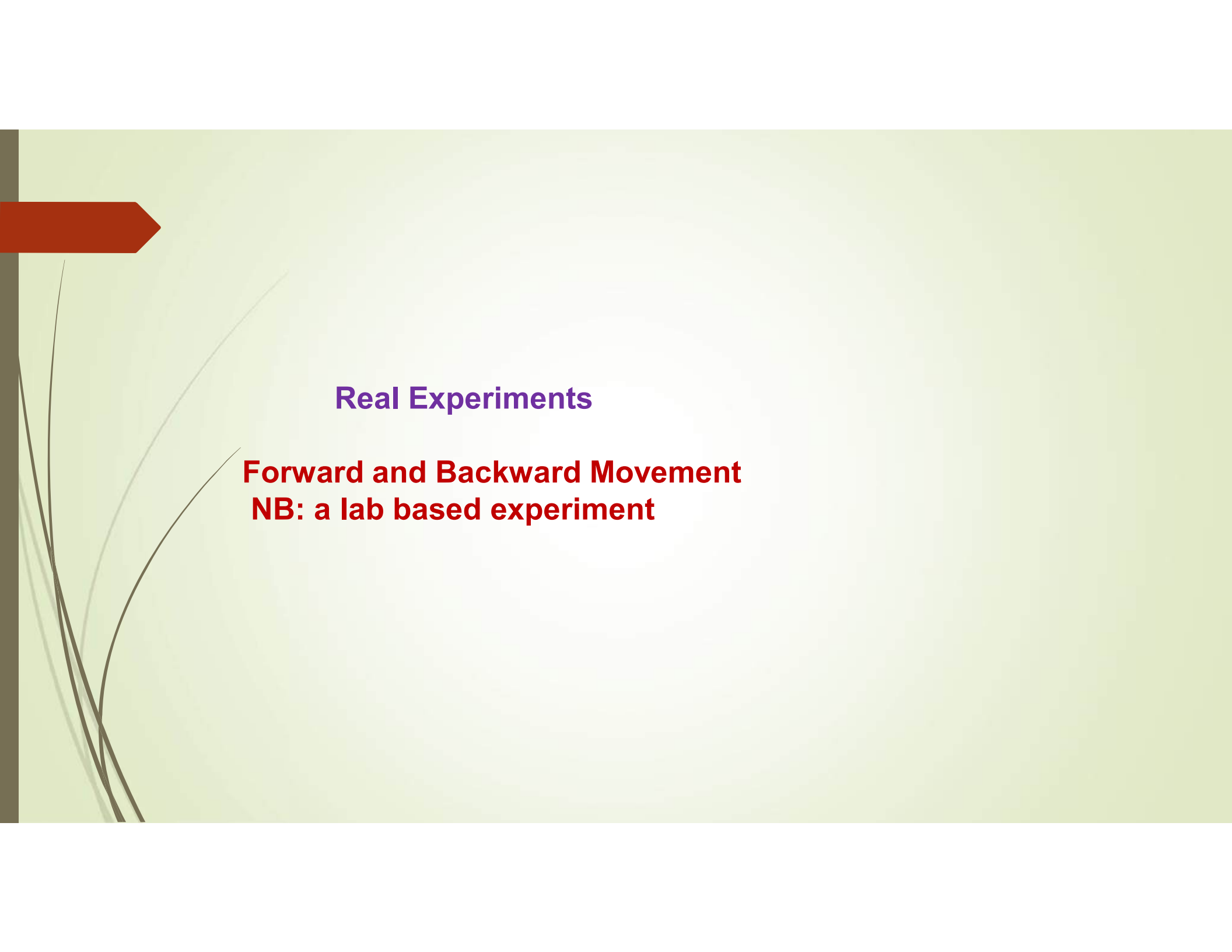
$$DBM_{system} = \frac{S_w - x_1 + x_2 + L_7}{2} - |X_{ZMP,system}|$$

Results of single support phase during ascending the staircases



Results of Double support phase during ascending the staircases





Real Experiments

Forward and Backward Movement
NB: a lab based experiment



Conclusion

- DSP plays a vital role in biped walking by ensuring stability and smooth motion transitions.
 - Effective ZMP control, force distribution, and trajectory planning are essential for robust walking.
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